Spatial Matching with Heterogeneous Demand: Coordinated Key Matching Indices for Ride-Hailing

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Abstract. Ride-hailing platforms may serve heterogeneous demand classes, whether differentiated by service tier or dynamic pricing, from a shared pool of drivers. Waiting customers may abandon before being matched, and even after matching, lengthy pickups can lead to cancellation. The challenge is that matching decisions interact across classes: serving one class consumes idle drivers, degrading pickup efficiency for the other. This coupling means that the decision of when to match each class cannot be optimized independently. Building on spatial matching models for single-class systems, we extend the framework to heterogeneous demand and formulate a joint steady-state optimization problem. The formulation applies to either welfare or revenue maximization, where each class's matching threshold interacts with the other through the shared driver pool. We derive Key Matching Indices (KMI), one for each class, whose form depends on whether the platform maximizes welfare or revenue. Each KMI decomposes into three interpretable terms: abandonment reduction, direct opportunity cost, and a cross-market term that captures how each class's threshold affects the other's service quality. Despite this difference, optimal thresholds are characterized by driving both indices to unity simultaneously. Based on this characterization, we propose a coordinated control algorithm that uses the Jacobian of the KMI map to adjust thresholds dynamically, with convergence from any initial state. The cross-market coupling term is essential: ignoring it leads to suboptimal coordination, while explicitly accounting for it enables the algorithm to outperform static priority rules and decoupled heuristics across a range of operating conditions. Numerical experiments reveal that the alignment between welfare and revenue objectives is load-dependent: under low load, both can be optimized simultaneously, but under congestion, revenue maximization requires disproportionate rationing of the lower-priority class.

Key words: Ride-hailing operations; Spatial matching; Service differentiation; Fluid approximation; Abandonment; Threshold policies

1. Introduction

The rapid expansion of on-demand ride-hailing platforms has transformed urban transportation. By algorithmically matching passengers with nearby drivers, platforms such as Uber, Lyft, and DiDi have reduced the search costs that traditionally plagued taxi markets. As these platforms mature, they increasingly serve heterogeneous demand classes from a shared pool of drivers. Such heterogeneity arises naturally: dynamic pricing creates segments with different willingness to pay, service tiers (e.g., UberX vs. Uber Comfort) attract distinct customer populations, and subscription programs differentiate members from casual users. This heterogeneity introduces a tension. Platform managers must balance revenue from high-margin customers

against maintaining service quality for price-sensitive segments. The shared driver pool means that prioritizing one class necessarily affects the other.

The operational complexity stems from a distinctive feature of ride-hailing: pickup times are endogenous. Unlike traditional queues where service times are exogenous, the time required to pick up a matched passenger depends jointly on the spatial density of idle drivers and waiting passengers. When demand surges or driver supply contracts, pickup times lengthen for all classes, leading to two forms of customer loss: passengers may abandon before being matched (pre-match abandonment), or cancel after matching if the quoted pickup time is too long (post-match cancellation). The key challenge is that matching decisions interact across classes. When the platform matches a passenger from one class, an idle driver is consumed, reducing the pool available for other classes. This capacity spillover degrades pickup efficiency for the remaining classes through the same spatial mechanism. Consequently, the optimal matching policy for one class depends on the state and policy of the other. Static priority rules or decoupled optimization methods do not capture this interdependence.

This paper develops an analytical framework for coordinated matching control. We use pickup rate thresholds as control levers: for each class, the platform specifies a minimum pickup rate below which matches are deferred. The central question is how to set these thresholds optimally, accounting for the coupling between classes.

We consider a ride-hailing system with two demand classes sharing a pool of drivers. Passengers arrive according to class-specific Poisson processes and join separate queues. The matching rate for each class follows a Cobb-Douglas spatial production function, where the rate increases in both the number of waiting passengers and the count of idle drivers. This formulation, established in prior work on single-class systems, captures the economies of spatial density: more passengers and more drivers lead to faster matching. Passengers exhibit two-stage impatience: they may abandon while waiting in queue, or cancel after matching if the pickup takes too long. The platform controls matching through pickup rate thresholds: a passenger is matched only when the current pickup rate exceeds a class-specific threshold.

Our main theoretical contribution is the derivation of the *Key Matching Indices* (KMI), denoted (ζ_1, ζ_2), which provide observable criteria for optimal threshold selection. Each index decomposes into three interpretable terms: (i) *abandonment reduction*, the benefit from reduced queue length and shorter waits; (ii) *direct opportunity cost*, the cost of consuming an idle driver; and (iii) a *cross-market term*, the effect on the other class through the shared driver pool. The third term is crucial: it captures the capacity spillover that single-class models and decoupled methods ignore. We prove that optimal thresholds are characterized by driving both indices to unity: $(\zeta_1, \zeta_2) = (1, 1)$. This condition holds for both welfare maximization and revenue maximization, though the precise form of the indices differs between objectives.

Building on the single-class spatial framework of Wang et al. (2024), we extend the model to heterogeneous demand and derive fluid approximations that remain tractable for optimization. The steady-state analysis

yields the KMI characterization, which reduces the high-dimensional optimization problem to a system of coupled nonlinear equations. Based on the KMI structure, we design a coordinated control algorithm that uses the Jacobian of the KMI map to adjust thresholds dynamically. Because the cross-market terms create genuine coupling, the Jacobian captures capacity spillovers that decoupled gradient methods miss. Numerical experiments demonstrate that the algorithm converges to optimal thresholds from any initial state. We systematically characterize the welfare-revenue frontier across different load conditions. Under low load, both objectives can be optimized simultaneously with generous thresholds. Under congestion, however, revenue maximization requires disproportionate rationing of the regular class to preserve service quality for premium customers. The KMI framework quantifies this trade-off and explains why heuristic tuning methods perform poorly in congested conditions.

1.1. Contributions

Our main contributions are as follows:

1. We derive Key Matching Indices (KMI), (ζ_1, ζ_2), which are observable, state-dependent indicators for optimal threshold selection. Each index decomposes into three interpretable terms: abandonment reduction, direct opportunity cost, and a cross-market term that captures how each class's threshold affects the other. We prove that the condition $(\zeta_1, \zeta_2) = (1, 1)$ characterizes optimal thresholds for both welfare and revenue objectives, reducing steady-state optimization to a tractable system of coupled equations. The welfare and revenue indices are structurally distinct: the former measures opportunity costs in physical units (throughput per hour), the latter in value units (revenue per hour), yet the unity condition holds for both.
2. We design a coordinated control algorithm that uses the Jacobian of the KMI map to adjust thresholds dynamically. The Jacobian is diagonally dominant (each threshold's own effect exceeds its cross-market effect), which ensures invertibility and local stability of the Newton-type iteration. By internalizing the cross-market coupling, the algorithm converges to optimal thresholds from any initial state and outperforms decoupled or myopic tuning rules that ignore cross-class externalities.
3. We characterize the welfare-revenue frontier across load conditions. While both objectives align under low load, they diverge significantly under congestion, where revenue maximization requires disproportionate rationing of the regular class to preserve spatial density for premium users. Specifically, the revenue-optimal policy reduces the premium queue by over 40% relative to the welfare-optimal policy but more than doubles the regular queue. The KMI framework reveals asymmetric sensitivities that explain why heuristic tuning methods perform poorly in congested regimes.
4. Relative to static priority rules and decoupled heuristics, our KMI-based control delivers substantial performance gains, particularly under congestion: the welfare-optimal KMI policy improves welfare by 19% and revenue by 9% over static dispatch. The primary source of these gains is the internalization of

cross-market externalities that decoupled policies ignore. From a managerial perspective, the results support a shift from static or heuristic dispatch rules toward model-based threshold management, with the KMI serving as real-time indicators for service differentiation.

1.2. Related Literature

Our work connects to four streams of research: spatial matching models, service systems with abandonment, dynamic matching and priority mechanisms, and platform objectives.

Spatial Matching and Endogenous Pickup Times. The defining feature of ride-hailing is that pickup times are endogenous, depending on the spatial density of drivers and passengers. This concept has been rigorously formalized by Besbes et al. (2021, 2022), who establish key results for single-class settings. Our work builds directly on the stochastic-fluid framework of Wang et al. (2024), which characterizes matching rates via Cobb-Douglas production functions and derives optimal thresholds under abandonment. Building on this foundation, we extend the framework to heterogeneous demand, where capacity spillovers arise from the shared driver pool that is absent in single-class models. An influential stream of recent research incorporates strategic driver behavior into spatial matching models. The seminal work of Afèche et al. (2023) develops a rigorous game-theoretic fluid model that captures the interplay between platform control and self-interested driver decisions, yielding a key insight: platforms may optimally reject demand at low-demand locations to induce strategic repositioning toward high-demand areas. Their characterization of equilibrium behavior under different control regimes provides essential benchmarks for evaluating platform policies. Garg and Nazerzadeh (2022) further explore driver strategic responses, while other extensions include pricing for matching failures (Castillo et al. 2025, 2017) and routing considerations (Braverman et al. 2019). Complementary to this line of work, which focuses on spatial repositioning across locations, our work examines how threshold-based matching controls can manage service differentiation across heterogeneous customer classes sharing a common driver pool.

Service Systems with Abandonment. Customer impatience is a central feature of on-demand services. Fluid approximations for many-server queues with abandonment began to emerge following the work of Whitt (2006); see also Zhang (2013), Liu and Whitt (2012, 2014), Kang and Ramanan (2012), Long and Zhang (2014) and the survey by Dai and He (2012). Bassamboo and Randhawa (2010) further show that fluid-based capacity prescriptions achieve $O(1)$ accuracy in the presence of customer abandonment. Kohlenberg and Gurvich (2025) derive scaling laws for the cost of impatience in dynamic matching, while Bai et al. (2019) and Feng et al. (2021) incorporate waiting time into customer utility. Building on these models, our work provides stochastic integral equations that capture two-stage customer loss (pre-match abandonment and post-match cancellation) within a spatial priority framework, yielding tractable fluid limits for optimization.

Dynamic Matching and Priority Mechanisms. Dynamic matching networks constitute an active research area. Gurvich and Ward (2014) prove the asymptotic optimality of discrete review matching policies under large market assumptions, while Özkan and Ward (2020) develop revenue-maximizing dispatch control through fluid limit analysis. In a related many-server setting, Bassamboo et al. (2005) establish the asymptotic optimality of joint admission and routing control using fluid limits. Hu and Zhou (2022) analyze market thickness in dynamic type matching, and further studies explore market thickness effects (Akbarpour et al. 2020, Baccara et al. 2020), shared-ride efficiency (Taylor 2024), and two-stage stochastic matching (Feng et al. 2024). Index-based priority rules have a rich history in multi-class queueing. The generalized $c\mu$ rule (van Mieghem 1995, Mandelbaum and Stolyar 2004) assigns priority through a product of service rate and marginal delay cost; Long et al. (2020) extend this to many-server queues with abandonment via a $c\mu/h$ rule that divides by the patience hazard rate, building on the virtual allocation framework of Long and Zhang (2019). Related work on priority scheduling in overloaded systems includes Puha and Ward (2019), Atar et al. (2014), Mandelbaum (1995), Mandelbaum and Pats (1998). Our approach implements priority through state-dependent pickup rate thresholds rather than discrete class prioritization. Like the $c\mu/h$ index, each KMI incorporates abandonment costs, but it additionally includes a cross-market externality term that captures capacity spillovers through the shared driver pool. Zhou et al. (2025) also employ prioritization mechanisms, though their focus is on peak/off-peak balancing whereas we address service differentiation across heterogeneous classes.

Platform Objectives and Economics. A substantial body of research examines how ride-hailing platforms balance revenue, welfare, and labor objectives. Pricing and wage mechanisms have been extensively studied (Taylor 2018, Benjaafar et al. 2022, Chen et al. 2022, Hu and Zhou 2020), as have dynamic pricing with demand learning (den Boer 2014, Keskin and Zeevi 2014, Besbes and Zeevi 2015) and surge pricing strategies (Hu et al. 2022, Yan et al. 2020). Özkan (2020) and Banerjee et al. (2015) analyze joint pricing and matching decisions, while Banerjee et al. (2022), Iglesias et al. (2019), Gurvich et al. (2019) study spatial capacity planning. Complementary to this pricing-centric literature, we treat prices as exogenous and focus on matching control through threshold policies. We employ the concept of marginal value conservation of idle drivers, inspired by state-dependent control approaches in closed queueing networks (Banerjee et al. 2018, Powell and Schultz 2004), to derive unified optimality criteria that apply to both welfare and revenue objectives.

The rest of the paper is organized as follows. Section 2 formulates the stochastic model for the two-class system with endogenous pickup times and two-stage customer loss. We construct and analyze the corresponding fluid model in Section 3. In Section 4, we derive the Key Matching Indices and characterize optimal thresholds for both welfare and revenue objectives. Section 5 presents numerical experiments demonstrating the performance of our coordinated control algorithm, and Section 6 concludes with managerial insights and future research directions.

2. Stochastic Model

We formulate a stochastic model for a ride-hailing system serving two heterogeneous customer classes from a shared pool of drivers. The key feature distinguishing this system from traditional queueing models is that pickup times are endogenous, depending on the spatial density of both waiting customers and idle drivers. Waiting customers may abandon before being matched (pre-match abandonment), and even after matching, lengthy pickups can lead to cancellation (post-match cancellation). The platform controls matching through threshold policies that specify minimum pickup rates required for each class.

The system operates with K homogeneous drivers who transition among five states. A driver can be idle and available for dispatch (count $Z_0(t)$ at time t), assigned to pick up a Class $i \in \{1,2\}$ customer (count $Z_{1i}(t)$), or actively transporting a Class i customer (count $Z_{2i}(t)$), where the driver population is conserved: $Z_0(t) + \sum_{i=1}^{2} Z_{1i}(t) + \sum_{i=1}^{2} Z_{2i}(t) = K$ for all $t \geq 0$.

Customers form class-specific queues, with $Q_i(t)$ denoting the number of Class i customers awaiting a match. The two classes are differentiated by service tier (e.g., UberX versus Uber Comfort) or dynamic pricing, which creates customer segments with different willingness to pay. What matters for our analysis is that both classes draw from the same idle-driver pool $Z_0(t)$, creating competition for scarce capacity. This coupling through the shared driver pool is the key source of interaction between the two classes and will play a central role in our optimization analysis.

Class i customers arrive according to a Poisson process $A_i(t)$ with rate $\lambda_i > 0$. While waiting in queue, customers are impatient. Each Class i customer has an exponentially distributed patience with rate $\theta_{0i} \geq 0$. If a customer's patience expires before being matched, they abandon. The cumulative abandonment count is $R_{0i}(t) = W_{0i}(\int_0^t \theta_{0i} Q_i(s)ds)$, where W_{0i} is an independent unit-rate Poisson process.

The defining feature of ride-hailing systems is that pickup times depend on spatial density. When more customers and more drivers are present, the average distance between a matched pair is shorter, leading to faster pickups. For a customer-driver pair matched at time t, we model the pickup time as a random variable following an exponential distribution with rate $\mu_{1i}(t)$. Based on the spatial model introduced in Wang et al. (2024), we make Assumption 1.

ASSUMPTION 1 **(Spatial Matching Function)**. *When a Class i customer is matched at time t, the pickup time for that match is exponentially distributed with rate*

$$\mu_{1i}(t) = C(Q_i(t))^{\alpha_1}(Z_0(t))^{\alpha_2}, \quad i = 1,2, \tag{1}$$

where $C, \alpha_1, \alpha_2 > 0$ are constants. Conditional on the system state at the moment of matching, the pickup time is independent of all other random variables.

We refer to $\mu_{1i}(t)$ as the pickup rate for Class i. The parameters C, α_1, α_2 capture the spatial characteristics of the operating region. Higher values of C correspond to denser urban areas with shorter distances. The

exponents α_1 and α_2 reflect returns to density: more waiting customers increase the likelihood that one is located near an idle driver, and more idle drivers increase coverage. A critical feature is that the rate $\mu_{1i}(t)$ is determined at the instant of matching and remains fixed throughout the subsequent pickup phase.

Threshold-Based Matching Policy. To maintain service quality, the platform applies a threshold-based matching policy by imposing class-specific thresholds μ_{11}^* and μ_{12}^*. A Class i customer is matched only when the pickup rate exceeds the threshold μ_{1i}^*, i.e., when $\mu_{1i}(t) \geq \mu_{1i}^*$. When $\mu_{1i}(t) < \mu_{1i}^*$, the platform defers matching and allows the number of waiting customers and idle drivers to accumulate until the pickup rate improves. Define the eligibility indicator $E_i(t) = \mathbf{1}\{\mu_{1i}(t) \geq \mu_{1i}^*\}$ and the activity indicator $I_i(t) = E_i(t)\mathbf{1}\{Q_i(t) > 0\}$. Class i is active at time t if both eligibility and queue nonemptiness conditions hold.

The Matching Process. Matches occur at event times triggered by either (i) customer arrivals or (ii) drivers becoming idle. Since a matching can only happen when a customer arrives or a driver becomes idle, we denote $N_D(t) = \sum_{j=1}^{2} R_{1j}(t) + \sum_{j=1}^{2} D_{2j}(t)$ as the cumulative count of idle driver releases, where R_{1j} and D_{2j} are pickup cancellations and trip completions defined below. We now describe the matching logic at these event times which leads $\mu_{1i}(t)$ to become bigger than the threshold μ_{1i}^*.

When a Class i customer arrives, a match is formed immediately if an idle driver is available ($Z_0(t^-) > 0$) and Class i is eligible ($E_i(t^-) = 1$). Otherwise, the customer joins the queue.

When a driver becomes idle, the matching logic depends on which classes are active at that moment:

- If exactly one class k is active ($I_k(t^-) = 1$ and $I_j(t^-) = 0$ for $j \neq k$), the driver is matched to Class k.
- If both classes are active ($I_1(t^-) = I_2(t^-) = 1$), the driver is assigned to Class 1 with probability $p_1(t^-) = Q_1(t^-)/(Q_1(t^-) + Q_2(t^-))$ and to Class 2 with probability $p_2(t^-) = Q_2(t^-)/(Q_1(t^-) + Q_2(t^-))$.
- If no class is active, the driver remains idle.

This matching rule leads to the following stochastic integral representations for cumulative matches:

$$\begin{aligned} M_1(t) &= \int_0^t \mathbf{1}\{E_1(s^-), Z_0(s^-) > 0\} dA_1(s) \\ &\quad + \int_0^t \left[I_1(s^-)(1 - I_2(s^-)) + I_1(s^-) I_2(s^-) p_1(s^-) \right] dN_D(s), \end{aligned} \tag{2}$$

$$\begin{aligned} M_2(t) &= \int_0^t \mathbf{1}\{E_2(s^-), Z_0(s^-) > 0\} dA_2(s) \\ &\quad + \int_0^t \left[I_2(s^-)(1 - I_1(s^-)) + I_1(s^-) I_2(s^-) p_2(s^-) \right] dN_D(s). \end{aligned} \tag{3}$$

Here s^- denotes the left limit, capturing the system state immediately before each event.

Once a Class i customer is matched at time τ, the assigned driver attempts pickup. Two events compete: pickup completion (exponentially distributed with the rate $\mu_{1i}(\tau)$ fixed at matching) and customer cancellation (exponentially distributed with rate $\theta_{1i} \geq 0$). If pickup completes first, the pair transitions to the trip phase (state Z_{2i}). If the customer cancels first, the driver returns to the idle pool and the customer exits the system.

We denote cumulative pickup completions by $D_{1i}(t)$ and cumulative cancellations by $R_{1i}(t)$. The construction of these processes accounts for pairs initially in the pickup state and pairs matched during $(0, t]$. For the k-th matched pair with matching time τ_k, let $Y_k \sim \text{Exp}(\mu_{1i}(\tau_k))$ be the pickup time and $V_k \sim \text{Exp}(\theta_{1i})$ be the remaining patience. The cumulative outcomes are formally given by sums of indicator functions over all relevant pairs.

Trip durations are exponentially distributed with a class-independent rate $\mu_2 > 0$. Upon trip completion D_{2i}, the driver returns to the idle state. The cumulative trip completions evolve as $D_{2i}(t) = W_{2i}(\int_0^t \mu_2 Z_{2i}(s)ds)$, where W_{2i} are independent unit-rate Poisson processes.

The queue dynamics follow from arrivals, abandonments, and matches:

$$Q_i(t) = Q_i(0) + A_i(t) - R_{0i}(t) - M_i(t), \quad i = 1, 2. \tag{4}$$

Driver state transitions are governed by matching, pickup outcomes, and trip completions. The idle pool evolves as

$$Z_0(t) = Z_0(0) + \sum_{j=1}^{2} R_{1j}(t) + \sum_{j=1}^{2} D_{2j}(t) - \sum_{j=1}^{2} M_j(t). \tag{5}$$

Pickup-state populations satisfy

$$Z_{1i}(t) = Z_{1i}(0) + M_i(t) - D_{1i}(t) - R_{1i}(t), \quad i = 1, 2, \tag{6}$$

and trip-state populations evolve as

$$Z_{2i}(t) = Z_{2i}(0) + D_{1i}(t) - D_{2i}(t), \quad i = 1, 2. \tag{7}$$

Together with the definitions of the cumulative processes, these flow balance equations fully specify the stochastic dynamics of the system.

In summary, we have introduced a stochastic model for a ride-hailing system with heterogeneous customer classes and two-stage customer loss (pre-match abandonment and post-match cancellation). The primitive parameters of the system are $\phi = (K, \lambda_1, \lambda_2, \theta_{01}, \theta_{02}, \theta_{11}, \theta_{12}, \mu_2, C, \alpha_1, \alpha_2)$, where the platform's decision variables are the matching thresholds (μ_{11}^*, μ_{12}^*).

From an economic perspective, the platform earns revenue from completed trips. Assume that the fixed fee charged per trip is p_f and that the price charged per unit of time is p_s. Expected revenue by time t can then be written as

$$\mathbb{E}\left[\sum_{i=1}^{2}\left(p_f D_{2i}(t) + p_s \int_0^t Z_{2i}(s)ds\right)\right]. \tag{8}$$

The prices and revenue-sharing arrangements between the platform and drivers are treated as exogenous. Our focus is on how the choice of thresholds (μ_{11}^*, μ_{12}^*) affects system performance metrics such as throughput, queue lengths, and service quality. In this sense, our analytical results can be applied with any revenue allocation between the platform and the drivers.

3. Performance Approximation: The Fluid Model

The stochastic model in Section 2 captures the microscopic dynamics of the system but is analytically intractable, particularly in large-scale settings. To obtain a tractable characterization, we develop a deterministic fluid model that approximates the stochastic system in a many-server, heavy-traffic regime. The fluid model reduces the analysis to a system of ordinary differential equations and admits a unique steady-state equilibrium that can be computed in closed form.

Because both customer classes draw from the same pool of idle drivers, any matching policy for one class affects the pickup efficiency of the other. The fluid equilibrium makes this coupling explicit: the steady-state queue lengths, pickup rates, and driver allocations are jointly determined by both thresholds (μ_{11}^*, μ_{12}^*). This structure provides the foundation for the optimization analysis in Section 4.

To develop this approximation, we analyze a sequence of stochastic systems indexed by the total number of drivers N. In the N-th system, passenger arrival rates grow proportionally with system size, $\lambda_i^N = N\lambda_i$, while all service and patience rates $(\mu_2, \theta_{0i}, \theta_{1i})$ remain of order one.

A subtlety arises in the spatial pickup process: as N increases, the average distance between idle drivers and waiting passengers would shrink roughly as $N^{-(\alpha_1+\alpha_2)}$ under standard spatial intensity assumptions. Without proper rescaling, the pickup process would become instantaneous in the limit, eliminating the critical role of endogenous pickup times that shape system performance. To preserve this nontrivial spatial effect, we therefore scale the matching constant as

$$C^N = \frac{C}{N^{\alpha_1+\alpha_2}},$$

which keeps the effective pickup rates $\mu_{1i}(t)$ of constant order.

Under this regime, the scaled stochastic processes (e.g., $Q_i(t)/N$, $Z_{ji}(t)/N$) converge in probability to their deterministic fluid counterparts, as formally established in Appendix EC.5 for the symmetric-exponent case $\alpha_1 = \alpha_2$.[1] The CN scaling thus provides a consistent bridge between the stochastic dynamics and the fluid limit: as the market grows, the expected pickup distance shrinks at rate $N^{-(\alpha_1+\alpha_2)}$, and the rescaling of C compensates for this contraction so that endogenous pickup times remain of first-order importance in the limiting dynamics.

3.1. Fluid Model Formulation

The fluid model describes the evolution of the proportions of passengers and drivers in various states. Let $q_i(t)$ be the fluid-scaled number of waiting class-i customers, $i \in \{1, 2\}$, and let $z_0(t)$, $z_{1i}(t)$, and $z_{2i}(t)$ be the proportions of drivers who are idle, assigned to class i, and busy with class i, respectively. As these are proportions of the total driver pool, they must satisfy the conservation law for all $t \geq 0$:

$$z_0(t) + z_{11}(t) + z_{12}(t) + z_{21}(t) + z_{22}(t) = 1. \tag{9}$$

[1] The convergence proof in Appendix EC.5 is carried out under $\alpha_1 = \alpha_2$, which is the empirically relevant case (Wang et al. 2024). All numerical experiments in this paper use $\alpha_1 = \alpha_2 = 0.5$.

The dynamics of the system are captured by a set of ordinary differential equations (ODEs). The core idea is to replace stochastic processes with their mean rates of change. Let $\pi_i(t)$ denote the instantaneous matching rate for class-i customers in the fluid limit. The cumulative number of class-i matches up to time t, denoted by $m_i(t)$, is given by $m_i(t) = \int_0^t \pi_i(s)ds$. The evolution of the system state is governed by the following equations, which describe how the number of waiting customers, idle drivers, and drivers in various states change over time:

$$q_1'(t) = \lambda_1 - \theta_{01}q_1(t) - \pi_1(t), \tag{10}$$

$$q_2'(t) = \lambda_2 - \theta_{02}q_2(t) - \pi_2(t), \tag{11}$$

$$z_0'(t) = \theta_{11}z_{11}(t) + \theta_{12}z_{12}(t) + \mu_2(z_{21}(t) + z_{22}(t)) - (\pi_1(t) + \pi_2(t)), \tag{12}$$

$$z_{11}'(t) = \pi_1(t) - (\mu_{11}(t) + \theta_{11})z_{11}(t), \tag{13}$$

$$z_{12}'(t) = \pi_2(t) - (\mu_{12}(t) + \theta_{12})z_{12}(t), \tag{14}$$

$$z_{21}'(t) = \mu_{11}(t)z_{11}(t) - \mu_2 z_{21}(t), \tag{15}$$

$$z_{22}'(t) = \mu_{12}(t)z_{12}(t) - \mu_2 z_{22}(t). \tag{16}$$

Here, the pickup rates $\mu_{11}(t)$ and $\mu_{12}(t)$ are endogenously determined by the system state through the spatial matching functions:

$$\mu_{11}(t) = Cq_1(t)^{\alpha_1}z_0(t)^{\alpha_2}, \tag{17}$$

$$\mu_{12}(t) = Cq_2(t)^{\alpha_1}z_0(t)^{\alpha_2}. \tag{18}$$

The matching rates, $\pi_1(t)$ and $\pi_2(t)$, are control variables determined by the platform's policy. We analyze a threshold-based policy, where the platform sets target pickup rates, μ_{11}^* and μ_{12}^*, for each customer class. Matching for a given class occurs only to prevent its pickup rate from exceeding the corresponding threshold. This policy can be stated formally as:

$$\mu_{1i}(t) \le \mu_{1i}^*, \quad \forall t \ge 0, \quad i \in \{1,2\}, \tag{19}$$

$$\int_0^t \mathbf{1}_{\{\mu_{1i}(s) < \mu_{1i}^*\}} d\pi_i(s) = 0, \quad \forall t \ge 0, \quad i \in \{1,2\}. \tag{20}$$

Condition (19) is a state constraint ensuring the pickup rate never surpasses the threshold. Condition (20) is a complementarity condition, stating that the matching rate $\pi_i(t)$ can be positive only when the pickup rate $\mu_{1i}(t)$ is precisely at its threshold μ_{1i}^*. This policy is intuitive: the platform defers matching until the pickup rate, which is inversely related to the expected pickup time, reaches its class-specific target.

To provide an alternative view of the system dynamics, we express the state variables in integral form. This representation clarifies the link between stochastic and deterministic dynamics. Let $d_{1i}(t)$ and $r_{1i}(t)$ be

the cumulative number of successful pickups and post-matching cancellations for class-i customers up to time t, respectively. These quantities depend on the entire history of the matching process $m_i(t)$.

For drivers who are in the "assigned to class i" state at time $t = 0$, with initial quantity $z_{1i}(0)$, the number who have successfully picked up their passenger by time t is given by $\frac{\mu_{1i}(0)}{\mu_{1i}(0)+\theta_{1i}} z_{1i}(0)(1 - e^{-(\mu_{1i}(0)+\theta_{1i})t})$. For drivers matched at a later time s in the interval $0 < s \leq t$, the probability of a successful pickup is $\frac{\mu_{1i}(s)}{\mu_{1i}(s)+\theta_{1i}}$. Integrating over all past matches, we obtain the complete expression for the cumulative successful pickups $d_{1i}(t)$ and cancellations $r_{1i}(t)$:

$$
\begin{aligned}
d_{1i}(t) &= \frac{\mu_{1i}(0)}{\mu_{1i}(0)+\theta_{1i}} z_{1i}(0)(1 - e^{-(\mu_{1i}(0)+\theta_{1i})t}) \\
&\quad + \int_0^t \frac{\mu_{1i}(s)}{\mu_{1i}(s)+\theta_{1i}} (1 - e^{-(\mu_{1i}(s)+\theta_{1i})(t-s)}) \pi_i(s) ds, \qquad (21) \\
r_{1i}(t) &= \frac{\theta_{1i}}{\mu_{1i}(0)+\theta_{1i}} z_{1i}(0)(1 - e^{-(\mu_{1i}(0)+\theta_{1i})t}) \\
&\quad + \int_0^t \frac{\theta_{1i}}{\mu_{1i}(s)+\theta_{1i}} (1 - e^{-(\mu_{1i}(s)+\theta_{1i})(t-s)}) \pi_i(s) ds. \qquad (22)
\end{aligned}
$$

Here, we assume for simplicity that the pickup rate for the initially assigned drivers $z_{1i}(0)$ is determined by the initial state $\mu_{1i}(0)$. Under our threshold policy, for any s where matching occurs ($\pi_i(s) > 0$), we have $\mu_{1i}(s) = \mu^*_{1i}$.

With these definitions, the system dynamics can be expressed as a set of integral equations:

$$
\begin{aligned}
q_i(t) &= q_i(0) + \lambda_i t - \theta_{0i} \int_0^t q_i(s) ds - m_i(t), \\
z_{1i}(t) &= z_{1i}(0) + m_i(t) - d_{1i}(t) - r_{1i}(t), \\
z_{2i}(t) &= z_{2i}(0) + d_{1i}(t) - \mu_2 \int_0^t z_{2i}(s) ds.
\end{aligned}
$$

By taking the time derivative of these integral equations (and applying the Leibniz rule to equations (21) and (22)), we recover the system of ordinary differential equations (ODEs) (10)-(16). For instance, $z'_{1i}(t) = m'_i(t) - d'_{1i}(t) - r'_{1i}(t) = \pi_i(t) - (\mu_{1i}(t) + \theta_{1i}) z_{1i}(t)$. This demonstrates that our ODE model is a direct and more tractable representation of these underlying integral dynamics.

With all components defined, we can now formally state the fluid model.

DEFINITION 1 (FLUID PROCESS). For a given set of parameters and decision thresholds (μ^*_{11}, μ^*_{12}), a process $x(t) = (q_1(t), q_2(t), z_0(t), z_{11}(t), z_{12}(t), z_{21}(t), z_{22}(t))$ is a fluid solution if it satisfies the conservation law (9) and the dynamic equations (10)-(20).

PROPOSITION 1. *The fluid model for the two-class customer system, as defined by equations* (9)-(20) *with the threshold matching policy where matching for class $i \in \{1, 2\}$ occurs if its potential matching intensity $\mu_{1i}(t) = C q_i(t)^{\alpha_1} z_0(t)^{\alpha_2}$ meets or exceeds a threshold μ^*_{1i}, has a unique solution trajectory $(q_i(t), z_0(t), z_{1i}(t), z_{2i}(t))$ for $t \geq 0$ given initial conditions.*

A rigorous proof of existence and uniqueness of the fluid model solution is provided in Appendix EC.5. The argument proceeds by first establishing that the system of ODEs under the threshold policy forms a well-posed initial value problem with locally Lipschitz right-hand sides, ensuring local existence and uniqueness via the Picard–Lindelöf theorem (Coddington and Levinson 1955). Global existence and boundedness follow from the conservation law and non-negativity constraints. Uniqueness of the solution trajectory is further established by showing that the mapping defined by the integral form of the dynamics is a contraction under suitable norms, exploiting the monotonicity and continuity properties of the system.

3.2. Equilibrium Analysis and Approximation

A key advantage of the fluid model is that it often converges to a unique, stable equilibrium. This equilibrium represents the long-run average behavior of the system and can be found by solving a system of algebraic equations, which is far more tractable than analyzing the stochastic model's steady-state distribution.

At equilibrium, the system is in steady state, so all time derivatives of the state variables are zero: $q_i'(t) = 0, z_{ji}'(t) = 0$. Setting the left-hand side of equations (10)-(16) to zero yields a set of flow-balance equations. Before stating the main convergence result, we impose feasibility requirements for a non-degenerate equilibrium: the pre-match abandonment rates must be strictly positive ($\theta_{0i} > 0$) and the thresholds must be attainable. Because $\theta_{0i} > 0$, the queue length is bounded in the absence of matching by $q_i(t) \le \lambda_i/\theta_{0i}$, and since $z_0(t) \le 1$, the pickup rate $\mu_{1i}(t) = Cq_i(t)^{\alpha_1} z_0(t)^{\alpha_2}$ has a natural upper bound $\mu_{1i}(t) \le C(\lambda_i/\theta_{0i})^{\alpha_1}$. Thus we restrict attention to policies with $\mu_{1i}^* \in (0, C(\lambda_i/\theta_{0i})^{\alpha_1}]$; setting a larger threshold would preclude matching for class i.

THEOREM 1 **(Convergence to Equilibrium)**. *Consider the fluid model where matching for class $i \in \{1,2\}$ occurs if its potential matching intensity $\mu_{1i}(t) = Cq_i(t)^{\alpha_1} z_0(t)^{\alpha_2}$ meets or exceeds a positive threshold $\mu_{1i}^* > 0$. If system parameters allow for a physical equilibrium where matching is active for both classes, that is, the derived matching rates π_i^* are non-negative and $z_0^* > 0$, and the conditions $\mu_{1i}^* \in \left(0, C\left(\frac{\lambda_i}{\theta_{0i}}\right)^{\alpha_1}\right]$ hold for $i = 1,2$, then there exists a unique non-negative solution $(q_1^*, q_2^*, z_0^*, z_{11}^*, z_{12}^*, z_{21}^*, z_{22}^*)$ with associated non-negative matching rates (π_1^*, π_2^*) to the equilibrium equations:*

$$\lambda_1 = q_1^*\theta_{01} + z_{11}^*\theta_{11} + z_{21}^*\mu_2, \tag{23}$$

$$\lambda_2 = q_2^*\theta_{02} + z_{12}^*\theta_{12} + z_{22}^*\mu_2, \tag{24}$$

$$z_{11}^*\mu_{11}^* = z_{21}^*\mu_2, \tag{25}$$

$$z_{12}^*\mu_{12}^* = z_{22}^*\mu_2, \tag{26}$$

$$1 = z_0^* + z_{11}^* + z_{12}^* + z_{21}^* + z_{22}^*, \tag{27}$$

$$\mu_{11}^* = C(q_1^*)^{\alpha_1}(z_0^*)^{\alpha_2}, \tag{28}$$

$$\mu_{12}^* = C(q_2^*)^{\alpha_1}(z_0^*)^{\alpha_2}. \tag{29}$$

This solution is the unique physical equilibrium point of the fluid model when both matching thresholds μ^*_{1i} are active. In addition, the fluid process $(q_i(t), z_0(t), z_{1i}(t), z_{2i}(t))$ for $i = 1, 2$, converges to this equilibrium point exponentially fast as $t \to \infty$.

The intuition behind these equilibrium equations is straightforward. Equations (23) and (24) represent the passenger flow balance for each class: the arrival rate must equal the total rate of departure through abandonment before matching ($q^*_i\theta_{0i}$), cancellation after matching ($z^*_{1i}\theta_{1i}$), and successful trip completion ($z^*_{2i}\mu_2$). Equations (25) and (26) represent the driver flow balance between the "assigned" and "busy" states: the rate at which drivers become busy, i.e., successful pickups ($z^*_{1i}\mu^*_{1i}$), must equal the rate at which they leave the busy state, i.e., trip completions ($z^*_{2i}\mu_2$). Equation (27) is the driver conservation law at steady state. Finally, equations (28) and (29) enforce the matching policy at equilibrium. The proof of this theorem is detailed in Appendix EC.5.

3.3. Numerical Validation and Strategic Insights

This section evaluates the accuracy of the fluid approximation and illustrates how the model reveals operational implications for platforms managing heterogeneous customer classes. We assess the approximation along two dimensions (process-level convergence and steady-state accuracy) and use the equilibrium structure to examine the effects of threshold-based differentiation.

We consider a system with $N = 1000$ drivers and the baseline parameters summarized in Table 1. Service differentiation is implemented by varying the matching thresholds (μ^*_{11}, μ^*_{12}) across three representative policies:

$$\text{A} : (6, 6), \qquad \text{B} : (4, 8), \qquad \text{C} : (8, 4),$$

where smaller thresholds correspond to higher service standards.

Table 1 Baseline Parameters for Numerical Experiments

Parameter	Description	Value
λ_1, λ_2	Arrival rates (Class 1, Class 2)	2.0, 3.0
θ_{01}, θ_{02}	Pre-match abandonment rates	2.0, 1.0
θ_{11}, θ_{12}	Post-match cancellation rates	3.0, 4.0
μ_2	Trip completion rate	10.0
C	Spatial matching constant	50.0
α_1, α_2	Spatial exponents	0.5, 0.5

Figure 1 plots a sample path of the scaled stochastic process and the corresponding fluid trajectory under Scenario A. The two dynamics nearly coincide, including their transient behavior and convergence rate. Similar agreement holds under Scenarios B and C, as well as at higher load levels, indicating that the CN-scaling fluid model captures the essential system evolution beyond lightly loaded regimes.

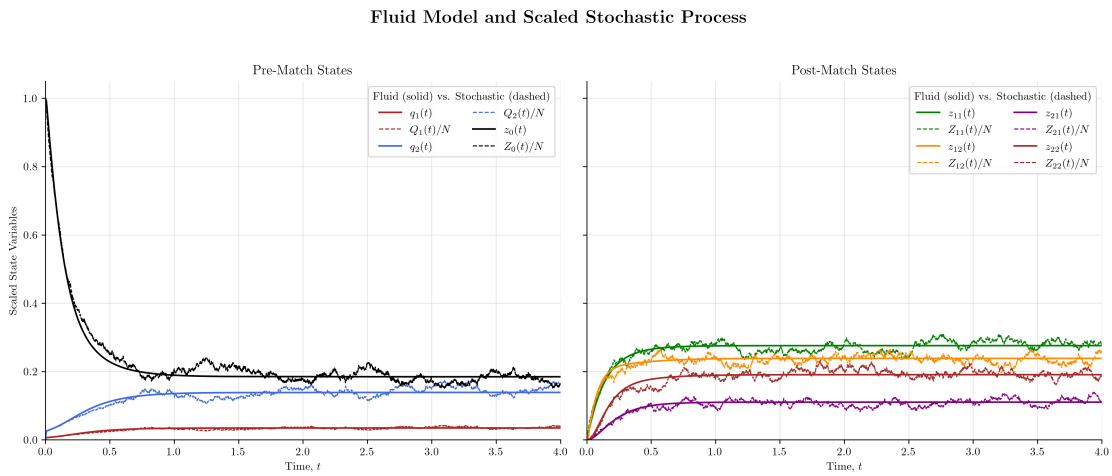


Figure 1 Fluid and Stochastic Trajectories under Scenario A ($N = 1000$).

Table 2 compares steady-state simulations for increasing system sizes with the fluid equilibrium from Theorem 1. Two patterns consistently emerge: (i) scaled stochastic averages converge numerically to the fluid predictions, and (ii) the variance diminishes as N grows. For instance, in Scenario A (high load) the simulated ratios $E[Q_1]/N = 0.0710$ and $E[Q_2]/N = 0.0712$ at $N = 1000$ differ from the fluid values by less than 1%. Across all scenarios and load conditions, queue-length discrepancies remain within 6%, supporting the model's reliability for strategic analysis.

Table 2 Comparison of Fluid Approximation and Stochastic System in Steady State under Different Policies and Loads.

Scenario	**Load**	**N**	$E[Q_1]/N$	$E[Q_2]/N$	$E[Z_0]/N$	$E[Z_{11}]/N$	$E[Z_{12}]/N$	$E[Z_{21}]/N$	$E[Z_{22}]/N$
A (Symmetric)	$\lambda \times 0.5$	100	0.0100 ± 0.0000	0.0102 ± 0.0000	0.7971 ± 0.0010	0.0539 ± 0.0004	0.0736 ± 0.0006	0.0317 ± 0.0004	0.0437 ± 0.0005
		500	0.0168 ± 0.0000	0.0169 ± 0.0000	0.8012 ± 0.0005	0.0513 ± 0.0003	0.0729 ± 0.0003	0.0308 ± 0.0002	0.0437 ± 0.0003
		1000	0.0174 ± 0.0000	0.0175 ± 0.0000	0.8015 ± 0.0004	0.0510 ± 0.0002	0.0732 ± 0.0002	0.0306 ± 0.0001	0.0438 ± 0.0002
		Fluid	**0.0180**	**0.0180**	**0.8004**	**0.0516**	**0.0732**	**0.0309**	**0.0439**
	$\lambda \times 2.0$	100	0.0718 ± 0.0011	0.0729 ± 0.0011	0.2038 ± 0.0026	0.2082 ± 0.0011	0.2894 ± 0.0013	0.1242 ± 0.0008	0.1744 ± 0.0012
		500	0.0705 ± 0.0004	0.0707 ± 0.0004	0.2075 ± 0.0011	0.2051 ± 0.0004	0.2904 ± 0.0007	0.1225 ± 0.0004	0.1745 ± 0.0004
		1000	0.0710 ± 0.0002	0.0712 ± 0.0002	0.2066 ± 0.0005	0.2058 ± 0.0003	0.2907 ± 0.0003	0.1228 ± 0.0002	0.1742 ± 0.0003
		Fluid	**0.0715**	**0.0715**	**0.2013**	**0.2063**	**0.2928**	**0.1238**	**0.1757**
B (Prioritize VIPs)	$\lambda \times 0.5$	100	0.0000 ± 0.0000	0.0287 ± 0.0001	0.7929 ± 0.0016	0.0716 ± 0.0009	0.0599 ± 0.0007	0.0282 ± 0.0004	0.0474 ± 0.0006
		500	0.0071 ± 0.0000	0.0311 ± 0.0000	0.7969 ± 0.0006	0.0689 ± 0.0004	0.0593 ± 0.0003	0.0274 ± 0.0002	0.0475 ± 0.0002
		1000	0.0076 ± 0.0000	0.0316 ± 0.0000	0.7966 ± 0.0006	0.0690 ± 0.0003	0.0595 ± 0.0002	0.0276 ± 0.0001	0.0473 ± 0.0002
		Fluid	**0.0080**	**0.0322**	**0.7955**	**0.0691**	**0.0598**	**0.0277**	**0.0479**
	$\lambda \times 2.0$	100	0.0322 ± 0.0006	0.1449 ± 0.0023	0.1863 ± 0.0027	0.2761 ± 0.0019	0.2371 ± 0.0010	0.1118 ± 0.0008	0.1887 ± 0.0009
		500	0.0338 ± 0.0002	0.1383 ± 0.0007	0.1908 ± 0.0010	0.2745 ± 0.0005	0.2365 ± 0.0003	0.1093 ± 0.0004	0.1888 ± 0.0004
		1000	0.0341 ± 0.0001	0.1382 ± 0.0004	0.1901 ± 0.0006	0.2746 ± 0.0004	0.2374 ± 0.0003	0.1090 ± 0.0003	0.1890 ± 0.0003
		Fluid	**0.0347**	**0.1386**	**0.1847**	**0.2758**	**0.2384**	**0.1103**	**0.1908**
C (Prioritize Regulars)	$\lambda \times 0.5$	100	0.0276 ± 0.0001	0.0000 ± 0.0000	0.7977 ± 0.0018	0.0406 ± 0.0004	0.0923 ± 0.0010	0.0317 ± 0.0006	0.0376 ± 0.0006
		500	0.0308 ± 0.0000	0.0070 ± 0.0000	0.8000 ± 0.0004	0.0393 ± 0.0002	0.0921 ± 0.0003	0.0316 ± 0.0001	0.0370 ± 0.0002
		1000	0.0314 ± 0.0000	0.0075 ± 0.0000	0.7996 ± 0.0005	0.0395 ± 0.0001	0.0925 ± 0.0003	0.0317 ± 0.0001	0.0368 ± 0.0001
		Fluid	**0.0320**	**0.0080**	**0.7988**	**0.0396**	**0.0927**	**0.0317**	**0.0371**
	$\lambda \times 2.0$	100	0.1324 ± 0.0018	0.0304 ± 0.0005	0.1969 ± 0.0023	0.1574 ± 0.0008	0.3721 ± 0.0017	0.1255 ± 0.0009	0.1481 ± 0.0012
		500	0.1301 ± 0.0006	0.0320 ± 0.0002	0.2010 ± 0.0009	0.1564 ± 0.0004	0.3694 ± 0.0007	0.1251 ± 0.0004	0.1481 ± 0.0004
		1000	0.1297 ± 0.0005	0.0321 ± 0.0001	0.2015 ± 0.0007	0.1565 ± 0.0003	0.3695 ± 0.0005	0.1252 ± 0.0003	0.1473 ± 0.0002
		Fluid	**0.1305**	**0.0326**	**0.1962**	**0.1581**	**0.3709**	**0.1265**	**0.1484**

Notes. Base parameters: $\lambda_1 = 1.0$, $\lambda_2 = 1.5$, $\theta_{01} = 2.0$, $\theta_{02} = 1.0$, $\theta_{11} = 3.0$, $\theta_{12} = 4.0$, $C = 50.0$, $\alpha_1 = \alpha_2 = 0.5$, $\mu_2 = 10.0$. Matching thresholds (μ_{11}^*, μ_{12}^*) vary by scenario: A (6,6), B (4,8), C (8,4).

The numerical experiments reveal how the shared pool of idle drivers mediates cross-class externalities. As indicated by the equilibrium matching conditions (28)–(29), both customer classes rely on the same

endogenous idle-driver level z_0^*; consequently, raising the service standard for one segment inevitably constrains the effective matching rate for the other. This trade-off is clearly visible under high load when shifting from the symmetric benchmark (Scenario A) to a policy prioritizing VIPs (Scenario B). In this case, the VIP queue is reduced by more than half ($0.0715 \to 0.0347$, a 51% reduction), yet this improvement comes at a significant cost to the regular queue ($0.0715 \to 0.1386$, a 94% increase), demonstrating the cross-class externalities inherent in shared-capacity systems.

Beyond simple trade-offs, the system exhibits distinct structural behaviors under asymmetric prioritization. For instance, under Scenario C at high load, the aggressive threshold set for regular customers ($\mu_{12}^* = 4$) reduces the idle-driver pool ($z_0^* = 0.1962$) while substantially increasing the VIP queue ($q_1^* = 0.1305$, compared to 0.0715 in Scenario A). This illustrates how prioritizing one class can significantly impact waiting times for the other class. The convergence of stochastic simulations toward these fluid predictions confirms that these trade-offs are intrinsic properties of the system dynamics, not artifacts of the approximation.

From a strategic perspective, these findings suggest that threshold-based policies (μ_{11}^*, μ_{12}^*) offer greater flexibility than binary priority rules, providing a continuous mechanism to regulate the allocation of idle supply. Small adjustments in thresholds translate into predictable shifts in equilibrium queue lengths and pickup rates, enabling platforms to quantify the trade-off between revenue and customer retention. The fluid model thus provides both an accurate approximation for transient and steady-state performance and a tractable foundation for optimizing service levels across heterogeneous customer classes.

4. Optimal Matching Policy Design

Having established the fluid model and its steady-state properties in Section 3, we now use this framework to design optimal matching policies. The fluid approximation provides a tractable analytical basis for characterizing the interplay between matching thresholds, driver supply, and passenger waiting times. This section formulates the threshold optimization problem and derives the necessary optimality conditions.

The central prescriptive question is: *how should the platform dynamically adjust the thresholds* (μ_{11}^*, μ_{12}^*) *to optimize a given objective?* The answer involves a careful balancing of three competing forces: *(i)* the benefit of reducing customer abandonment by matching earlier, *(ii)* the opportunity cost of committing a driver to one class, and *(iii)* the externality imposed on the other class through the shared idle-driver pool. These trade-offs are encoded in the *Key Matching Indices* (KMI), from which we derive a coordinated control algorithm capable of maximizing platform objectives in real time.

4.1. Problem Formulation

We study the prescriptive design of matching policies by characterizing the optimal thresholds in the fluid limit. To streamline the exposition, we introduce the vector notation $\boldsymbol{\mu} = (\mu_{11}, \mu_{12})^\top \in \mathbb{R}_{++}^2$ for the threshold pair, where each component μ_{1i} corresponds to the threshold μ_{1i}^* introduced in Section 3. Let

$\mathbf{x}^* = (z_0^*, z_{11}^*, z_{12}^*, z_{21}^*, z_{22}^*, q_1^*, q_2^*)$ denote the unique steady state induced by a given $\boldsymbol{\mu}$, as characterized in Theorem 1. The platform seeks to maximize a weighted throughput objective:

$$\max_{\boldsymbol{\mu}>0} \quad J(\boldsymbol{\mu}) := w_1 z_{21}^*(\boldsymbol{\mu}) + w_2 z_{22}^*(\boldsymbol{\mu}), \tag{30}$$

where w_i denotes the economic weight of class i. Setting $w_1 = w_2 = 1$ yields a *social welfare* objective that maximizes the total rate of successfully completed trips. Alternatively, setting $w_i = p_i$, where p_i is the price charged to class i, yields a *platform revenue* objective. Since the trip completion rate for class i is $\mu_2 z_{2i}^*$, the objective $\sum_i w_i z_{2i}^*$ is proportional to weighted throughput; setting $w_i = p_i$ thus weights each class's contribution by its per-trip revenue.

The challenge in solving (30) stems from the implicit dependence of $\mathbf{x}^*$ on $\boldsymbol{\mu}$ through the nonlinear equilibrium conditions (23)–(29). Standard gradient-based methods require evaluating the sensitivity $\nabla_{\boldsymbol{\mu}} \mathbf{x}^*$, which is not available in closed form. Our approach circumvents this difficulty by deriving compact *optimality indicators*, the Key Matching Indices, that encapsulate the first-order conditions in terms of observable steady-state quantities.

4.2. Optimality Conditions and Key Matching Indices

We derive the first-order necessary conditions for problem (30) using the KKT framework applied to the fluid equilibrium. The derivation, detailed in Appendix EC.3.1, constructs a Lagrangian incorporating three classes of constraints: the passenger flow balance for each class, the driver flow balance between pickup and trip states, and the overall driver conservation law. The Lagrange multipliers associated with these constraints admit natural economic interpretations as shadow prices, and their manipulation yields the optimality conditions stated below.

The first result establishes a necessary aggregate condition that any optimal solution must satisfy.

PROPOSITION 2 **(Idle-Driver Marginal Value Conservation)**. *Any interior optimal solution $\boldsymbol{\mu}$ to problem (30) must satisfy*

$$\sum_{i=1}^{2} \frac{\alpha_2 (z_{1i}^*/z_0^*)}{s_i} = 1, \tag{31}$$

where

$$s_i \triangleq 1 - \alpha_1 \frac{z_{1i}^* \theta_{1i}}{q_i^* \theta_{0i}}$$

captures the elasticity of the queue length with respect to matching volume.

Condition (31) has a natural interpretation: at optimality, the marginal value of an idle driver must be fully allocated across both matching routes. The term $\alpha_2 (z_{1i}^*/z_0^*)/s_i$ can be understood as the shadow contribution of idle-driver supply to matching class i, adjusted by the queue elasticity s_i. This is a spatial matching analogue of the classical equi-marginal principle: at optimality, the shadow value of an idle driver must be

equalized across both matching routes. When the conservation law is violated, the platform can improve its objective by reallocating idle-driver capacity between the two classes.

While Proposition 2 provides a necessary aggregate condition, it does not prescribe *how* to set the individual thresholds μ_{11}^* and μ_{12}^*. To resolve this, we derive the *Key Matching Indices* (KMI), which provide class-specific optimality criteria. These indices emerge from eliminating the Lagrange multipliers between the first-order conditions and encode the trade-off between three competing effects: *(i)* the benefit of reducing customer abandonment by matching earlier, *(ii)* the direct opportunity cost of committing an idle driver, and *(iii)* the externality imposed on the other class through the shared driver pool.

DEFINITION 2 (KEY MATCHING INDICES). For each class $i \in \{1, 2\}$, the Key Matching Index is defined as follows, with $j \neq i$ denoting the other class.

1. **Welfare objective** ($w_1 = w_2 = 1$):

$$\zeta_i^W = \alpha_1 \frac{z_{1i}^* \theta_{1i}}{q_i^* \theta_{0i}} + \alpha_2 \frac{z_{1i}^*}{z_0^*} + \frac{\alpha_2 \left(\frac{1}{\mu_{1i}^*} + \alpha_1 \frac{z_{1i}^*}{q_i^* \theta_{0i}} \right)}{\frac{z_0^*}{\mu_{1j}^* z_{1j}^*} + \alpha_1 \frac{z_0^*}{q_j^* \theta_{0j}}}. \tag{32}$$

2. **Revenue objective** ($w_i = p_i$):

$$\zeta_i^R = \alpha_1 \frac{z_{1i}^* \theta_{1i}}{q_i^* \theta_{0i}} + \alpha_2 \frac{z_{1i}^*}{z_0^*} + \alpha_2 \frac{p_j}{p_i} \frac{\mu_{1j}^*}{\mu_{1i}^*} \frac{z_{1j}^*}{z_0^*} \frac{\mu_{1i}^* s_i + \mu_2 A_i}{\mu_{1j}^* s_j + \mu_2 A_j}, \tag{33}$$

where $A_i \triangleq 1 + \alpha_1 \frac{\mu_{1i}^* z_{1i}^*}{q_i^* \theta_{0i}}$ is an auxiliary elasticity factor.

The structure of both indices reflects the trade-offs in threshold optimization, with each term capturing one of the three competing forces:

- **Abandonment Reduction** (first term): The ratio $\alpha_1 z_{1i}^* \theta_{1i} / (q_i^* \theta_{0i})$ measures the marginal benefit of reducing customer loss. Lowering the threshold increases matching intensity, which shortens the queue and reduces both pre-match abandonment (at rate θ_{0i}) and post-match cancellation (at rate θ_{1i}). This term is large when the queue is long relative to the matching flow, indicating high potential gains from more aggressive matching.
- **Direct Opportunity Cost** (second term): The ratio $\alpha_2 z_{1i}^* / z_0^*$ captures the cost of consuming an idle driver from the shared pool. Each match commits a driver to the pickup-then-trip cycle, temporarily removing capacity from the system. This term is large when picking-up drivers are numerous relative to idle drivers, signaling that the pool is already depleted.
- **Cross-Market Externality** (third term): The remaining term accounts for the spillover effect on the other class. When the platform matches more aggressively for class i, it draws down the idle-driver pool z_0^*, degrading pickup efficiency for class j. This externality is the key source of coupling between the two classes and distinguishes our framework from single-class models. Examining the welfare

index (32), the cross-market term is large when class i matches aggressively (low threshold μ_{1i}^*, so that the numerator term $1/\mu_{1i}^*$ is large) relative to the absorptive capacity of class j, as measured by the idle-driver pool available per unit of class j matching flow. Operationally, a large cross-market term signals that further threshold reduction for class i would come at a disproportionate cost to class j's service quality.

At optimality, these three effects must be balanced such that each index equals unity: $\zeta_i = 1$.

REMARK 1 (STRUCTURAL DIVERGENCE BETWEEN WELFARE AND REVENUE). The revenue index ζ_i^R does not reduce to the welfare index ζ_i^W even under symmetric pricing ($p_1 = p_2 = 1$), as established in Appendix EC.3.2. This divergence reflects an economic distinction: ζ_i^W measures opportunity costs in *physical units* (throughput per hour), whereas ζ_i^R measures them in *value units* (revenue per hour). In the welfare formulation, the cross-market term penalizes the reduction in matching volume for class j caused by serving class i. In the revenue formulation, the Lagrange multipliers incorporate prices directly, yielding $\gamma_i \propto p_i/(\mu_{1i}^* s_i + \mu_2 A_i)$, which embeds the marginal revenue substitution rate into the index. Consequently, even with equal prices, the revenue-optimal operating point differs from the welfare-optimal one: revenue maximization values the *rate of revenue generated per driver-hour* (incorporating turnover and cancellation risk) rather than the raw *volume* of completed trips.

The following theorem establishes that the KMI provide a complete characterization of stationary points.

THEOREM 2 **(Stationarity Characterization)**. *A threshold vector $\boldsymbol{\mu}$ is a stationary point of problem* (30) *if and only if:*

1. *the idle-driver conservation condition* (31) *holds, and*
2. *both indices satisfy*

$$\zeta_1(\boldsymbol{\mu}) = 1 \qquad \textit{and} \qquad \zeta_2(\boldsymbol{\mu}) = 1,$$

where $\zeta_i = \zeta_i^W$ for welfare or $\zeta_i = \zeta_i^R$ for revenue.

The proof is provided in Appendix EC.3.1.

REMARK 2 (ON GLOBAL OPTIMALITY). Since the spatial matching function is non-convex, the KMI stationarity condition is necessary but not sufficient for global optimality. Uniqueness of the fluid equilibrium (Theorem 1) and numerical evidence of unimodality (Figures 5–7) support the conclusion that the stationary point is the global maximizer; see Appendix EC.3.1.4 for details.

To state the result more compactly, we introduce the vector notation $\boldsymbol{\zeta} \triangleq (\zeta_1, \zeta_2)^\top$ and recall the threshold vector $\boldsymbol{\mu}$ defined in Section 4.1. Theorem 2 then transforms the threshold optimization problem into a root-finding problem: the optimal thresholds are characterized by the system of equations $\boldsymbol{\zeta}(\boldsymbol{\mu}) = \mathbf{1}$. This reformulation is useful for two reasons. First, the indices ζ_i are computed directly from observable steady-state quantities $(z_0^*, z_{1i}^*, z_{2i}^*, q_i^*)$, obviating the need to compute implicit gradients $\nabla_{\boldsymbol{\mu}} \mathbf{x}^*$. Second, the condition

$\zeta_i = 1$ has a natural interpretation: at optimality, the marginal benefit of tightening the threshold for class i (reducing abandonment, improving pickup speed) must exactly equal its marginal cost (consuming driver capacity, degrading service for class j). Since the sensitivity analysis reveals $\partial\zeta_i/\partial\mu_{1i} < 0$, when $\zeta_i > 1$, the marginal benefit exceeds the marginal cost, indicating that the threshold μ^*_{1i} should be *raised* to restore balance; when $\zeta_i < 1$, the threshold should be *lowered*.

4.3. Sensitivity Analysis and Numerical Properties

Theorem 2 characterizes the optimal thresholds as roots of the system $\boldsymbol{\zeta}(\boldsymbol{\mu}) = \mathbf{1}$. However, designing an algorithm to find these roots requires understanding the sensitivity of $\boldsymbol{\zeta}$ to changes in $\boldsymbol{\mu}$, namely the Jacobian matrix $\mathbf{J} = \nabla_{\boldsymbol{\mu}}\boldsymbol{\zeta}$. Deriving closed-form expressions for $\mathbf{J}$ is analytically intractable because each steady-state component $(z^*_0, z^*_{1i}, z^*_{2i}, q^*_i)$ depends implicitly on the threshold vector $\boldsymbol{\mu}$ through the equilibrium equations. To circumvent this difficulty, we characterize the Jacobian structure through extensive numerical experiments, which reveal consistent patterns.

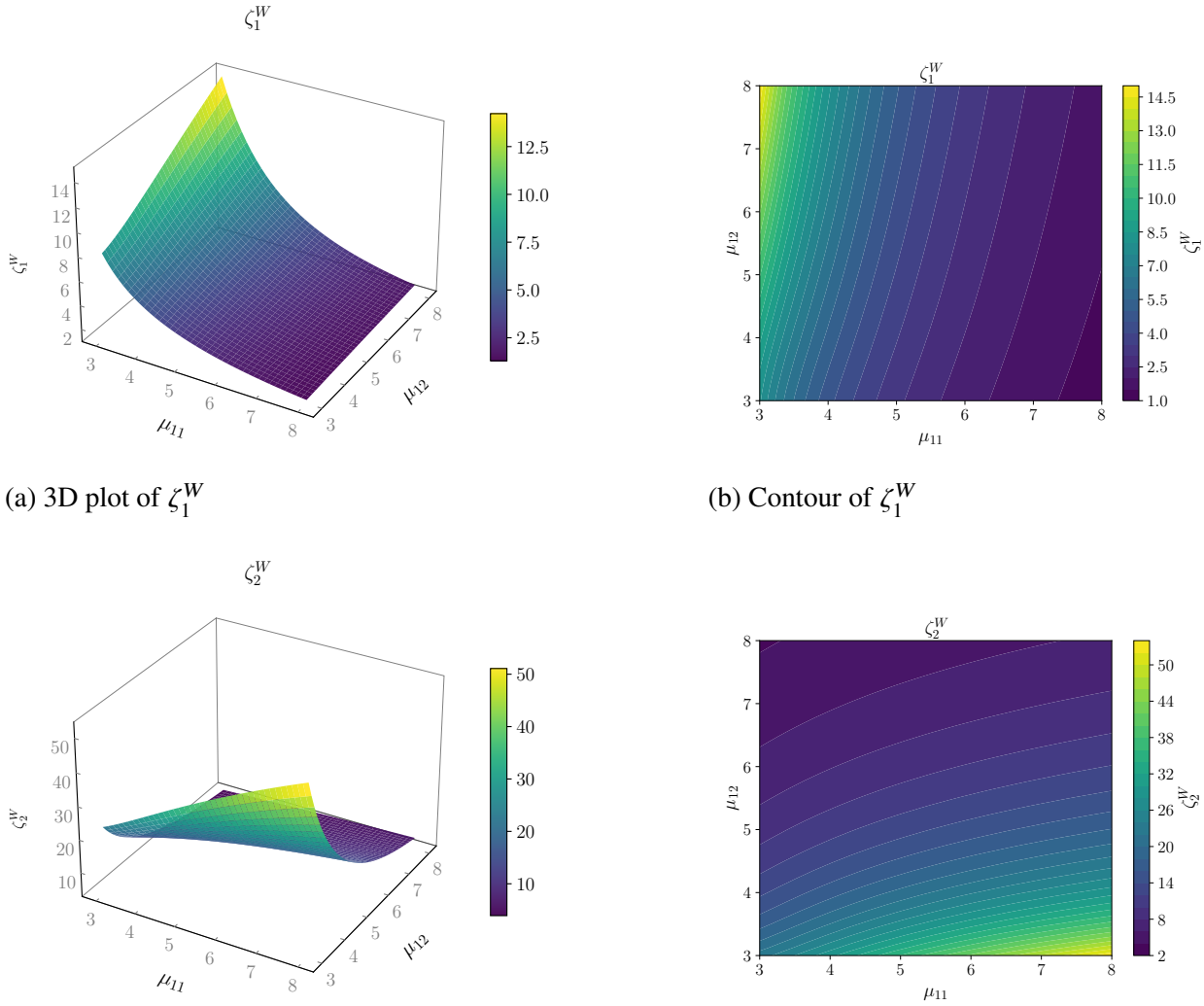
Figures 2 and 3 display the response surfaces of ζ^W_i and ζ^R_i to variations in the threshold pair (μ_{11}, μ_{12}). These surfaces reveal two critical structural properties that persist across objectives and parameter regimes:

1. **Direct monotonicity:** $\partial\zeta_i/\partial\mu_{1i} < 0$. Increasing the threshold μ_{1i} restricts matching opportunities for class i: fewer passengers are matched, the queue q^*_i grows, and drivers accumulate in the idle pool z^*_0. All three terms of ζ_i decrease as a result: the abandonment-reduction benefit shrinks (shorter pickup-to-queue ratio), the direct opportunity cost falls (more idle drivers available), and the cross-market externality weakens. Thus, raising μ_{1i} mechanically reduces ζ_i.
2. **Cross-market coupling:** $\partial\zeta_i/\partial\mu_{1j} > 0$ for $j \neq i$. Raising the threshold for class j (tightening matching) reduces the matching volume for class j, which alters the equilibrium in favor of class i. As a result, ζ_i increases. This positive cross-sensitivity means that threshold adjustments in one market spill over to affect the optimality conditions in the other, creating the coupling that necessitates coordinated control.

A crucial finding is that the *direct effect dominates the cross-market effect*: $\left|\frac{\partial\zeta_i}{\partial\mu_{1i}}\right| > \left|\frac{\partial\zeta_i}{\partial\mu_{1j}}\right|$ for all $i \neq j$. This inequality ensures that the Jacobian matrix $\mathbf{J}$ is diagonally dominant, which has two important consequences. First, diagonal dominance guarantees that $\mathbf{J}$ is invertible, so the system $\boldsymbol{\zeta}(\boldsymbol{\mu}) = \mathbf{1}$ admits a locally unique solution. Second, it ensures the local stability of Newton-type iterations, preventing oscillatory or divergent behavior.

However, the non-zero off-diagonal entries underscore a critical limitation of independent tuning strategies. Consider a naive coordinate-descent approach that adjusts μ_{11} and μ_{12} in isolation, ignoring cross-market effects. Suppose $\zeta_1 > 1$ initially; the controller raises μ_{11} to reduce ζ_1 toward unity. Due to cross-market coupling ($\partial\zeta_2/\partial\mu_{11} > 0$), this adjustment simultaneously raises ζ_2. If ζ_2 now exceeds unity, the controller responds by raising μ_{12}, which in turn raises ζ_1 back above unity via $\partial\zeta_1/\partial\mu_{12} > 0$. This ping-pong dynamic can produce coupled oscillations that fail to converge. The *coordinated* control algorithm developed in the next subsection explicitly accounts for the full Jacobian structure to avoid such instabilities.

Figure 2 Sensitivity analysis of Welfare-based Key Matching Indices (ζ^W)



(a) 3D plot of ζ_1^W

(b) Contour of ζ_1^W

(c) 3D plot of ζ_2^W

(d) Contour of ζ_2^W

Notes. The plots show the response of ζ_1^W and ζ_2^W to variations in matching thresholds μ_{11} and μ_{12}. The 3D surfaces and contour plots illustrate the direct and cross-market effects, confirming that ζ_i^W decreases monotonically with its own threshold μ_{1i} and increases with the competing threshold μ_{1j} ($j \neq i$). Base parameters: $\lambda_1 = 2.0$, $\lambda_2 = 3.0$, $\theta_{01} = 2.0$, $\theta_{02} = 1.0$, $\theta_{11} = 3.0$, $\theta_{12} = 4.0$, $C = 50.0$, $\alpha_1 = \alpha_2 = 0.5$, $\mu_2 = 10.0$.

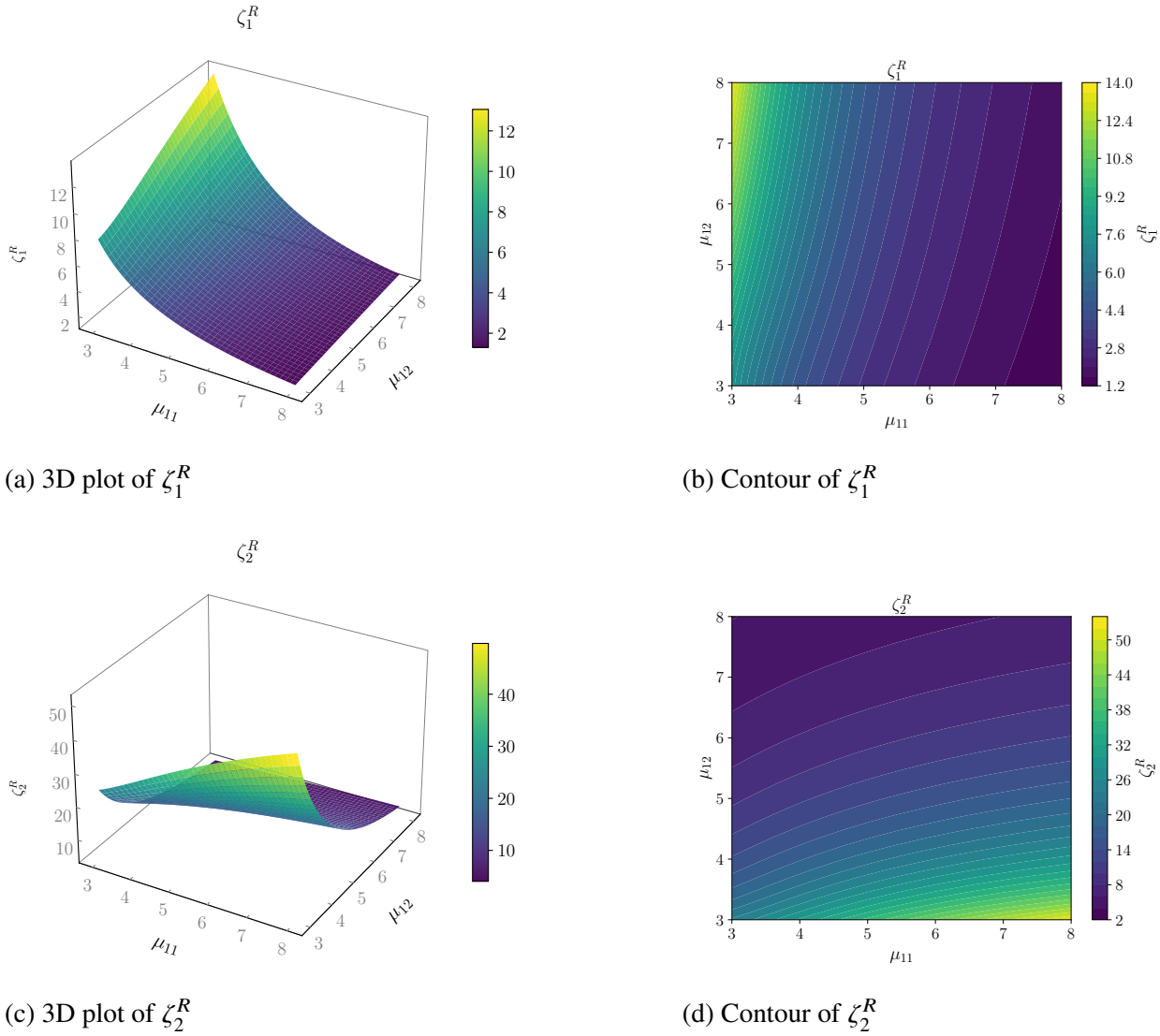
4.4. Coordinated Control Algorithm

The stationarity characterization in Theorem 2 reduces the threshold optimization problem to solving the nonlinear system

$$\boldsymbol{\zeta}(\boldsymbol{\mu}) = \mathbf{1}.$$

Standard root-finding algorithms can be applied, but the diagonal-dominant Jacobian structure established in Section 4.3 suggests that Newton-type methods are particularly well-suited.

Figure 3 Sensitivity analysis of Revenue-based Key Matching Indices (ζ^R)



(a) 3D plot of ζ_1^R

(b) Contour of ζ_1^R

(c) 3D plot of ζ_2^R

(d) Contour of ζ_2^R

Notes. Similar to the welfare case, ζ_i^R exhibits monotonic decay with respect to μ_{1i} and coupled increase with respect to μ_{1j}, confirming the robustness of the Jacobian structure across different objectives.

We employ a damped Newton–Raphson iteration that exploits the full Jacobian:

$$\boldsymbol{\mu}_{k+1} = \boldsymbol{\mu}_k - \eta \hat{\mathbf{J}}^{-1}\big(\boldsymbol{\zeta}(\boldsymbol{\mu}_k) - \mathbf{1}\big), \tag{34}$$

where $\eta \in (0, 1]$ is a step-size parameter that ensures stability, and $\hat{\mathbf{J}}$ denotes a numerical estimate of the Jacobian matrix $\mathbf{J} = \nabla_{\boldsymbol{\mu}} \boldsymbol{\zeta}$. The damping parameter η can be tuned via line search or set to a conservative value (e.g., $\eta = 0.5$) for robustness.

At each iteration, the algorithm estimates the steady-state variables by time-averaging, computes the current KMI values via (32) or (33), estimates the Jacobian via finite differences, and applies the Newton update (34) with projection onto $\boldsymbol{\mu} \geq \mathbf{0}$. The complete procedure is given in Algorithm 1 in the Appendix.

5. Numerical Experiments

We conduct numerical experiments to validate the KMI-guided control, quantify the welfare-revenue frontier, and assess the value of coordination. For concreteness, we refer to Class 1 as the *premium* segment (e.g., Uber Comfort) and Class 2 as the *regular* segment (e.g., UberX) throughout this section. The analysis focuses on three canonical load regimes: (i) *low-load*, where supply exceeds demand; (ii) *balanced-load*, where queues coexist with idle capacity; and (iii) *congested*, where demand outstrips supply. Three main findings emerge: the KMI intersection reliably identifies the optimal operating point, the coordinated controller substantially outperforms decoupled and static benchmarks, and the welfare-revenue trade-off intensifies with congestion. All quantitative results rely on steady-state statistics obtained via long-run tail averages. Detailed simulation parameters and algorithmic tuning details are provided in Appendix EC.2.

5.1. Validation and Structural Geometry

We first verify the optimality conditions derived in Section 4. First we define the First-Level Judgment (FLJ) index:

$$\text{FLJ} \triangleq \sum_{i\in\{1,2\}} \frac{\alpha_2(z_{1i}^*/z_0^*)}{s_i}, \quad \text{where} \quad s_i = 1 - \alpha_1 \frac{z_{1i}^*\theta_{1i}}{q_i^*\theta_{0i}}. \tag{35}$$

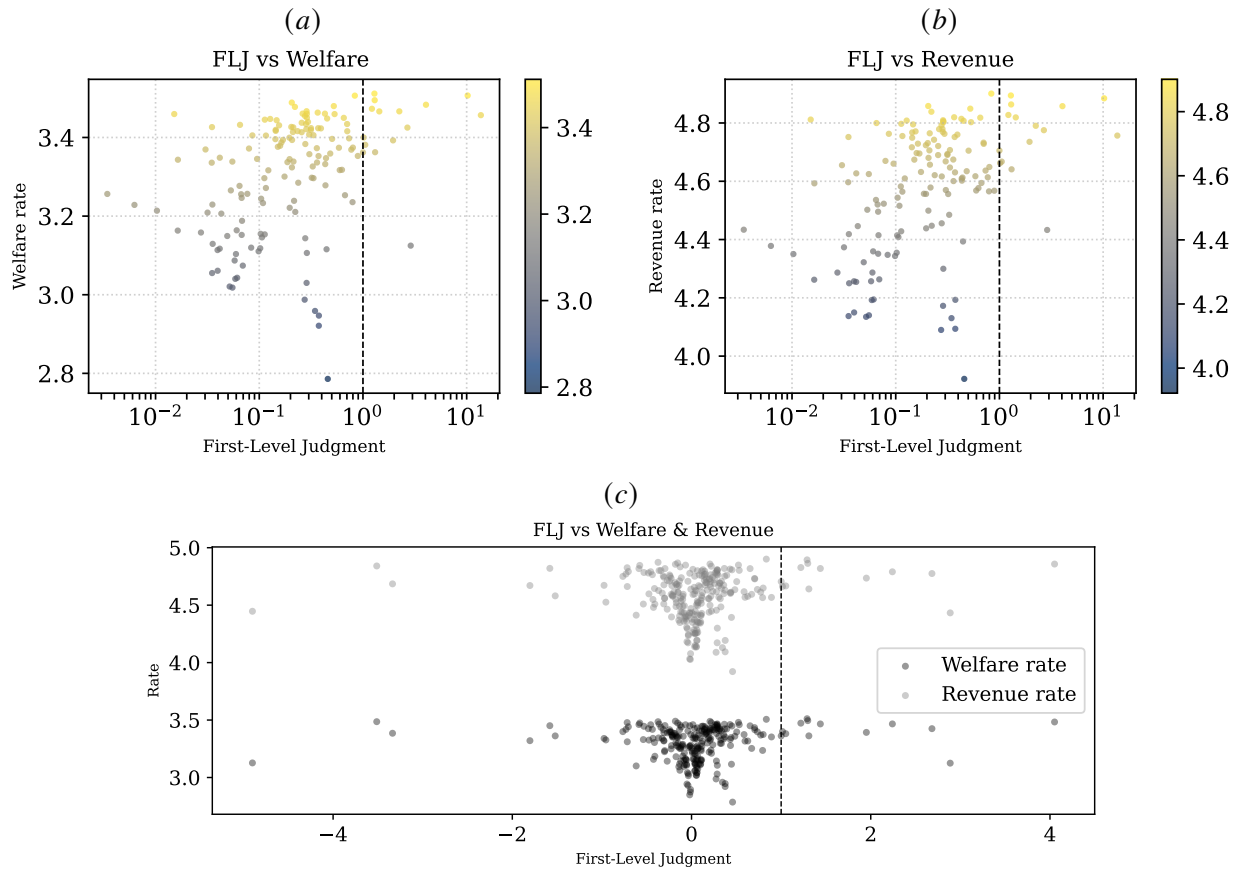
Theory posits that FLJ = 1 is necessary for optimality. Figure 4 plots the welfare and revenue rates against the FLJ index. While the performance envelope peaks near FLJ = 1, suboptimal configurations also satisfy this condition, confirming that the aggregate condition is necessary but not sufficient for optimality.

5.1.1. Objective Function Topography and Solution Structure To resolve the sufficiency gap and understand the solution geometry, we examine the topography of the objective in the (μ_{11}, μ_{12}) plane using stochastic steady-state averages. For welfare, Figure 5 shows a unimodal landscape with an isolated peak; the contour set is smooth and approximately convex. Revenue displays a comparable geometry (Figure 7), with a single global maximizer.

These plots also reveal pronounced anisotropy and curvature heterogeneity: slopes vary sharply across directions, and under high load, one observes ridges and plateaus. Directly tuning (μ_{11}, μ_{12}) is therefore difficult; the mapping from thresholds to the objective is nonlinear and regime-dependent. The KMI structure addresses this difficulty. By monitoring the state through (ζ_1, ζ_2) and steering these indices toward one, control actions align with the unique maximizer and remain stable despite local irregularities. Intuitively, while the objective surface in (μ_{11}, μ_{12}) has complex curvature, the KMI provides a stable gradient field that guides the search toward the optimum.

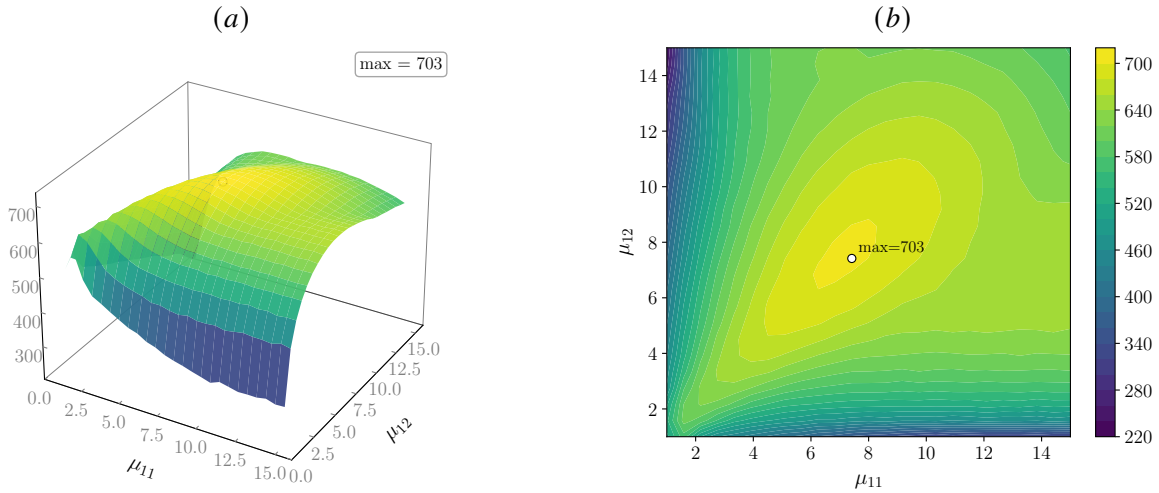
We next superimpose the KMI iso-structure. Figures 6 and 8 plot the $\zeta_1 = 1$ and $\zeta_2 = 1$ contours. The numerical evidence shows that their unique intersection coincides with the objective peak in both cases. This affirms the optimality condition and clarifies the operational role of KMIs as a state-dependent indicator: thresholds should be steered so that $(\zeta_1, \zeta_2) \to (1, 1)$. Given the surface complexity in (μ_{11}, μ_{12}), blind

Figure 4 Necessity of the FLJ condition



Notes. Vertical line marks FLJ = 1. High values concentrate near this locus, yet suboptimal rates also occur at FLJ = 1, highlighting necessity rather than sufficiency.

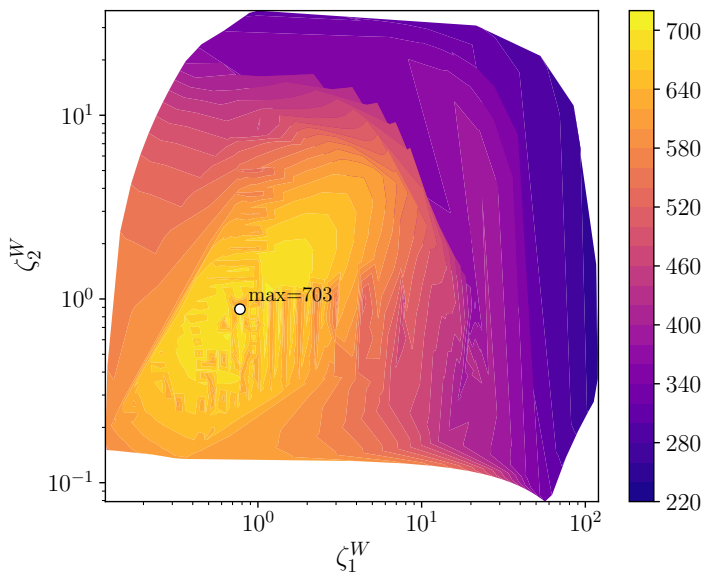
Figure 5 Welfare objective topography (stochastic steady state)



Notes. (a) 3D surface of the threshold-based welfare proxy over (μ_{11}, μ_{12}). (b) Corresponding contour plot. The landscape has an isolated peak with smooth gradients yet pronounced anisotropy.

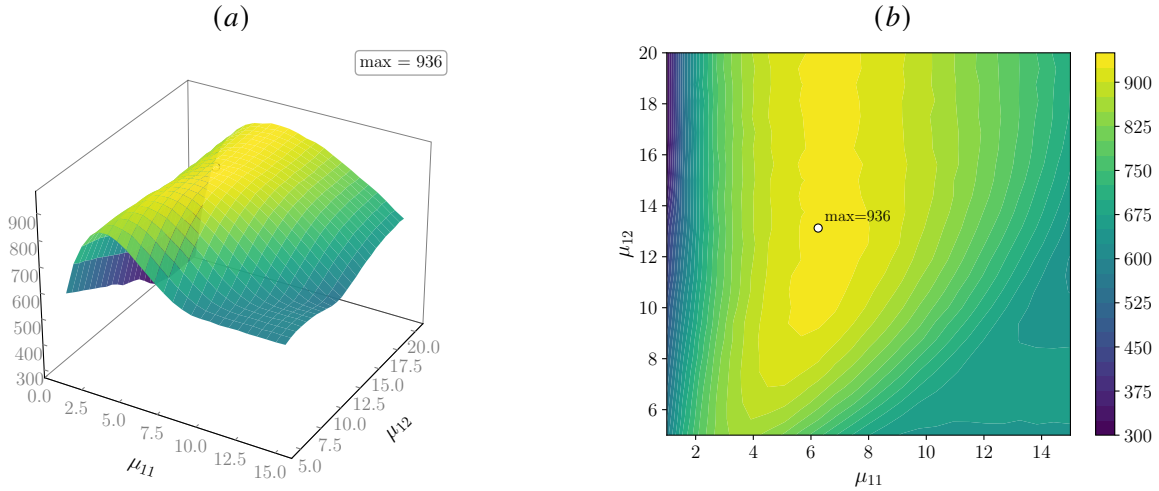
tuning is fragile; KMI guidance regularizes search and delivers the maximizer reliably. For completeness, the corresponding 3D landscapes of the KMI surfaces are provided in Appendix EC.2.2.

Figure 6 Welfare validation: KMI intersection aligns with the optimum



Notes. Contours of $\zeta_1^W = 1$ and $\zeta_2^W = 1$ whose unique intersection aligns with the welfare-optimal thresholds.

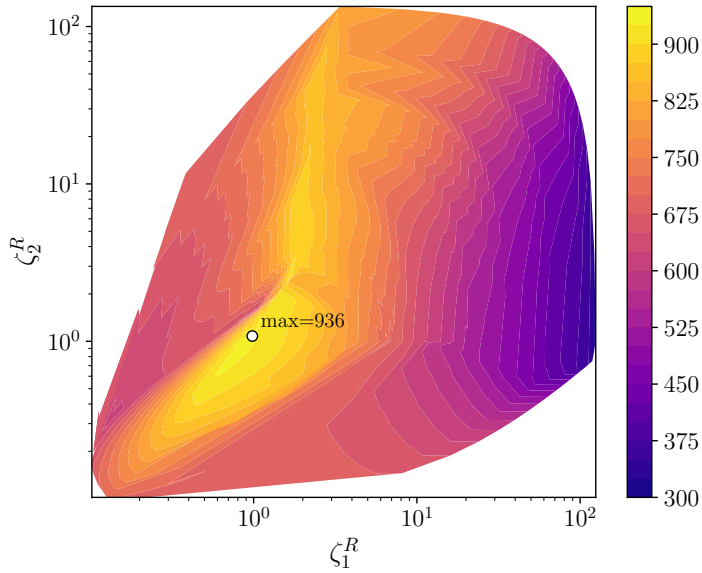
Figure 7 Revenue objective topography (stochastic steady state)



Notes. (a) 3D surface of the threshold-based revenue proxy. (b) Corresponding contour plot.

5.1.2. Quantitative Comparison of Optimal Policies We now quantify the trade-off between welfare and revenue maximization under the baseline parameterization. Table 3 reports the optimal KMI-based policies and their corresponding steady-state performance metrics.

Figure 8 Revenue validation: KMI intersection aligns with the optimum



Notes. Contours of $\zeta_1^R = 1$ and $\zeta_2^R = 1$ whose unique intersection aligns with the revenue-optimal thresholds.

To maximize revenue, the platform shifts its policy from the welfare-optimal thresholds of $(\mu_{11}^*, \mu_{12}^*) \approx (9.99, 15.18)$ to a more permissive stance at $(10.58, 15.64)$. This adjustment yields a marginal revenue gain of 0.14% but comes at the cost of degraded service quality, as evidenced by increases in the premium and regular customer queues (q_1^* and q_2^*) of 10.4% and 4.1%. The revenue-optimal policy achieves 99.8% of the maximum possible welfare, indicating a tight welfare-revenue frontier in the balanced-load regime.

Table 3 Optimal thresholds and steady-state metrics (normalized by K; baseline regime)

Policy	μ_{11}^*	μ_{12}^*	Welfare rate	Revenue rate	q_1^*	q_2^*	z_0^*
Welfare-opt	9.99	15.18	3.512	4.894	0.087	0.230	0.490
Revenue-opt	10.58	15.64	3.506	4.901	0.096	0.239	0.497

Notes: Rates are per unit time in fluid scale. Welfare rate is $\mu_2(z_{21}^*+z_{22}^*)$; revenue rate is $p_1\mu_2 z_{21}^*+p_2\mu_2 z_{22}^*$.

5.2. Benchmarking and the Value of Coordination

We benchmark the KMI-based control against two rational alternatives: (i) a *decoupled proxy* that optimizes thresholds sequentially, ignoring cross-segment externalities; and (ii) a *static-priority* rule fixed at $(\mu_{11}, \mu_{12}) \approx (4, 8)$.

Table 4 summarizes performance in the congested regime. The KMI-based control dominates both benchmarks. While the revenue gain over the decoupled proxy is marginal (0.2%), the coordination gain in *social welfare* is substantial (2.9%). This asymmetry arises because the decoupled proxy can approximately locate the revenue-maximizing point (since revenue is concentrated in the premium class), but it ignores

the externality imposed on the regular class, leading to a larger welfare loss. The primary value of the KMI framework therefore lies not merely in revenue, but in internalizing externalities to prevent inefficient cannibalization of the regular class. By explicitly accounting for the shadow cost of capacity, the coordinated controller achieves a superior Pareto outcome.

We then quantify the **value of dynamism** by contrasting KMI control with the static-priority rule. The state-dependent KMI policies deliver substantial gains: the welfare-optimal KMI increases welfare by 19.4% and revenue by 8.7%, while the revenue-optimal KMI boosts revenue by 12.0% and welfare by 17.2%. These results underscore the importance of adapting matching criteria to real-time system state, especially under high load.

The benchmarks also enrich our understanding of the welfare-revenue trade-off. Under congestion, the revenue-optimal KMI policy reduces the premium queue (q_1^*) by 40.6% relative to its welfare-optimal counterpart, but at the cost of a 55.0% increase in the regular queue (q_2^*). This trade-off shows how profit maximization reallocates scarce capacity toward the high-margin segment. The KMI framework effectively manages this trade-off, substantially outperforming both the myopic decoupled policy and the non-adaptive static rule.

Table 4 Performance benchmarking in the congested regime (normalized by K)

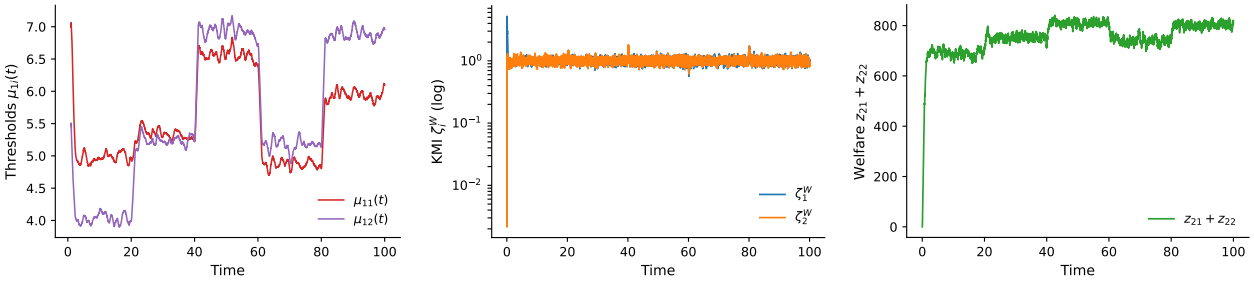
Metric	Welfare-Opt (KMI)	Revenue-Opt (KMI)	Decoupled	Static Priority
Welfare Rate	**6.1003**	5.9880	5.9275	5.1089
Revenue Rate	8.3013	**8.5484**	8.5328	7.6345
Premium Queue (q_1^*)	0.6216	0.3690	0.3219	0.0880
Regular Queue (q_2^*)	1.1334	1.7568	1.8142	2.7246
Idle Fraction (z_0^*)	0.2168	0.2188	0.1895	0.1456

5.3. Dynamic Control and Convergence

This section evaluates the performance of the ζ-guided controller, focusing on convergence under time-varying demand. Technical details regarding the convergence rate and hyperparameter tuning are deferred to Appendix EC.2.3.

5.3.1. Convergence under Time-Varying Demand We demonstrate convergence under non-stationary demand by initializing the system with suboptimal thresholds and subjecting it to piecewise-constant arrival rates that shift every 20 time units. Figure 9 illustrates the response. Matching thresholds $\mu_{1i}(t)$ adapt rapidly to load changes, converging to stable equilibria within each interval. Concurrently, the KMIs $\zeta_i^W(t)$ are steered toward unity, stabilizing the welfare rate near its theoretical maximum. This confirms that the controller effectively tracks the optimum under dynamic conditions.

Figure 9 Dynamic performance under time-varying arrivals



(a) Thresholds $\mu_{1i}(t)$ (b) KMI $\zeta_i^W(t)$ (c) Welfare Rate

Notes. Arrival rates (λ_1, λ_2) shift every 20 time units: $(2.0, 1.0) \to (2.5, 2.0) \to (4.0, 4.0) \to (2.0, 2.0) \to (3.0, 4.0)$. The controller maintains stability across transitions.

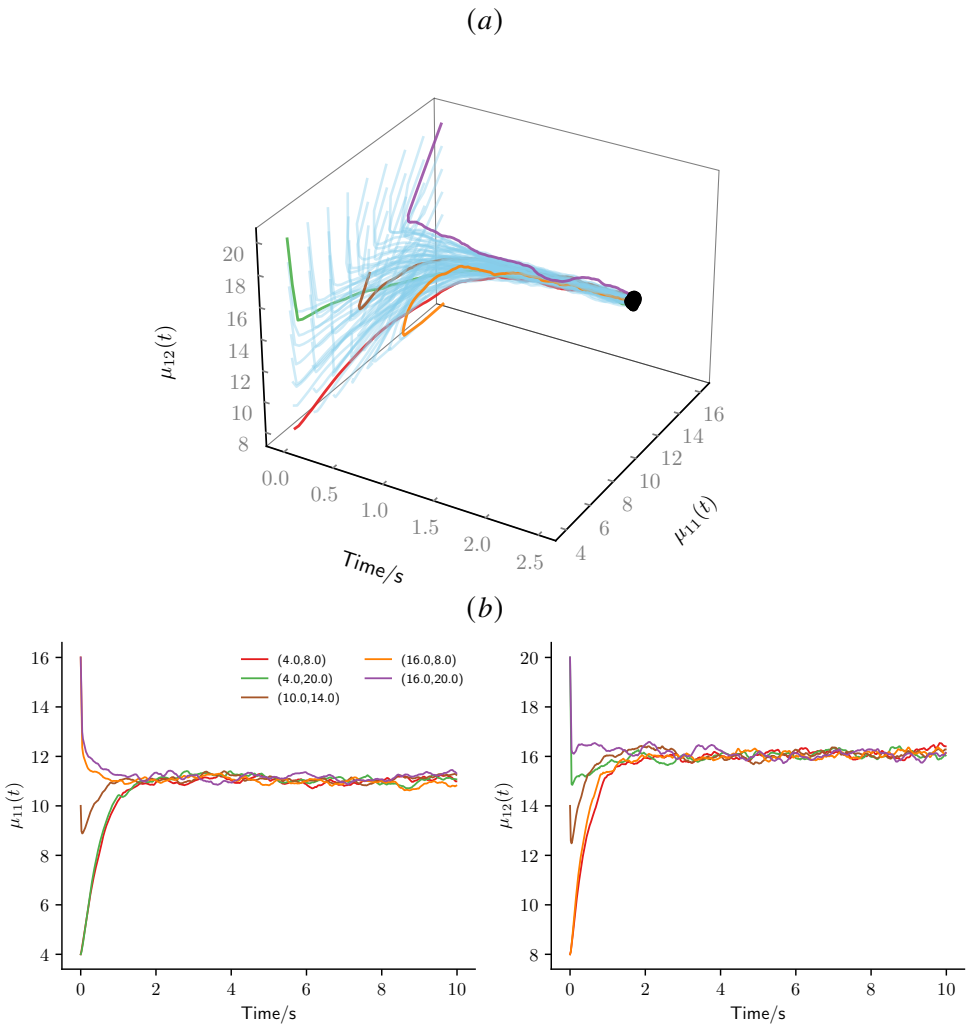
5.3.2. Global Convergence To assess the controller's robustness to initial conditions, we conduct a multi-start experiment. The thresholds (μ_{11}, μ_{12}) are initialized from a grid of 81 distinct points, including those far from the optimum. Figure 10 shows the resulting trajectories in the threshold space and their corresponding time series. All paths, regardless of their origin, converge to a single fixed point. This empirically identified attractor coincides with the revenue-optimal policy reported in Section 4, providing strong evidence for the controller's global convergence property within the tested parameter space.

5.4. Sensitivity Analysis and Regime-Dependent Insights

We now quantify the sensitivity of the optimal policy to three key operational dimensions: aggregate system load, driver supply, and customer post-match patience. The analysis relies on global policy scans over the discrete grid $(\mu_{11}, \mu_{12}) \in [1, 20]^2$ within our stochastic simulation framework to identify the welfare- and revenue-maximizing policies under each parametric setting.

5.4.1. Impact of Aggregate Load Table 5 documents the optimal policies and corresponding system performance across low, balanced, and high-load regimes. A clear structural pattern emerges: the value of service differentiation is contingent on system congestion. In the low-load regime, where driver supply is abundant relative to demand, queues are negligible, and the welfare- and revenue-optimal policies coincide. As the system transitions to a high-load, congested state, the objectives diverge. The revenue-maximizing policy intensifies service differentiation by substantially tightening the matching threshold for the regular class (a higher μ_{12}^*). This strategy prioritizes the high-margin premium segment, reducing its queue length (q_1^*) at the expense of a significantly longer queue for regular customers (q_2^*). In contrast, the welfare-optimal policy adopts a more balanced approach, mitigating the negative externality imposed on the regular class. The KMI-based framework provides the tool to precisely quantify the optimal degree of this trade-off, determining exactly how much service to the regular class must be curtailed to maximize revenue. This result formally demonstrates how profit-seeking behavior reallocates scarce capacity in congested systems.

Figure 10 Multi-start convergence analysis



Notes. (a) Phase Plane Trajectories. (b) Time Series.

Table 5 Optimal thresholds and steady-state KPIs across load regimes (normalized by K).

Regime	Policy	μ_{11}^*	μ_{12}^*	Welfare	Revenue	q_1^*	q_2^*
Low	Revenue-opt	8.00	15.00	1.71	2.39	0.04	0.13
Low	Welfare-opt	8.00	15.00	1.71	2.39	0.04	0.13
Balanced	Revenue-opt	8.00	16.00	3.51	4.93	0.06	0.27
Balanced	Welfare-opt	9.00	15.00	3.53	4.91	0.07	0.24
High	Revenue-opt	13.00	19.00	6.06	8.61	0.40	1.73
High	Welfare-opt	14.00	13.00	6.13	8.45	0.53	1.26

Notes: Each row reports global optima over $(\mu_{11}, \mu_{12}) \in [1, 20]^2$ on a unit grid for the specified regime. KPIs are steady-state tail averages from the stochastic simulator.

5.4.2. Impact of Driver Supply Table 6 isolates the effect of driver supply, K, holding total customer arrival rates constant. The results align with theoretical expectations for queueing systems. As driver supply becomes scarcer (lower K), the system becomes more congested, leading to longer queues for both customer classes. In response, the optimal policies become stricter, particularly for the lower-priority regular class

Table 6 Sensitivity to driver supply K: optimal thresholds and KPIs (normalized by K).

K	Policy	μ_{11}^*	μ_{12}^*	Welfare	Revenue	q_1^*	q_2^*
600	Revenue-opt	11.00	20.00	5.44	7.68	0.22	1.11
600	Welfare-opt	13.00	16.00	5.49	7.64	0.31	0.76
800	Revenue-opt	13.00	15.00	4.36	6.05	0.19	0.32
800	Welfare-opt	13.00	15.00	4.36	6.05	0.19	0.32
1000	Revenue-opt	8.00	16.00	3.51	4.93	0.06	0.27
1000	Welfare-opt	9.00	15.00	3.53	4.91	0.07	0.24
1200	Revenue-opt	9.00	14.00	2.96	4.14	0.06	0.16
1200	Welfare-opt	9.00	14.00	2.96	4.14	0.06	0.16
1500	Revenue-opt	10.00	12.00	2.33	3.25	0.06	0.10
1500	Welfare-opt	9.00	16.00	2.33	3.25	0.05	0.17

Table 7 Sensitivity to post-match patience $(\theta_{11}, \theta_{12})$: optimal thresholds and KPIs.

θ_{11}	θ_{12}	Policy	μ_{11}^*	μ_{12}^*	Welfare	Revenue	q_1^*	q_2^*
2.00	3.00	Revenue-opt	9.00	15.00	3.72	5.22	0.08	0.25
2.00	3.00	Welfare-opt	11.00	15.00	3.73	5.20	0.11	0.24
3.00	4.00	Revenue-opt	8.00	16.00	3.51	4.93	0.06	0.27
3.00	4.00	Welfare-opt	9.00	15.00	3.53	4.91	0.07	0.24
4.00	5.00	Revenue-opt	10.00	13.00	3.33	4.64	0.09	0.17
4.00	5.00	Welfare-opt	10.00	16.00	3.35	4.63	0.08	0.24
6.00	6.00	Revenue-opt	13.00	20.00	3.11	4.27	0.12	0.32
6.00	6.00	Welfare-opt	13.00	20.00	3.11	4.27	0.12	0.32

under the revenue objective, to ration the limited service capacity more effectively. Conversely, as driver supply increases, the system's congestion level eases, customer queues diminish, and the matching thresholds become more permissive. The divergence between the welfare- and revenue-optimal policies narrows as supply grows, underscoring that the conflict between social good and platform profit is most acute when capacity is constrained.

5.4.3. Impact of Post-Match Patience We examine the role of post-match patience, parameterized by the abandonment hazard rates $(\theta_{11}, \theta_{12})$, where higher values signify less patient customers. As shown in Table 7, when customers are less patient, the optimal policy responds by tightening matching thresholds. This strategic adjustment reduces average pickup times, thereby mitigating the risk of costly post-match abandonments. This defensive posture, however, comes at the cost of lower overall welfare and revenue. When customers are more patient (lower θ values), the system can tolerate longer pickup times, allowing for more relaxed thresholds and achieving superior performance. Across all patience levels, the revenue-optimal policy is consistently more aggressive in restricting access for the regular class than its welfare-oriented counterpart, a pattern that is amplified when patience is low.

5.5. Managerial Insights

Our numerical investigation yields several insights for platform operations. The KMI framework, validated as a reliable state-dependent indicator, provides the foundation for the control approach.

- **Differentiation Driven by Scarcity:** The optimal policy adapts to market conditions. Under scarcity (high load or low supply), the conflict between welfare and revenue widens, necessitating increased service differentiation to protect high-margin segments. Under abundance, objectives align, and differentiation diminishes. This formalizes the intuition that prioritization intensity should dynamically track resource scarcity.
- **Model-Based Control via KMI:** Targeting the $(\zeta_1, \zeta_2) = (1, 1)$ condition provides a systematic threshold adjustment rule. This approach replaces reactive experimentation (such as A/B testing) with a model-based search that exploits the structure of the optimality conditions. Because the stationarity condition $\boldsymbol{\zeta} = \mathbf{1}$ defines a state-dependent efficient frontier rather than a fixed operating point, the optimal policy continuously adapts to changing load conditions, whereas static priority rules correspond to corner solutions that may be optimal under one demand pattern but severely suboptimal when conditions shift.
- **Stability and Externality Internalization:** The gains from our control framework stem from two sources: (1) *coordination*, by internalizing the cross-segment externalities (e.g., the impact of regular class admission on premium queues) that decoupled policies ignore, and (2) *dynamism*, by adapting to the state-dependent system response.

6. Conclusion

This paper addresses the control problem of managing heterogeneous demand classes within a shared-capacity ride-hailing network, where matching decisions for one class affect service quality for the other through endogenous pickup times that depend jointly on driver and passenger densities. By formulating a spatial queueing model with two-stage customer loss (pre-match abandonment and post-match cancellation), we established a tractable fluid framework that captures the endogenous feedback between matching decisions, idle driver density, and service quality. The central theoretical contribution is the derivation of the Key Matching Indices (KMI), (ζ_1, ζ_2), which transform complex state-dependent interactions into observable dimensionless metrics. Each index decomposes into three interpretable components: abandonment reduction, direct opportunity cost, and a cross-market term, with the third component explicitly coupling the two classes through the shared driver pool. We proved that driving both indices to unity characterizes the stationary points of the steady-state optimization problem for both welfare and revenue objectives, reducing a high-dimensional policy design problem to a tractable system of two coupled nonlinear equations. This structural property provides an operational bridge between abstract fluid limits and real-time algorithmic control: the Jacobian-based update logic ensures convergence to optimal thresholds even under changing demand.

Our numerical investigations reveal that the alignment between social welfare and platform revenue is load-dependent. Under low load, both objectives can be optimized simultaneously with generous matching

thresholds for both classes. Under congestion, however, revenue maximization requires disproportionate rationing of the regular class to preserve spatial density for premium users. The divergence of optimal thresholds across objectives quantifies the high shadow cost of shared capacity when the two classes compete for idle drivers. Compared to decoupled heuristics or static priority rules, the KMI-guided coordination explicitly internalizes these cross-market costs and delivers better outcomes across all load regimes tested. The controller's ability to recover the optimum from any initial state further supports the well-behaved structure of the underlying objective.

Our model relies on several structural idealizations that suggest natural directions for future work. First, our analysis presumes steady-state operation with exogenous pricing and arrivals; incorporating endogenous price sensitivity and short-horizon transients would bridge the gap between operational control and demand management. Second, extending the framework to networked operations with multi-region imbalances would require characterizing the contraction properties of the KMI map in higher dimensions. Third, deploying threshold controls in practice would benefit from learning-theoretic enhancements, such as confidence-adaptive step sizes, to handle model misspecification and data sparsity. Finally, as platforms face increasing regulatory scrutiny regarding algorithmic fairness, integrating equity constraints directly into the KMI formulation would allow mapping a three-way frontier among welfare, revenue, and distributional fairness.

Overall, this work provides a methodological foundation for moving from reactive, heuristic dispatching to model-based, state-dependent control in ride-hailing operations.

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Proofs

EC.1. Coordinated Control Algorithm

Algorithm 1 details the coordinated threshold adjustment procedure described in Section 4.4. At each iteration, the algorithm estimates the steady-state system variables $(z_0^*, z_{1i}^*, z_{2i}^*, q_i^*)$ by time-averaging observations from the matching system, computes the current KMI values $\boldsymbol{\zeta}^{(k)}$ using the closed-form expressions (32) or (33), and estimates the Jacobian via finite differences. It then inverts the 2×2 Jacobian matrix and applies the Newton update with projection onto $\boldsymbol{\mu} \geq \mathbf{0}$.

Algorithm 1 Coordinated Threshold Adjustment (Newton–Raphson Type)

Initialize: $\boldsymbol{\mu}^{(0)}$, tolerance ϵ, step size η.
while $\|\boldsymbol{\zeta}(\boldsymbol{\mu}^{(k)}) - \mathbf{1}\| > \epsilon$ **do**
  Estimate steady-state variables (z_0, z_{1i}, q_i) by time averaging.
  Compute $\boldsymbol{\zeta}^{(k)}$ via (32) or (33).
  Estimate Jacobian $\hat{\mathbf{J}}$ via finite differences:

$$\hat{J}_{ij} \approx \frac{\zeta_i(\mu_{1j} + \delta) - \zeta_i(\mu_{1j})}{\delta}.$$

  Compute direction

$$\mathbf{d}_k = -\hat{\mathbf{J}}^{-1}(\boldsymbol{\zeta}^{(k)} - \mathbf{1}).$$

  Update $\boldsymbol{\mu}^{(k+1)} \leftarrow \max(\mathbf{0}, \boldsymbol{\mu}^{(k)} + \eta \mathbf{d}_k)$.
  $k \leftarrow k + 1$.
end while

The computational burden is minimal: each Jacobian estimation requires only two additional evaluations of the two-dimensional fluid system (one per threshold direction), and the matrix inversion involves only a 2×2 system. The entire update step executes in sub-second time, making the algorithm suitable for online deployment. The algorithm handles both welfare and revenue objectives by selecting the appropriate KMI formula.

EC.2. Experiment Implementation Details and Additional Results

EC.2.1. Baseline Parameters

Unless otherwise noted, the numerical experiments utilize the baseline parameters detailed in Table EC.1. The system operates in a stylized two-tier market with driver supply $K = 1000$.

Table EC.1 Baseline Simulation Parameters

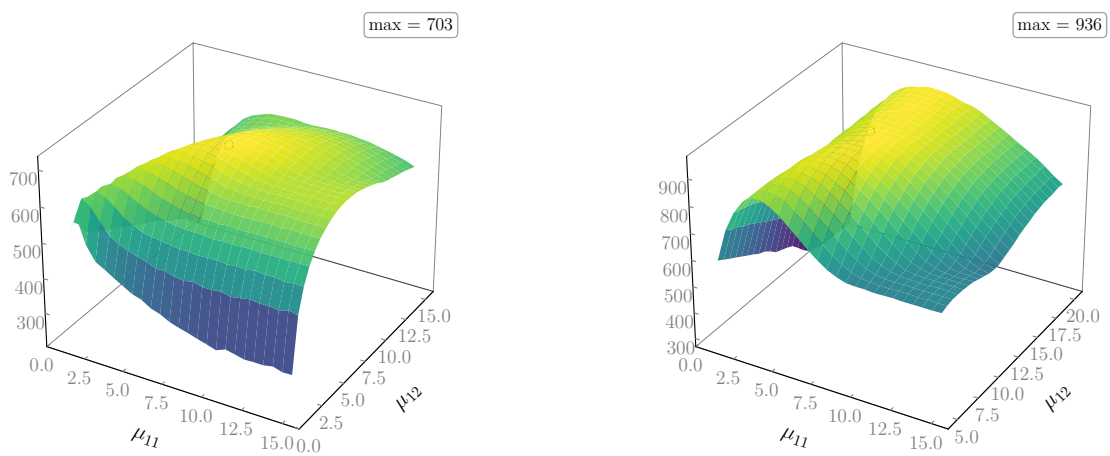
Parameter	Symbol	Value
Driver Supply	K	1000
Arrival Rates (Balanced)	λ_1, λ_2	$2.0K, 3.0K$
Prices	p_1, p_2	2, 1
Pre-match Patience	θ_{01}, θ_{02}	2, 1
Post-match Patience	θ_{11}, θ_{12}	3, 4
Spatial Parameters	C	50
Matching Exponents	α_1, α_2	0.5, 0.5
Trip Completion Rate	μ_2	10

EC.2.2. 3D Objective and KMI Landscapes

To complement the contour plots in Section 5.1, we provide 3D visualizations of both the objective functions and the KMI surfaces.

Figure EC.1 illustrates the anisotropy of the optimization landscape, highlighting why direct gradient ascent on thresholds can be unreliable. Figure EC.2 displays the KMI surfaces, demonstrating how the ζ indices provide a regularized gradient field that robustly guides the system toward the optimum.

Figure EC.1 3D Visualization of Welfare and Revenue Landscapes



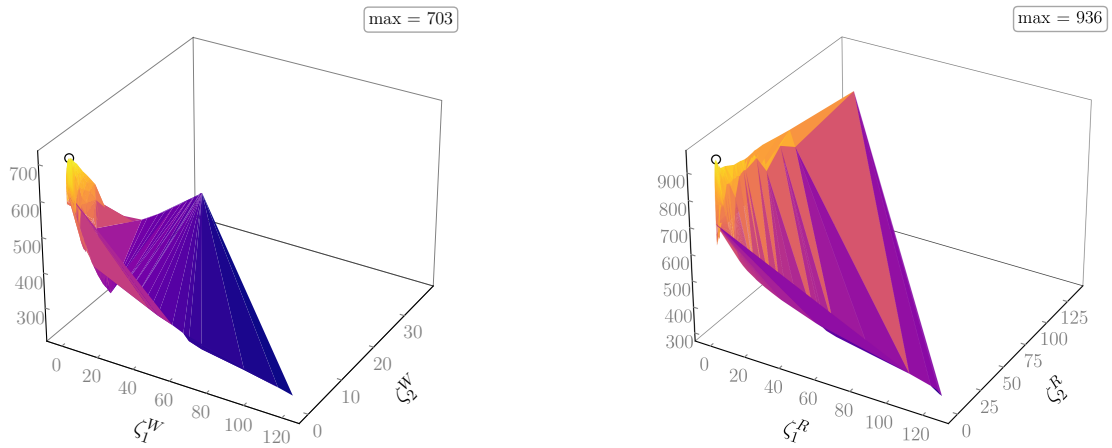
(a) Welfare Surface

(b) Revenue Surface

EC.2.3. Convergence Rate and Hyperparameter Tuning

EC.2.3.1. Convergence Rate Analysis To guide practical implementation, we analyze the controller's convergence rate as a function of the learning rate, η. We define the performance metric as the **time-to-ε**, the time required for the combined log-error, $\sqrt{(\log \zeta_1)^2 + (\log \zeta_2)^2}$, to fall below a threshold of $\varepsilon = 0.01$. For statistical robustness, we average results over 100 replications for each η value.

Figure EC.2 3D Visualization of Key Matching Indices (KMI)

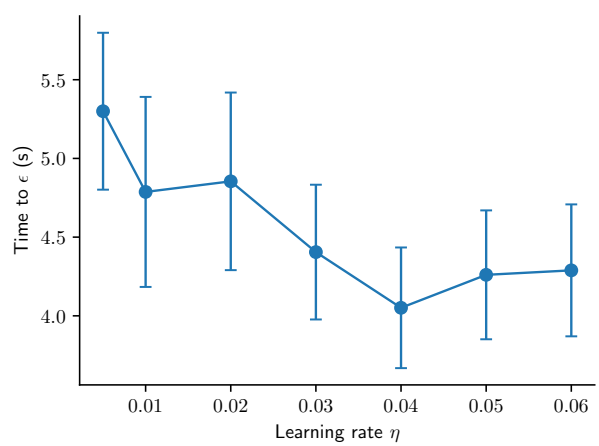


(a) Welfare KMI Surface (ζ^W)

(b) Revenue KMI Surface (ζ^R)

Figure EC.3 plots the mean time-to-ε against η, with error bars representing 95% confidence intervals. The results reveal a distinct trade-off that defines an optimal tuning window. For small η, convergence is slow, as the small step size leads to insufficient update magnitudes. As η increases, the convergence time drops sharply, identifying a "sweet spot" where the controller is both fast and stable, evidenced by narrow confidence intervals that indicate low variance across replications. Beyond this region, further increases in η yield diminishing returns; the curve flattens and eventually rises, while the widening confidence intervals signal growing instability and oscillatory behavior near the stability boundary. This analysis provides an actionable guide for selecting an effective η that balances convergence speed with robustness.

Figure EC.3 Convergence rate diagnostics

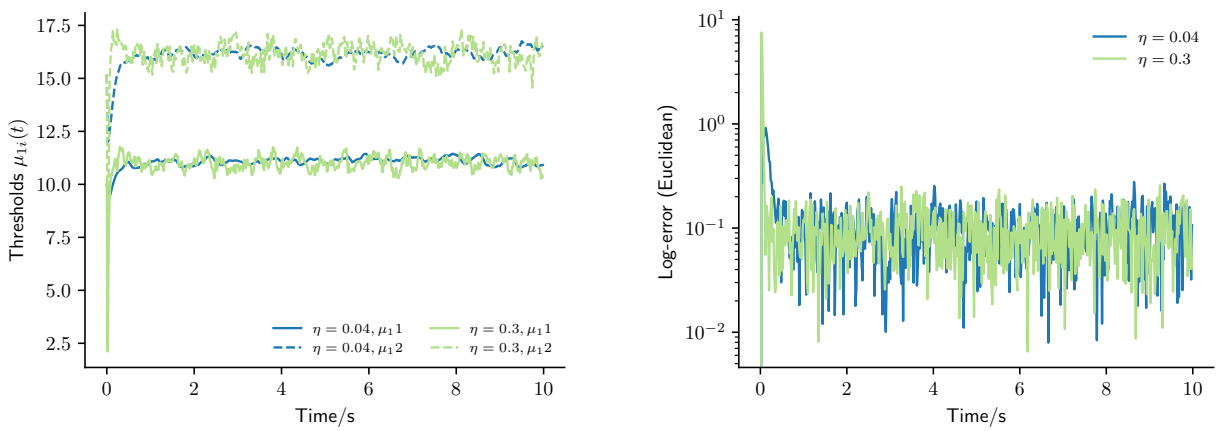


Notes. Mean time-to-ε vs. learning rate η (with 95% confidence intervals).

EC.2.3.2. Stability Boundary and Implementation The choice of learning rate η is critical for practical deployment. While a larger η can hasten convergence, an excessively high value may lead to instability. Figure EC.4 demonstrates this by comparing a conservative rate ($\eta = 0.04$) with an aggressive one ($\eta = 0.30$). The aggressive rate causes the thresholds to overshoot the optimal values, and the log-error exhibits oscillations, indicating proximity to the stability boundary. This suggests that η should be tuned to balance speed and stability, possibly with an adaptive schedule that reduces the rate as the system approaches the optimum.

From an implementation perspective, the ζ-guided controller is efficient. It relies on summary statistics that are typically available in real-time platform monitoring systems. The Jacobian of the ζ mapping can be estimated infrequently using small perturbations, while the thresholds are updated more frequently. This two-timescale approach ensures that the computational overhead remains low, making the controller suitable for large-scale, dynamic environments.

Figure EC.4 Stability boundary demonstration



(a) Threshold Overshoot

(b) Log-Error Oscillations

EC.3. Proof of Results

EC.3.1. Proof of Theorem 2 and Proposition 2

This section derives the first-order necessary conditions for the optimization problem (30). The objective is to maximize the weighted throughput $J(\boldsymbol{\mu}) = w_1 z_{21}^* + w_2 z_{22}^*$ subject to the fluid equilibrium constraints derived in Section 3.

EC.3.1.1. Lagrangian Formulation Let $\lambda_i, \theta_{0i}, \theta_{1i}, \mu_2$ be the system parameters. The equilibrium is governed by the flow balance and state constraints. We construct the Lagrangian $\mathcal{L}$ associated with the optimization problem. Note that while the fluid dynamics involve z_{1i} and q_i, the control variables are the thresholds μ_{1i} which impose the constraint $C(q_i^*)^{\alpha_1}(z_0^*)^{\alpha_2} = \mu_{1i}^*$. For simplicity, we use the notation

$(z_{2i}, z_{1i}, q_i, z_0; i = 1,2)$ to denote the equilibrium state variables $(z^*_{2i}, z^*_{1i}, q^*_i, z^*_0; i = 1,2)$ in the remainder of Section EC.3.1.

The Lagrangian is

$$\begin{aligned}\mathcal{L} = {} & w_1 z_{21} + w_2 z_{22} \\ & + \sum_{i=1}^{2} \beta_i (\lambda_i - q_i \theta_{0i} - z_{1i}\theta_{1i} - z_{2i}\mu_2) \quad \text{(Passenger Flow Balance)} \\ & + \sum_{i=1}^{2} \gamma_i \left(z_{1i}\mu_{1i} - z_{2i}\mu_2\right) \quad \text{(Service Rate Balance)} \\ & + \gamma_3 \left(1 - z_0 - \sum_{i=1}^{2} (z_{1i} + z_{2i})\right) \quad \text{(Driver Conservation).}\end{aligned} \tag{EC.1}$$

Here, $\beta_i, \gamma_i, \gamma_3$ are the Lagrange multipliers (shadow prices) associated with the respective constraints.

EC.3.1.2. First-Order Conditions (FOCs) Assuming an interior solution exists where all state variables are strictly positive, the Karush-Kuhn-Tucker (KKT) conditions require the partial derivatives of $\mathcal{L}$ with respect to the state variables $(z_{2i}, z_{1i}, q_i, z_0)$ to satisfy stationarity:

$$\frac{\partial \mathcal{L}}{\partial z_{2i}} = w_i - \beta_i \mu_2 - \gamma_i \mu_2 - \gamma_3 = 0 \implies \gamma_3 = w_i - (\beta_i + \gamma_i)\mu_2, \quad i \in \{1,2\} \tag{EC.2}$$

$$\frac{\partial \mathcal{L}}{\partial z_{1i}} = -\beta_i \theta_{1i} + \gamma_i \mu_{1i} - \gamma_3 = 0 \implies \gamma_i \mu_{1i} = \beta_i \theta_{1i} + \gamma_3, \quad i \in \{1,2\} \tag{EC.3}$$

$$\frac{\partial \mathcal{L}}{\partial q_i} = -\beta_i \theta_{0i} + \gamma_i z_{1i} \frac{\partial \mu_{1i}}{\partial q_i}$$

Note: Treating μ_{1i} as state constraint $C q_i^{\alpha_1} z_0^{\alpha_2} - \mu_{1i} = 0$ yields equivalent dual relations.

Standard form derivation yields: $\beta_i \theta_{0i} = \gamma_i z_{1i} \alpha_1 \dfrac{\mu_{1i}}{q_i} = \gamma_i \alpha_1 \dfrac{\mu_{1i} z_{1i}}{q_i}$, (EC.4)

$$\frac{\partial \mathcal{L}}{\partial z_0} = -\gamma_3 + \sum_{i=1}^{2} \gamma_i z_{1i} \frac{\partial \mu_{1i}}{\partial z_0} = -\gamma_3 + \sum_{i=1}^{2} \gamma_i \alpha_2 \frac{\mu_{1i} z_{1i}}{z_0} = 0 \implies \gamma_3 = \alpha_2 \sum_{i=1}^{2} \gamma_i \frac{\mu_{1i} z_{1i}}{z_0}. \tag{EC.5}$$

EC.3.1.3. Derivation of the Optimality Conditions We now manipulate the FOCs to derive the key matching index (KMI) conditions stated in Theorem 2.

Deriving the Marginal Value Conservation (Proposition 2)

We begin by substituting the expression for β_i derived in (EC.4) into (EC.3). Recalling from (EC.4) that $\beta_i = \frac{\gamma_i}{\theta_{0i}} \alpha_1 \frac{\mu_{1i} z_{1i}}{q_i}$ (assuming unit consistency in differentials), the substitution into (EC.3) yields the relation $\gamma_i \mu_{1i} = \left(\gamma_i \alpha_1 \frac{\mu_{1i} z_{1i}}{q_i \theta_{0i}}\right) \theta_{1i} + \gamma_3$. Rearranging this equation to isolate γ_3 reveals that $\gamma_3 = \gamma_i \mu_{1i} \left(1 - \alpha_1 \frac{z_{1i}\theta_{1i}}{q_i \theta_{0i}}\right)$. By defining the elasticity term $s_i \triangleq 1 - \alpha_1 \frac{z_{1i}\theta_{1i}}{q_i \theta_{0i}}$, we simplify the relationship to

$$\gamma_3 = \gamma_i \mu_{1i} s_i. \tag{EC.6}$$

Substituting (EC.6) into the driver conservation condition (EC.5) leads to the expression $\gamma_3 = \alpha_2 \left(\frac{\gamma_3}{\mu_{11} s_1} \frac{\mu_{11} z_{11}}{z_0} + \frac{\gamma_3}{\mu_{12} s_2} \frac{\mu_{12} z_{12}}{z_0} \right)$. Since the shadow price of driver conservation γ_3 is non-zero, dividing through by γ_3 yields

$$1 = \frac{\alpha_2 z_{11}/z_0}{s_1} + \frac{\alpha_2 z_{12}/z_0}{s_2}, \tag{EC.7}$$

which recovers Eq. (31), representing the conservation of the marginal value of idle drivers.

Deriving the KMI Unity Conditions

We subsequently distinguish between the two objective cases to establish the KMI unity conditions. For the case of Societal Welfare where $w_1 = w_2 = 1$, the condition derived from (EC.2) implies $\gamma_3 = 1 - (\beta_i + \gamma_i)\mu_2$. Substituting β_i from (EC.4) into this expression establishes a direct relationship between γ_1 and γ_2. When combined with (EC.6), this yields the condition for $\zeta_i^W = 1$ as defined in Definition 2. The cross-term in ζ_i^W arises directly from the term γ_j appearing in the summation of (EC.5), representing the opportunity cost of allocating an idle driver to type j instead of i.

Alternatively, for the case of Platform Revenue where $w_i = p_i$, the condition from (EC.2) becomes $\gamma_3 = p_i - (\beta_i + \gamma_i)\mu_2$. The presence of p_i introduces asymmetry into the system. Letting $A_i = 1 + \alpha_1 \frac{\mu_{1i} z_{1i}}{q_i \theta_{0i}}$, the multiplier relationship transforms into

$$\gamma_i = \frac{p_i}{\mu_{1i} s_i + \mu_2 A_i}. \tag{EC.8}$$

Substituting this result into the conservation law (EC.5) and rearranging terms yields the Revenue KMI expression ζ_i^R. Here, the ratio p_j/p_i and the terms A_i, A_j in ζ_i^R explicitly capture the substitution of value (revenue) rather than volume (matches).

EC.3.1.4. Remark on Sufficiency and Global Optimality The conditions derived in Theorem 2 characterize a KKT stationary point $(\boldsymbol{\mu}^*, \mathbf{x}^*)$. In general nonlinear optimization, stationarity is a necessary but not sufficient condition for global optimality, particularly given the non-convex nature of the spatial matching function $\mu_i(z_0, q_i) = C q_i^{\alpha_1} z_0^{\alpha_2}$.

While the KMI condition $\boldsymbol{\zeta}(\boldsymbol{\mu}) = \mathbf{1}$ formally characterizes a stationary point, the structural properties of the system suggest this solution corresponds to the global optimum in the relevant operational regime. Central to this argument is the uniqueness of the fluid equilibrium x^* for any fixed set of thresholds $\boldsymbol{\mu}$, as established in Theorem EC.2, which effectively precludes the emergence of multiple steady states for a single policy. This theoretical foundation is reinforced by extensive numerical evidence, visualized in Figures EC.1 and EC.2, indicating that the objective function $J(\boldsymbol{\mu})$ behaves unimodally, specifically displaying quasi-concavity, with respect to $\boldsymbol{\mu}$ in the region where $z_0 > 0$. Furthermore, the Jacobian $\mathbf{J}_\zeta$ maintains full rank and diagonal dominance, a characteristic that ensures the gradient-based control law follows a unique ascent path. Consequently, despite the formal limitation to stationarity, the convergence of theoretical uniqueness and empirical unimodality strongly supports the conclusion that the solution is the global maximizer for the platform's objective.

EC.3.2. Structural Divergence of Welfare and Revenue Indices

This section shows why the Revenue KMI ζ_i^R does not algebraically reduce to the Welfare KMI ζ_i^W even in the symmetric price case ($p_1 = p_2 = 1$). This divergence is not an artifact of derivation but a reflection of structural economic differences in how opportunity costs are valued.

The distinction arises primarily from the contrast between physical and economic opportunity costs. While ζ_i^W optimizes the aggregate rate of service ($z_{21} + z_{22}$), the Lagrange multipliers associated with this objective represent the shadow price of system constraints in units of throughput per hour. Consequently, the cross-market term in ζ_i^W penalizes the reduction in matching volume for class j caused by serving class i. In contrast, ζ_i^R optimizes accumulated value ($p_1 z_{21} + p_2 z_{22}$), effectively treating each match as a unit of revenue generation rather than a raw throughput event. Even when prices are symmetric, the multipliers γ_i in the revenue case incorporate the price p_i directly into the numerator, such that $\gamma_i \propto \frac{p_i}{\mu_{1i} s_i + \mu_2 A_i}$, thereby embedding the marginal revenue substitution rate directly into the KMI.

Mathematically, this structural difference is evident in the multiplier relationships derived in the proof. For the welfare objective, the relation between γ_1 and γ_2 depends purely on the ratio of service completion rates and queue elasticities. Conversely, for the revenue objective, the relation is governed by $\frac{\gamma_1}{\gamma_2} = \frac{p_1(\mu_{12} s_2 + \mu_2 A_2)}{p_2(\mu_{11} s_1 + \mu_2 A_1)}$. The terms $A_i = 1 + \alpha_1 \frac{\mu_{1i} z_{1i}}{q_i \theta_{0i}}$ appear specifically in the revenue formulation because the cost of a driver is weighed against the revenue p_i, which is coupled with the trip duration and turnover rate. This creates a structurally distinct denominator in ζ_i^R that persists regardless of whether $p_i = p_j$. Therefore, the two objectives imply distinct operational frontiers; optimizing for revenue, even with equal prices, requires a different balancing of queue accumulation compared to maximizing raw throughput, as the platform implicitly values the reliability of revenue generation differently from the volume of passenger flow.

EC.4. Proof of Stochastic Model Converges to Fluid Model

We establish the convergence of stochastic model to fluid model when $\alpha_1 = \alpha_2 = \alpha$. Recall that we apply a threshold rule to control the matching process, i.e., a matching happens if and only if $\mu_{11}(t) \geq \mu_{11}$ or $\mu_{12}(t) \geq \mu_{12}$ for some thresholds value μ_{11}, μ_{12}. By Assumption 1 , when $\alpha_1 = \alpha_2$, a matching happens if and only if $Z_0(t) Q_i(t) \geq \bar{\mu}_{1i}$, where $\bar{\mu}_{1i} = \sqrt[\alpha]{\frac{\mu_{1i}}{C}}$ for some $\mu_{1i}, i = 1, 2$. When $Z_0(t) Q_i(t) \leq \bar{\mu}_{1i}$, the system dynamics of the stochastic model are described by equations (4)–(7), and the matching equations (9)–(10) can be rewritten in the following equivalent form:

$$\begin{aligned} M(t) = M_1(t) + M_2(t) = &\int_0^t \mathbf{1}_{\{(Q_1(s-)+1) Z_0(s-) \geq \bar{\mu}_{11}\}} dA_1(s) + \int_0^t \mathbf{1}_{\{(Q_2(s-)+1) Z_0(s-) \geq \bar{\mu}_{12}\}} dA_2(s) \\ &+ \int_0^t \mathbf{1}_{\{Q_1(s-)(Z_0(s-)+1) \geq \bar{\mu}_{11} \text{ or } Q_2(s-)(Z_0(s-)+1) \geq \bar{\mu}_{12}\}} d\Big(\sum_{i=1}^{2} (R_{1i}(s) + D_{2i}(s))\Big), \end{aligned} \tag{EC.9}$$

where the first two terms correspond to a matching triggered by a passenger arrival and the third term corresponds to a matching triggered by a driver becoming idle.

Before presenting the detailed proof, we provide a roadmap. As a first step, we need to characterize the state evolution dynamics of the stochastic model using the system input processes and parameters. The main difficulty lies in the nontrivial dependence of the matching process $M(t)$ on the system inputs, which is tackled by Lemma EC.1. Having established the dynamic equations, we observe that the stochastic and fluid model both satisfy differential equations of form (EC.19)-(EC.22), with difference in residual terms $B_i(t), i = 1,2,3,4$. For given $B_i(t), i = 1,2,3,4$, we first show that (EC.19)-(EC.22) can be regarded as a contraction mapping (Proposition EC.1), thus validating the existence and uniqueness of the stochastic model. Next, we establish using the Law of Large Numbers that the difference in residual terms $B_i(t), i = 1,2,3,4$, is asymptotically negligible (Lemma EC.2). Thus, combined with the fact that (EC.19)-(EC.22) is Lipschitz continuous, we complete the proof of convergence (Theorem EC.1).

We begin with Lemma EC.1, which characterizes the matching process under the threshold control policy; the proof proceeds by induction. Here $\lceil x \rceil$ denotes the smallest integer greater than or equal to x.

LEMMA EC.1. *Let $X_i(t) = Q_i(0) + A_i(t) - R_{0i}(t)$, $X(t) = X_1(t) + X_2(t)$ and $Y(t) = Z_0(0) + R_{11}(t) + D_{21}(t) + R_{12}(t) + D_{22}(t)$, then*

$$M(t) = M_1(t) + M_2(t) = \sup_{0 \le s \le t} \left\lceil \frac{X(s) + Y(s) - \sqrt{(X(s) - Y(s))^2 + 4(\bar{\mu}_{11} + \bar{\mu}_{12})}}{2} \right\rceil_+ . \tag{EC.10}$$

Proof of Lemma EC.1 The proof uses sample-path arguments. First we introduce the functions of calculating the amount of the matched two different class passengers. Let x_1 and x_2 be the number of matched passengers of class 1 and class 2, respectively, and $x = x_1 + x_2$, the total number of matched passengers. We can write the matching process as follows:

$$(X_1(t) - x_1)(Y(t) - x_1 - x_2) = \bar{\mu}_{11},$$
$$(X_2(t) - x_2)(Y(t) - x_1 - x_2) = \bar{\mu}_{12}.$$

Adding these two equations yields

$$\begin{aligned}(X_1(t) - x_1)(Y(t) - x_1 - x_2) &+ (X_2(t) - x_2)(Y(t) - x_1 - x_2) \\ &= (X(t) - x)(Y(t) - x) = \bar{\mu}_{11} + \bar{\mu}_{12}.\end{aligned}$$

Observe that $\frac{X(s)+Y(s)-\sqrt{(X(s)-Y(s))^2+4(\bar{\mu}_{11}+\bar{\mu}_{12})}}{2}$ is the root of the equation $(X(t)-x)(Y(t)-x) = \bar{\mu}_{11} + \bar{\mu}_{12}$. Assume that the k-th matching happens at time t_k with $0 < t_1 < t_2 < \cdots < t_k < \ldots$. For any $t \in [0, t_1)$, we have $X_1(t) = Q_1(t), X_2(t) = Q_2(t), Y(t) = Z_0(t)$, and $X_i(t)Y(t) \le \bar{\mu}_{1i}, i = 1,2$, which implies that the $X(t)Y(t) \le \bar{\mu}_{11} + \bar{\mu}_{12}$, so the right-hand side of (EC.10) is 0. When $t = t_1$, since the first matching happened, we have $X(t_1) = Q_1(t_1) + Q_2(t_1) + 1, Y(t_1) = Z_0(t_1) + 1, X(t_1)Y(t_1) > \bar{\mu}_{11} + \bar{\mu}_{12}$, and $(X(t_1) - 1)(Y(t_1) - 1) \le \bar{\mu}_{11} + \bar{\mu}_{12}$. Since $\sup_{0 \le s \le t} \frac{X(s)+Y(s)-\sqrt{(X(s)-Y(s))^2+4(\bar{\mu}_{11}+\bar{\mu}_{12})}}{2}$ is the root to equation $(X(t)-x)(Y(t)-x) = \bar{\mu}_{11} + \bar{\mu}_{12}$, the right-hand side of (EC.10) is 1 when $t = t_1$. By induction, one can verify that when

$t \in [t_k, t_{k+1})$, $X(t) - k = Q_1(t) + Q_2(t), Y(t) - k = Z_0(t)$, $(X(t) - k + 1)(Y(t) - k + 1) > \bar{\mu}_{11} + \bar{\mu}_{12}$, and $(X(t) - k)(Y(t) - k) \le \bar{\mu}_{11} + \bar{\mu}_{12}$. These imply that the right-hand side of (EC.10) is k for $t \in [t_k, t_{k+1})$ and prove the result. □

To carry out our asymptotic analysis, we index the performance of the sequence of queues by a superscript n, denoted by $Q_i^n, Z_0^n, Z_{1i}^n, Z_{2i}^n, M_i^n, R_{0i}^n, R_{1i}^n, D_{1i}^n, D_{2i}^n, X_i^n, i = 1, 2$, and Y^n, where the scaling is defined in the beginning of Section 3. According to (EC.10), for the n-th system, the system dynamics can be characterized as follows,

$$i = 1, 2: \quad Q_i^n(t) = Q_i^n(0) + A_i^n(t) - R_{0i}^n(t) - M_i^n(t), \tag{EC.11}$$

$$Z_0^n(t) = Z_0^n(0) + \sum_{i=1}^{2} \{R_{1i}^n(t) + D_{2i}^n(t) - M_i^n(t)\}, \tag{EC.12}$$

$$Z_{1i}^n(t) = Z_{1i}^n(0) + M_i^n(t) - D_{1i}^n(t) - R_{1i}^n(t), \tag{EC.13}$$

$$Z_{2i}^n(t) = Z_{2i}^n(0) + D_{1i}^n(t) - D_{2i}^n(t). \tag{EC.14}$$

We add a bar to denote quantities of fluid-scaled processes, e.g., $\bar{Q}_1^n(t) = Q_1^n(t)/n$. By using centering operation, the fluid-scaled dynamics can be represented as follows

$$i = 1, 2: \quad \bar{Q}_i^n(t) = (\bar{X}_i^n(t) - \tilde{X}_i^n(t)) - (\bar{M}_i^n(t) - \tilde{M}_i^n(t)) + \tilde{X}_i^n(t) - \tilde{M}_i^n(t), \tag{EC.15}$$

$$\bar{Z}_0^n(t) = (\bar{Y}^n(t) - \tilde{Y}^n(t)) - \sum_{i=1}^{2} (\bar{M}_i^n(t) - \tilde{M}_i^n(t)) + \tilde{Y}^n(t) - \sum_{i=1}^{2} \tilde{M}_i^n(t), \tag{EC.16}$$

$$\begin{aligned} \bar{Z}_{1i}^n(t) = \bar{Z}_{1i}^n(0) &- (\bar{R}_{1i}^n(t) - \theta_{1i} \int_0^t \bar{Z}_{1i}^n(s)ds) - (\bar{D}_{1i}^n(t) - \mu_{1i} \int_0^t \bar{Z}_{1i}^n(s)ds) \\ &+ (\bar{M}_i^n(t) - \tilde{M}_i^n(t)) + \tilde{M}_i^n(t) - \theta_{1i} \int_0^t \bar{Z}_{1i}^n(s)ds - \mu_{1i} \int_0^t \bar{Z}_{1i}^n(s)ds, \end{aligned} \tag{EC.17}$$

$$\begin{aligned} \bar{Z}_{2i}^n(t) = \bar{Z}_{2i}^n(0) &+ (\bar{D}_{1i}^n(t) - \mu_{1i} \int_0^t \bar{Z}_{1i}^n(s)ds) - (\bar{D}_{2i}^n(t) - \mu_2 \int_0^t \bar{Z}_{2i}^n(s)ds) \\ &+ \mu_{1i} \int_0^t \bar{Z}_{1i}^n(s)ds - \mu_2 \int_0^t \bar{Z}_{2i}^n(s)ds, \end{aligned} \tag{EC.18}$$

where $\tilde{M}^n(t) = \tilde{M}_1^n(t) + \tilde{M}_2^n(t) = \sup_{0 \le s \le t} \left[\frac{\tilde{X}^n(s) + \tilde{Y}^n(s) - \sqrt{(\tilde{X}^n(s) - \tilde{Y}^n(s))^2 + 4(\bar{\mu}_{11} + \bar{\mu}_{12})}}{2} \right]_+$, $\tilde{X}_i^n(t) = \bar{Q}_i^n(0) + \lambda_i t - \theta_{0i} \int_0^t \bar{Q}_i^n(s)ds, i = 1, 2$, and $\tilde{X}^n(t) = \tilde{X}_1^n(t) + \tilde{X}_2^n(t), \tilde{Y}^n(t) = \bar{Z}_0^n(t) + \sum_{i=1}^2 (\theta_{1i} \int_0^t \bar{Z}_{1i}^n(s)ds + \mu_2 \int_0^t \bar{Z}_{2i}^n(s)ds)$. Note that $\tilde{M}^n(t)$ is the centering process of the matching process $\bar{M}^n(t)$. We have the following result.

PROPOSITION EC.1. *For any given* $(B_{1i}, B_2, B_{3i}, B_{4i}, i = 1, 2) \in D([0, T], \mathbb{R}^7)$, *there exists a unique* $(Q_i, Z_0, Z_{1i}, Z_{2i}, i = 1, 2) \in D([0, T], \mathbb{R}^7)$ *satisfying*

$$i = 1, 2: \quad Q_i(t) = \tilde{X}_i(t) + B_{1i}(t) - \tilde{M}_i(t), \tag{EC.19}$$

$$Z_0(t) = \tilde{Y}(t) + B_2(t) - \tilde{M}_1(t) - \tilde{M}_2(t), \tag{EC.20}$$

$$Z_{1i}(t) = Z_{1i}(0) + B_{3i}(t) - \theta_{1i}\int_0^t Z_{1i}(s)ds - \mu_{1i}\int_0^t Z_{1i}(s)ds + \tilde{M}_i(t), \tag{EC.21}$$

$$Z_{2i}(t) = Z_{2i}(0) + B_{4i}(t) + \mu_{1i}\int_0^t Z_{1i}(s)ds - \mu_2\int_0^t Z_{2i}(s)ds, \tag{EC.22}$$

where $\tilde{M}(t) = \tilde{M}_1(t) + \tilde{M}_2(t) = \sup_{0\le s\le t}\lceil\frac{\tilde{X}(s)+\tilde{Y}(s)-\sqrt{(\tilde{X}(s)-\tilde{Y}(s))^2+4(\bar{\mu}_{11}+\bar{\mu}_{12})}}{2}\rceil^+$, $\tilde{X}_i(t) = Q_i(0) + \lambda_i t - \theta_{0i}\int_0^t Q_i(s)ds, i = 1,2$, $\tilde{X}(s) = \tilde{X}_1(s) + \tilde{X}_2(s)$, *and* $\tilde{Y}(t) = Z_0(t) + \sum_{i=1}^2(\theta_{1i}\int_0^t Z_{1i}(s)ds + \mu_2\int_0^t Z_{2i}(s)ds)$. *Thus, these representations constitute a mapping* f *from* $(B_{1i}, B_2, B_{3i}, B_{4i}, i = 1,2) \in D([0,T],\mathbb{R}^7)$ *to* $(Q_i, Z_0, Z_{1i}, Z_{2i}, i = 1,2) \in D([0,T],\mathbb{R}^7)$. *In addition, the mapping* f *is Lipschitz continuous under the uniform norm.*

Proof of Proposition EC.1 The proof is based on the contraction mapping principle. For any $(Q_i, Z_0, Z_{1i}, Z_{2i}, i = 1,2) \in D([0,T],\mathbb{R}^7)$ and $b < \frac{1}{\sum_{i=1}^2(4\theta_{0i}+5\theta_{1i}+2\mu_{1i}+5\mu_2)}$, define a mapping Φ from $D([0,b],\mathbb{R}^4)$ to $D([0,b],\mathbb{R}^7)$ given by (EC.19)-(EC.22). We prove that Φ is a contraction mapping. For any $(Q_i, Z_0, Z_{1i}, Z_{2i}, i = 1,2) \in D([0,T],\mathbb{R}^7)$ and $(Q_i^*, Z_0^*, Z_{1i}^*, Z_{2i}^*, i = 1,2) \in D([0,b],\mathbb{R}^7)$, we have

$$\begin{aligned}
&\|\Phi(Q_i, Z_0, Z_{1i}, Z_{2i}, i = 1,2) - \Phi(Q_i^*, Z_0^*, Z_{1i}^*, Z_{2i}^*, i = 1,2)\|_b \\
&\le \sum_{i=1}^2\{\theta_{0i}\int_0^b |Q_i(s) - Q_i^*(s)|ds + 2(\theta_{1i}+\mu_{1i})\int_0^b |Z_{1i}(s) - Z_{1i}^*(s)|ds + 2\mu_2\int_0^b |Z_{2i}(s) - Z_{2i}^*(s)|ds\} \\
&+ 3\sup_{0\le s\le b}|\tilde{M}(s) - \tilde{M}^*(s)|.
\end{aligned} \tag{EC.23}$$

For any $s \in [0, b]$, we have

$$
\begin{aligned}
&|\tilde{M}(s) - \tilde{M}^*(s)| \\
&\le \sup_{0\le r\le s} \left| \left[\frac{\tilde{X}(r) + \tilde{Y}(r) - \sqrt{(\tilde{X}(r) - \tilde{Y}(r))^2 + 4(\bar{\mu}_{11} + \bar{\mu}_{12})}}{2} \right]^+ \right. \\
&\qquad \left. - \left[\frac{\tilde{X}^*(r) + \tilde{Y}^*(r) - \sqrt{(\tilde{X}^*(r) - \tilde{Y}^*(r))^2 + 4(\bar{\mu}_{11} + \bar{\mu}_{12})}}{2} \right]^+ \right| \\
&\le \sup_{0\le r\le s} \left| \frac{\tilde{X}(r) + \tilde{Y}(r) - \sqrt{(\tilde{X}(r) - \tilde{Y}(r))^2 + 4(\bar{\mu}_{11} + \bar{\mu}_{12})}}{2} \right. \\
&\qquad \left. - \frac{\tilde{X}^*(r) + \tilde{Y}^*(r) - \sqrt{(\tilde{X}^*(r) - \tilde{Y}^*(r))^2 + 4(\bar{\mu}_{11} + \bar{\mu}_{12})}}{2} \right| \\
&\le \sup_{0\le r\le s} \left| \frac{\tilde{X}(r) - \tilde{X}^*(r) + \tilde{Y}(r) - \tilde{Y}^*(r)}{2} \right. \\
&\qquad \left. + \frac{[\tilde{X}(r) - \tilde{Y}(r) - (\tilde{X}^*(r) - \tilde{Y}^*(r))][\tilde{X}(r) - \tilde{Y}(r) + \tilde{X}^*(r) - \tilde{Y}^*(r)]}{2[\sqrt{(\tilde{X}(r) - \tilde{Y}(r))^2 + 4(\bar{\mu}_{11} + \bar{\mu}_{12})} + \sqrt{(\tilde{X}^*(r) - \tilde{Y}^*(r))^2 + 4(\bar{\mu}_{11} + \bar{\mu}_{12})}]} \right| \\
&\le \sup_{0\le r\le s} \left(|\tilde{X}(r) - \tilde{X}^*(r)| + |\tilde{Y}(r) - \tilde{Y}^*(r)| \right) \\
&\le \sup_{0\le r\le s} \left(|\tilde{X}_1(r) - \tilde{X}_1^*(r)| + |\tilde{X}_2(r) - \tilde{X}_2^*(r)| + |\tilde{Y}(r) - \tilde{Y}^*(r)| \right) \\
&\le \sum_{i=1}^{2} (\theta_{0i} \int_0^s |Q_i(r) - Q_i^*(r)|\, dr + \theta_{1i} \int_0^s |Z_{1i}(r) - Z_{1i}^*(r)|\, dr + \mu_2 \int_0^s |Z_{2i}(r) - Z_{2i}^*(r)|\, dr) \\
&\le \sum_{i=1}^{2} (\theta_{0i} \int_0^b |Q_i(r) - Q_i^*(r)|\, dr + \theta_{1i} \int_0^b |Z_{1i}(r) - Z_{1i}^*(r)|\, dr + \mu_2 \int_0^b |Z_{2i}(r) - Z_{2i}^*(r)|\, dr).
\end{aligned}
$$

Thus we have

$$
\begin{aligned}
\sup_{0\le s\le b} |\tilde{M}(s) - \tilde{M}^*(s)| \le \sum_{i=1}^{2} \Bigg(&\theta_{0i} \int_0^b |Q_i(r) - Q_i^*(r)|\, dr \\
&+ \theta_{1i} \int_0^b |Z_{1i}(r) - Z_{1i}^*(r)|\, dr \\
&+ \mu_2 \int_0^b |Z_{2i}(r) - Z_{2i}^*(r)|\, dr \Bigg).
\end{aligned} \tag{EC.24}
$$

Substituting (EC.24) into (EC.23) yields

$$
\begin{aligned}
&\|\Phi(Q_i, Z_0, Z_{1i}, Z_{2i}, i = 1, 2) - \Phi(Q_i^*, Z_0^*, Z_{1i}^*, Z_{2i}^*, i = 1, 2)\|_b \\
&\le \sum_{i=1}^{2} \left\{ \theta_{0i} \int_0^b |Q_i(s) - Q_i^*(s)| ds + 2(\theta_{1i} + \mu_{1i}) \int_0^b |Z_{1i}(s) - Z_{1i}^*(s)| ds \right. \\
&\qquad \left. + 2\mu_2 \int_0^b |Z_{2i}(s) - Z_{2i}^*(s)| ds \right\} \\
&\quad + 3 \sum_{i=1}^{2} \Bigg(\theta_{0i} \int_0^b |Q_i(r) - Q_i^*(r)|\, dr + \theta_{1i} \int_0^b |Z_{1i}(r) - Z_{1i}^*(r)|\, dr
\end{aligned}
$$

$$
\begin{aligned}
&\qquad + \mu_2 \int_0^b |Z_{2i}(r) - Z_{2i}^*(r)|\, dr\Bigg) \\
&= \sum_{i=1}^{2} \Bigg\{ (4\theta_{0i}) \int_0^b |Q_i(s) - Q_i^*(s)| ds + (5\theta_{1i} + 2\mu_{1i}) \int_0^b |Z_{1i}(s) - Z_{1i}^*(s)| ds \\
&\qquad + (5\mu_2) \int_0^b |Z_{2i}(s) - Z_{2i}^*(s)| ds \Bigg\} \\
&\le \left(\sum_{i=1}^{2} (4\theta_{0i} + 5\theta_{1i} + 2\mu_{1i} + 5\mu_2) \right) b \| (Q_i, Z_0, Z_{1i}, Z_{2i}, i = 1,2) - (Q_i^*, Z_0^*, Z_{1i}^*, Z_{2i}^*, i = 1,2) \|_b \\
&< \| (Q_i, Z_0, Z_{1i}, Z_{2i}, i = 1,2) - (Q_i^*, Z_0^*, Z_{1i}^*, Z_{2i}^*, i = 1,2) \|_b. \qquad \text{(EC.25)}
\end{aligned}
$$

Thus, the mapping Φ is a contraction. By the Contraction Mapping Theorem, there exists a unique solution to (EC.19)–(EC.22) on $[0, b]$. It now remains to extend the existence and uniqueness result from $[0, b]$ to $[0, T]$.

To extend to the full interval $[0, T]$, we proceed iteratively. Replacing $(Q_i(t), Z_0(t), Z_{1i}(t), Z_{2i}(t))$ by $(Q_i(b+t), Z_0(b+t), Z_{1i}(b+t), Z_{2i}(b+t))$ for $i = 1, 2$, and following the previous argument, we obtain a unique solution for (EC.19)–(EC.22) on $[0, 2b]$. Repeating this approach for $\lceil T/b \rceil$ times gives a unique solution on the interval $[0, T]$.

Next we prove that the mapping f is Lipschitz continuous under the uniform norm. For any

$$
\| (B_{1i}, B_2, B_{3i}, B_{4i}, i = 1,2) - (B_{1i}^*, B_2^*, B_{3i}^*, B_{4i}^*, i = 1,2) \|_T < \epsilon,
$$

following the same steps as in the proof of Proposition EC.1, we obtain

$$
\begin{aligned}
&\| f(B_{1i}, B_2, B_{3i}, B_{4i}, i = 1,2) - f(B_{1i}^*, B_2^*, B_{3i}^*, B_{4i}^*, i = 1,2) \|_T \\
&\le 4 \| (B_{1i}, B_2, B_{3i}, B_{4i}, i = 1,2) - (B_{1i}^*, B_2^*, B_{3i}^*, B_{4i}^*, i = 1,2) \|_T \\
&\quad + \sum_{i=1}^{2} \Bigg\{ 4\theta_{0i} \int_0^T |Q_i(s) - Q_i^*(s)| ds + (5\theta_{1i} + 2\mu_{1i}) \int_0^T |Z_{1i}(s) - Z_{1i}^*(s)| ds \\
&\qquad + 5\mu_2 \int_0^T |Z_{2i}(s) - Z_{2i}^*(s)| ds \Bigg\} \\
&\le 4\epsilon + \left(\sum_{i=1}^{2} (4\theta_{0i} + 5\theta_{1i} + 2\mu_{1i} + 5\mu_2) \right) \\
&\quad \times \int_0^T \| f(B_{1i}, B_2, B_{3i}, B_{4i}, i = 1,2) - f(B_{1i}^*, B_2^*, B_{3i}^*, B_{4i}^*, i = 1,2) \|_s \, ds. \qquad \text{(EC.26)}
\end{aligned}
$$

Thus by Grönwall's inequality, we have

$$
\begin{aligned}
&\| f(B_{1i}, B_2, B_{3i}, B_{4i}, i = 1,2) - f(B_{1i}^*, B_2^*, B_{3i}^*, B_{4i}^*, i = 1,2) \|_T \\
&\le 4\epsilon \exp\left\{ \left(\sum_{i=1}^{2} (4\theta_{0i} + 5\theta_{1i} + 2\mu_{1i} + 5\mu_2) \right) T \right\}, \qquad \text{(EC.27)}
\end{aligned}
$$

which implies that the mapping f is Lipschitz continuous under the uniform norm. □

When $n \to \infty$, the weak convergence of the centered processes $\bar{A}_i^n(t) - \lambda_i t$, $\bar{R}_{0i}^n(t) - \theta_{0i}\int_0^t \bar{Q}_i^n(s)ds$, $\bar{R}_{1i}^n(t) - \theta_{1i}\int_0^t \bar{Z}_{1i}^n(s)ds$, and $\bar{D}_{2i}^n(t) - \mu_2\int_0^t \bar{Z}_{2i}^n(s)ds$ for $i = 1, 2$ follows from the functional strong law of large numbers. We further prove that $\bar{M}^n(t) - \tilde{M}^n(t)$ and $\bar{D}_{1i}^n(t) - \mu_{1i}\int_0^t \bar{Z}_{1i}^n(s)ds$ converge to 0 when $n \to \infty$ for $i = 1, 2$.

LEMMA EC.2. *For any $\epsilon > 0$ and any given T, we have*

$$\lim_{n\to\infty} \mathbb{P}\left(\sup_{t\le T} |\bar{M}^n(t) - \tilde{M}^n(t)| > \epsilon\right) = 0, \tag{EC.28}$$

and

$$\lim_{n\to\infty} \mathbb{P}\left(\sup_{t\le T}\left|\bar{D}_{1i}^n(t) - \mu_{1i}\int_0^t \bar{Z}_{1i}^n(s)ds\right| > \epsilon\right) = 0, \quad i = 1, 2. \tag{EC.29}$$

Proof of Lemma EC.2 We prove each result separately.

Part 1: Convergence of matching process

For any $t \le T$, by Lemma EC.1 and the definition of $\tilde{M}^n(t)$, we have

$$\begin{aligned}
|\bar{M}^n(t) - \tilde{M}^n(t)| &\le \frac{1}{n} + \left|\sup_{0\le s\le t}\left[\frac{\bar{X}^n(s) + \bar{Y}^n(s) - \sqrt{(\bar{X}^n(s) - \bar{Y}^n(s))^2 + 4(\bar{\mu}_{11} + \bar{\mu}_{12})}}{2}\right]^+\right. \\
&\quad \left. - \sup_{0\le s\le t}\left[\frac{\tilde{X}^n(s) + \tilde{Y}^n(s) - \sqrt{(\tilde{X}^n(s) - \tilde{Y}^n(s))^2 + 4(\bar{\mu}_{11} + \bar{\mu}_{12})}}{2}\right]^+\right| \\
&\le \frac{1}{n} + \sup_{0\le s\le t}|\bar{X}^n(s) - \tilde{X}^n(s)| + \sup_{0\le s\le t}|\bar{Y}^n(s) - \tilde{Y}^n(s)|.
\end{aligned} \tag{EC.30}$$

Now we bound each term:

$$\begin{aligned}
\sup_{0\le s\le t}|\bar{X}^n(s) - \tilde{X}^n(s)| &= \sup_{0\le s\le t}|\bar{X}_1^n(s) + \bar{X}_2^n(s) - \tilde{X}_1^n(s) - \tilde{X}_2^n(s)| \\
&\le \sup_{0\le s\le t}\sum_{i=1}^2\left|(\bar{A}_i^n(s) - \lambda_i s) - (\bar{R}_{0i}^n(s) - \theta_{0i}\int_0^s \bar{Q}_i^n(r)dr)\right|,
\end{aligned} \tag{EC.31}$$

and

$$\begin{aligned}
\sup_{0\le s\le t}|\bar{Y}^n(s) - \tilde{Y}^n(s)| \le \sup_{0\le s\le t}\sum_{i=1}^2\Bigg|&(\bar{R}_{1i}^n(s) - \theta_{1i}\int_0^s \bar{Z}_{1i}^n(r)dr) \\
&+ (\bar{D}_{2i}^n(s) - \mu_2\int_0^s \bar{Z}_{2i}^n(r)dr)\Bigg|.
\end{aligned} \tag{EC.32}$$

By the functional strong law of large numbers, each centered process converges to 0 in probability, which establishes (EC.28).

Part 2: Convergence of departure process

For the convergence of $\bar{D}_{1i}^n(t) - \mu_{1i}\int_0^t \bar{Z}_{1i}^n(s)ds$, we first establish that the pickup rates remain bounded. Under our threshold control policy, a matching occurs when

$$\bar{Q}_i^n(t^-)\bar{Z}_0^n(t^-) \ge \bar{\mu}_{1i}, \quad i = 1, 2. \tag{EC.33}$$

For any $\epsilon_1 > 0$, we show that

$$\lim_{n\to\infty} \mathbb{P}\left(\sup_{t\le T} \frac{(Q_i^n(t^-)+1)Z_0^n(t^-)}{n^2} \ge \bar{\mu}_{1i} + \epsilon_1\right) + \mathbb{P}\left(\sup_{t\le T} \frac{Q_i^n(t^-)(Z_0^n(t^-)+1)}{n^2} \ge \bar{\mu}_{1i} + \epsilon_1\right) = 0. \tag{EC.34}$$

The first term corresponds to a rider arrival triggering a match, while the second term corresponds to an empty car arrival triggering a match.

Since $\frac{Q_i^n(t^-)}{n} \cdot \frac{Z_0^n(t^-)}{n} \le \bar{\mu}_{1i}$ by our control policy, the first case would require $Z_0^n(t^-) > \epsilon_1 n^2$. However, this contradicts the constraint $Z_0^n(t^-) \le n$ for sufficiently large n.

Similarly, the second case would require $Q_i^n(t^-) > \epsilon_1 n^2$. Since $Q_i^n(t^-) \le A_i^n(t^-) + Q_i^n(0)$, we would need $A_i^n(t^-) > \epsilon_1 n^2 - \bar{Q}_i(0)n$. By the law of large numbers:

$$\lim_{n\to\infty} \mathbb{P}\left(\sup_{t\le T} A_i^n(t^-) > \epsilon_1 n^2 - \bar{Q}_i(0)n\right) = 0. \tag{EC.35}$$

This establishes (EC.34).

Given (EC.34), we can apply a coupling argument: for large n, the departure process $D_{1i}^n(t)$ is sandwiched between two non-homogeneous Poisson processes with rates $\mu_{1i} Z_{1i}^n(t)$ and $(\mu_{1i} + \epsilon_1) Z_{1i}^n(t)$ respectively, with probability approaching 1. As $\epsilon_1 \to 0$, this yields the desired convergence in (EC.29). □

THEOREM EC.1. *Suppose the initial condition holds. Then as $n \to \infty$, $(\bar{Q}_i^n, \bar{Z}_0^n, \bar{Z}_{1i}^n, \bar{Z}_{2i}^n, i = 1,2)$ converges to $(\bar{Q}_i, \bar{Z}_0, \bar{Z}_{1i}, \bar{Z}_{2i}, i = 1,2)$ in probability, where the limit satisfies the fluid model equations:*

$$\begin{aligned}
i = 1,2: \quad & \bar{Q}_i(t) = \bar{Q}_i(0) + \lambda_i t - \theta_{0i} \int_0^t \bar{Q}_i(s)ds - \bar{M}_i(t), \\
& \bar{Z}_0(t) = \bar{Z}_0(0) + \sum_{i=1}^{2} \{\theta_{1i} \int_0^t \bar{Z}_{1i}(s)ds + \mu_2 \int_0^t \bar{Z}_{2i}(s)ds - \bar{M}_i(t)\}, \\
& \bar{Z}_{1i}(t) = \bar{Z}_{1i}(0) + \bar{M}_i(t) - \theta_{1i} \int_0^t \bar{Z}_{1i}(s)ds - \mu_{1i} \int_0^t \bar{Z}_{1i}(s)ds, \\
& \bar{Z}_{2i}(t) = \bar{Z}_{2i}(0) + \mu_{1i} \int_0^t \bar{Z}_{1i}(s)ds - \mu_2 \int_0^t \bar{Z}_{2i}(s)ds,
\end{aligned}$$

where the matching process is given by

$$\bar{M}(t) = \bar{M}_1(t) + \bar{M}_2(t) = \sup_{0\le s\le t} \left[\frac{\tilde{X}(s) + \tilde{Y}(s) - \sqrt{(\tilde{X}(s) - \tilde{Y}(s))^2 + 4(\bar{\mu}_{11} + \bar{\mu}_{12})}}{2} \right]^+,$$

where $\tilde{X}_i(t) = \bar{Q}_i(0) + \lambda_i t - \theta_{0i} \int_0^t \bar{Q}_i(s)ds$ for $i = 1,2$, $\tilde{X}(t) = \tilde{X}_1(t) + \tilde{X}_2(t)$, and $\tilde{Y}(t) = \bar{Z}_0(t) + \sum_{i=1}^2 (\theta_{1i} \int_0^t \bar{Z}_{1i}(s)ds + \mu_2 \int_0^t \bar{Z}_{2i}(s)ds)$.

Proof of Theorem EC.1 For each n, we define the residual terms as follows:

$$\begin{aligned}
B_{1i}^n(t) &= (\bar{A}_i^n(t) - \lambda_i t) - (\bar{R}_{0i}^n(t) - \theta_{0i} \int_0^t \bar{Q}_i^n(s)ds) - (\bar{M}_i^n(t) - \tilde{M}_i^n(t)), \\
B_2^n(t) &= \sum_{i=1}^{2} \{(\bar{R}_{1i}^n(t) - \theta_{1i} \int_0^t \bar{Z}_{1i}^n(s)ds) + (\bar{D}_{2i}^n(t) - \mu_2 \int_0^t \bar{Z}_{2i}^n(s)ds)\}
\end{aligned}$$

$$
\begin{aligned}
&- \sum_{i=1}^{2} (\bar{M}_i^n(t) - \tilde{M}_i^n(t)), \\
B_{3i}^n(t) &= -(\bar{R}_{1i}^n(t) - \theta_{1i} \int_0^t \bar{Z}_{1i}^n(s)ds) - (\bar{D}_{1i}^n(t) - \mu_{1i} \int_0^t \bar{Z}_{1i}^n(s)ds) \\
&+ (\bar{M}_i^n(t) - \tilde{M}_i^n(t)), \\
B_{4i}^n(t) &= (\bar{D}_{1i}^n(t) - \mu_{1i} \int_0^t \bar{Z}_{1i}^n(s)ds) - (\bar{D}_{2i}^n(t) - \mu_2 \int_0^t \bar{Z}_{2i}^n(s)ds).
\end{aligned}
$$

With these definitions, the scaled stochastic processes $(\bar{Q}_i^n, \bar{Z}_0^n, \bar{Z}_{1i}^n, \bar{Z}_{2i}^n, i = 1, 2)$ and residual terms $(B_{1i}^n, B_2^n, B_{3i}^n, B_{4i}^n, i = 1, 2)$ jointly satisfy the system of equations (EC.19)–(EC.22).

By the functional strong law of large numbers, each centered arrival and service process converges to zero in probability:

$$
\begin{aligned}
\lim_{n\to\infty} \mathbb{P}\left(\sup_{t\le T} |\bar{A}_i^n(t) - \lambda_i t| > \epsilon\right) &= 0, \quad i = 1, 2, \\
\lim_{n\to\infty} \mathbb{P}\left(\sup_{t\le T} \left|\bar{R}_{0i}^n(t) - \theta_{0i} \int_0^t \bar{Q}_i^n(s)ds\right| > \epsilon\right) &= 0, \quad i = 1, 2, \\
\lim_{n\to\infty} \mathbb{P}\left(\sup_{t\le T} \left|\bar{R}_{1i}^n(t) - \theta_{1i} \int_0^t \bar{Z}_{1i}^n(s)ds\right| > \epsilon\right) &= 0, \quad i = 1, 2, \\
\lim_{n\to\infty} \mathbb{P}\left(\sup_{t\le T} \left|\bar{D}_{2i}^n(t) - \mu_2 \int_0^t \bar{Z}_{2i}^n(s)ds\right| > \epsilon\right) &= 0, \quad i = 1, 2.
\end{aligned}
$$

From Lemma EC.2, we have

$$
\begin{aligned}
\lim_{n\to\infty} \mathbb{P}\left(\sup_{t\le T} |\bar{M}^n(t) - \tilde{M}^n(t)| > \epsilon\right) &= 0, \\
\lim_{n\to\infty} \mathbb{P}\left(\sup_{t\le T} \left|\bar{D}_{1i}^n(t) - \mu_{1i} \int_0^t \bar{Z}_{1i}^n(s)ds\right| > \epsilon\right) &= 0, \quad i = 1, 2.
\end{aligned}
$$

Therefore, for any $\epsilon > 0$,

$$
\lim_{n\to\infty} \mathbb{P}\left(\sup_{t\le T} \|(B_{1i}^n, B_2^n, B_{3i}^n, B_{4i}^n, i = 1, 2)\|_T > \epsilon\right) = 0,
$$

which implies that $(B_{1i}^n, B_2^n, B_{3i}^n, B_{4i}^n, i = 1, 2)$ converges to $(0, 0, 0, 0, 0, 0, 0)$ in probability.

Since the mapping f defined by equations (EC.19)–(EC.22) is Lipschitz continuous (Proposition EC.1), and $(B_{1i}^n, B_2^n, B_{3i}^n, B_{4i}^n, i = 1, 2) \to (0, 0, 0, 0, 0, 0, 0)$ in probability, we conclude that

$$
(\bar{Q}_i^n, \bar{Z}_0^n, \bar{Z}_{1i}^n, \bar{Z}_{2i}^n, i = 1, 2) \to (\bar{Q}_i, \bar{Z}_0, \bar{Z}_{1i}, \bar{Z}_{2i}, i = 1, 2)
$$

in probability, where the limit satisfies the fluid model equations with the matching process. □

EC.5. Fluid Model: Existence, Uniqueness, and Equilibrium Convergence

$$q_i(t) = q_i(0) + \lambda_i t - \theta_{0i} \int_0^t q_i(s)ds - m_i(t), \quad i = 1, 2, \tag{EC.36}$$

$$z_0(t) = z_0(0) + \sum_{j=1}^{2} \int_0^t (\theta_{1j} z_{1j}(s) + \mu_2 z_{2j}(s))ds - \sum_{j=1}^{2} m_j(t), \tag{EC.37}$$

$$z_{1i}(t) = z_{1i}(0) + m_i(t) - (\theta_{1i} + \mu_{1i}) \int_0^t z_{1i}(s)ds, \quad i = 1, 2, \tag{EC.38}$$

$$z_{2i}(t) = z_{2i}(0) + \mu_{1i} \int_0^t z_{1i}(s)ds - \mu_2 \int_0^t z_{2i}(s)ds, \quad i = 1, 2. \tag{EC.39}$$

Throughout this section, we set the spatial matching parameters as $\alpha_1 = \alpha_2 = \alpha$. This specific condition is adopted to ensure consistency with the derivation of matching rates under threshold policies previously established in Section EC.4 (see Lemma EC.1 and Theorem EC.2). The fluid model for the two-type customer system, under this assumption, is defined by equations (EC.53)-(EC.56).

PROPOSITION EC.2. *The fluid model for the two-type customer system, as defined by equations (EC.53)-(EC.56) with the threshold matching policy where matching for type $i \in \{1, 2\}$ occurs if its potential matching intensity $\mu_{1i}(t) = Cq_i(t)^\alpha z_0(t)^\alpha$ meets or exceeds a threshold μ^*_{1i}, has a unique solution trajectory $(q_i(t), z_0(t), z_{1i}(t), z_{2i}(t))$ for $t \geq 0$ given initial conditions.*

Proof of Proposition EC.2 The proof analyzes the system dynamics in stages, based on whether the matching thresholds are met.

Stage 1: No matching thresholds are met. Assume the system starts in or enters a state where $\mu_{1i}(t) < \mu^*_{1i}$ for both $i = 1, 2$. In this stage, no matching occurs, so the matching rates $m_1(t) = 0$ and $m_2(t) = 0$. Assume $\mu_{1i}(t) = \mu_{1i}, i = 1, 2$ remains below the threshold, and the system dynamics (10)-(20) simplify to:

$$q_i(t) = q_i(0) + \lambda_i t - \theta_{0i} \int_0^t q_i(s)ds, \quad i = 1, 2, \tag{EC.40}$$

$$z_0(t) = z_0(0) + \int_0^t \sum_{j=1}^{2} (\theta_{1j} z_{1j}(s) + \mu_2 z_{2j}(s))ds, \tag{EC.41}$$

$$z_{1i}(t) = z_{1i}(0)e^{-(\theta_{1i}+\mu_{1i})t}, \quad i = 1, 2, \tag{EC.42}$$

$$z_{2i}(t) = z_{2i}(0) + \frac{\mu_{1i}}{\theta_{1i} + \mu_{1i}} z_{1i}(0)(1 - e^{-(\theta_{1i}+\mu_{1i})t}) - \mu_2 \int_0^t z_{2i}(s)ds, \quad i = 1, 2. \tag{EC.43}$$

The derivatives of the right-hand sides of these equations are linear in the state variables, hence Lipschitz continuous. By the Picard-Lindelöf theorem (Coddington and Levinson 1955), a unique solution exists for this stage. Typically, $q_i(t)$ evolves towards λ_i/θ_{0i} (if $\theta_{0i} > 0$), $z_{1i}(t) \to 0$, $z_{2i}(t) \to 0$, and $z_0(t) \to 1$ (due to driver conservation, assuming $\sum(z_0 + z_{1j} + z_{2j}) = 1$). Consequently, $\mu_{1i}(t)$ tends towards $C(\lambda_i/\theta_{0i})^\alpha$. If system parameters are such that $C(\lambda_i/\theta_{0i})^\alpha > \mu^*_{1i}$ for some i, then $\mu_{1i}(t)$ for that type i will eventually reach its threshold μ^*_{1i}. Let t_1 be the first time this occurs.

Stage 2: Both matching thresholds are met and active. We set the spatial matching parameters as $\alpha_1 = \alpha_2 = \alpha$ for simplicity, and the case that $\alpha_1 \neq \alpha_2$ can be proven similarly. Consider the scenario where both thresholds are met and active for $t \geq t_1$, i.e., $\mu_{1i}(t) = Cq_i(t)^\alpha z_0(t)^\alpha = \mu_{1i}^*$ for $i = 1, 2$, which implies $\frac{d}{dt}(Cq_i(t)^\alpha z_0(t)^\alpha) = 0$. The condition $\frac{d}{dt}(Cq_i(t)^\alpha z_0(t)^\alpha) = 0$ for $i = 1, 2$ implies that the logarithmic derivative is zero. Since C and α are constants, and assuming $q_i(t) > 0$ and $z_0(t) > 0$:

$$\frac{d}{dt}(\alpha \ln q_i(t) + \alpha \ln z_0(t)) = 0$$

$$\alpha \left(\frac{1}{q_i(t)} \frac{dq_i}{dt} + \frac{1}{z_0(t)} \frac{dz_0}{dt} \right) = 0$$

Multiplying by $q_i(t)z_0(t)/\alpha$ (assuming $\alpha \neq 0$) gives

$$z_0(t) \frac{dq_i}{dt} + q_i(t) \frac{dz_0}{dt} = 0, \quad i = 1, 2.$$

Now, we substitute the expressions for the derivatives $\frac{dq_i}{dt}$ and $\frac{dz_0}{dt}$ from equations (EC.53) and (EC.54), note that $m_i(t)$ is monotonically increasing and non-negative so $\pi_i(t) \triangleq m_i'(t)$ exists a.e., $i = 1, 2$:

$$\frac{dq_i}{dt} = \lambda_i - \theta_{0i} q_i(t) - \pi_i(t),$$

$$\frac{dz_0}{dt} = \sum_{j=1}^{2} (\theta_{1j} z_{1j}(t) + \mu_2 z_{2j}(t)) - \sum_{j=1}^{2} \pi_j(t).$$

Let $s(t) = \sum_{j=1}^{2} (\theta_{1j} z_{1j}(t) + \mu_2 z_{2j}(t))$. Then $\frac{dz_0}{dt} = s(t) - (\pi_1(t) + \pi_2(t))$.

Consider the case for $i = 1$:

$$z_0(t) \left(\lambda_1 - \theta_{01} q_1(t) - \pi_1(t) \right) + q_1(t) \left(s(t) - \pi_1(t) - \pi_2(t) \right) = 0.$$

Expanding,

$$z_0(t)\lambda_1 - z_0(t)\theta_{01} q_1(t) - z_0(t)\pi_1(t) + q_1(t)s(t) - q_1(t)\pi_1(t) - q_1(t)\pi_2(t) = 0.$$

Collecting terms involving $\pi_1(t)$ and $\pi_2(t)$ on the left-hand side,

$$z_0(t)\pi_1(t) + q_1(t)\pi_1(t) + q_1(t)\pi_2(t) = z_0(t)\lambda_1 - z_0(t)\theta_{01} q_1(t) + q_1(t)s(t).$$

Factoring yields

$$(z_0(t) + q_1(t))\pi_1(t) + q_1(t)\pi_2(t) = (\lambda_1 - \theta_{01} q_1(t)) z_0(t) + s(t) q_1(t).$$

This is the first equation of the linear system. Repeating the same calculation for $i = 2$ produces the second equation, giving the following linear system for $\pi_1(t)$ and $\pi_2(t)$:

$$(z_0(t) + q_1(t))\pi_1(t) + q_1(t)\pi_2(t) = (\lambda_1 - \theta_{01} q_1(t)) z_0(t) + s(t) q_1(t),$$

$$q_2(t)\pi_1(t) + (z_0(t) + q_2(t))\pi_2(t) = (\lambda_2 - \theta_{02} q_2(t)) z_0(t) + s(t) q_2(t).$$

The determinant of the coefficient matrix is $z_0(t)(z_0(t) + q_1(t) + q_2(t))$. Since $\mu_{1i}(t) = \mu^*_{1i} > 0$ implies $q_i(t) > 0$ and $z_0(t) > 0$, the determinant is strictly positive. Thus, we can uniquely solve for $\pi_1(t)$ and $\pi_2(t)$ as functions of the state variables, denoted $m^*_i(q_1, q_2, z_0, z_{11}, z_{12}, z_{21}, z_{22})$. Substituting these $\pi^*_i(\cdot)$ into (EC.53)-(EC.56) yields a system of ODEs whose right-hand sides are continuous and, on any compact subset of the state space where q_i, z_0 are bounded away from zero, Lipschitz continuous. The Picard-Lindelöf theorem then guarantees a unique solution for this stage.

If the system state evolves such that a matching condition $\mu_{1k}(t) < \mu^*_{1k}$ for some k, then $\pi_k(t)$ becomes 0, and the system dynamics adjust accordingly, potentially re-entering Stage 1 or a mixed stage.

Stage 3: Mixed stage - one threshold active, one not. Without loss of generality, assume $\mu_{11}(t) = Cq_1(t)^\alpha z_0(t)^\alpha = \mu^*_{11}$ (active) and $\mu_{12}(t) < \mu^*_{12}$ (inactive). In this case, $\pi_2(t) = 0$. The condition for the active threshold μ^*_{11} is

$$z_0(t)\frac{dq_1}{dt} + q_1(t)\frac{dz_0}{dt} = 0.$$

Substituting the derivatives,

$$\frac{dq_1}{dt} = \lambda_1 - \theta_{01}q_1(t) - \pi_1(t),$$

$$\frac{dz_0}{dt} = \sum_{j=1}^{2}(\theta_{1j}z_{1j}(t) + \mu_2 z_{2j}(t)) - \pi_1(t) \quad (\text{since } \pi_2(t) = 0).$$

Let $s(t) = \sum_{j=1}^{2}(\theta_{1j}z_{1j}(t) + \mu_2 z_{2j}(t))$. The condition becomes

$$z_0(t)(\lambda_1 - \theta_{01}q_1(t) - \pi_1(t)) + q_1(t)(s(t) - \pi_1(t)) = 0,$$

$$z_0(t)\lambda_1 - z_0(t)\theta_{01}q_1(t) - z_0(t)\pi_1(t) + q_1(t)s(t) - q_1(t)\pi_1(t) = 0.$$

Solving for $\pi_1(t)$,

$$(z_0(t) + q_1(t))\pi_1(t) = z_0(t)\lambda_1 - z_0(t)\theta_{01}q_1(t) + q_1(t)s(t),$$

$$\pi_1(t) = \frac{(\lambda_1 - \theta_{01}q_1(t))z_0(t) + s(t)q_1(t)}{z_0(t) + q_1(t)}.$$

Since $\mu^*_{11} > 0$, $q_1(t) > 0$ and $z_0(t) > 0$, so the denominator $z_0(t) + q_1(t) > 0$. Thus, $\pi_1(t)$ is uniquely determined as a function of the state variables. Substituting $\pi_1(t)$ and $\pi_2(t) = 0$ into (EC.53)-(EC.56) yields a system of ODEs. Similar to Stage 2, the right-hand sides are continuous and Lipschitz on relevant compact sets, guaranteeing a unique solution. The system remains in this stage as long as $\mu_{11}(t) = \mu^*_{11}$ and $\mu_{12}(t) < \mu^*_{12}$. Transitions occur if $\mu_{12}(t)$ reaches μ^*_{12} (to Stage 2) or if the conditions for maintaining $\mu_{11}(t) = \mu^*_{11}$ (e.g., $\pi_1(t) \geq 0$) are violated, potentially leading to Stage 1.

Overall Existence and Uniqueness. The unique solutions within each stage, combined with deterministic conditions for transitioning between stages (i.e., $\mu_{1i}(t)$ reaching or falling below μ^*_{1i}), establish the existence and uniqueness of the overall fluid trajectory. The matching rates $\pi_i(t)$ are defined piecewise based on these conditions, ensuring a well-defined path for the system state. □

THEOREM EC.2. *Consider the fluid model where matching for type $i \in \{1,2\}$ occurs if its potential matching intensity $\mu_{1i}(t) = Cq_i(t)^\alpha z_0(t)^\alpha$ meets or exceeds a positive threshold $\mu^*_{1i} > 0$. If system parameters allow for a physical equilibrium where matching is active for both types (meaning the derived matching rates π^*_i are non-negative and $z^*_0 > 0$), then there exists a unique non-negative solution $(q^*_1, q^*_2, z^*_0, z^*_{11}, z^*_{12}, z^*_{21}, z^*_{22})$ with associated non-negative matching rates (π^*_1, π^*_2) to the equilibrium equations:*

$$\lambda_i = \theta_{0i} q^*_i + \theta_{1i} z^*_{1i} + \mu_2 z^*_{2i}, \qquad i = 1,2, \tag{EC.44}$$

$$\mu^*_{1i} z^*_{1i} = \mu_2 z^*_{2i}, \qquad i = 1,2, \tag{EC.45}$$

$$1 = z^*_0 + \sum_{j=1}^{2} (z^*_{1j} + z^*_{2j}), \tag{EC.46}$$

$$\mu^*_{1i} = C(q^*_i)^\alpha (z^*_0)^\alpha, \qquad i = 1,2. \tag{EC.47}$$

This solution is the unique physical equilibrium point of the fluid model when both matching thresholds μ^*_{1i} are active. Furthermore, the fluid process $(q_i(t), z_0(t), z_{1i}(t), z_{2i}(t))$ for $i = 1,2$, converges exponentially fast to this equilibrium point as $t \to \infty$.

Proof of Theorem EC.2 This proof addresses the existence, uniqueness, and convergence to an equilibrium point $(q^*_1, q^*_2, z^*_0, z^*_{11}, z^*_{12}, z^*_{21}, z^*_{22})$ where both matching thresholds are active. This means the matching intensity for type i, $\mu_{1i}(t) = Cq_i(t)^\alpha z_0(t)^\alpha$, is equal to a predefined positive threshold μ^*_{1i} for $i = 1,2$.

1. Existence and Uniqueness of a Solution to Equilibrium Equations

At an equilibrium point, all time derivatives in the fluid model equations (EC.53)-(EC.56) must be zero. This yields the following system of algebraic equations:

$$\lambda_i = \theta_{0i} q^*_i + \pi^*_i, \quad i = 1,2, \tag{EC.48}$$

$$\sum_{j=1}^{2} \pi^*_j = \sum_{j=1}^{2} (\theta_{1j} z^*_{1j} + \mu_2 z^*_{2j}), \tag{EC.49}$$

$$\pi^*_i = (\theta_{1i} + \mu^*_{1i}) z^*_{1i}, \quad i = 1,2, \tag{EC.50}$$

$$\mu^*_{1i} z^*_{1i} = \mu_2 z^*_{2i}, \quad i = 1,2. \tag{EC.51}$$

Additionally, the conservation of drivers (total proportion is 1) requires:

$$1 = z^*_0 + \sum_{j=1}^{2} (z^*_{1j} + z^*_{2j}) \tag{EC.52}$$

For an equilibrium with active matching thresholds as assumed:

$$C(q^*_i)^\alpha (z^*_0)^\alpha = \mu^*_{1i}, \quad i = 1,2 \tag{EC.53}$$

where $\mu^*_{1i} > 0$ are the predefined thresholds for the matching intensities and we assume $\alpha_1 = \alpha_2 = \alpha$ for simplicity. Equation (EC.53) implies $q^*_i > 0$ and $z^*_0 > 0$.

From (EC.53), we express q_i^* as $q_i^* = (\mu_{1i}^*/C)^{1/\alpha}(z_0^*)^{-1}$. From (EC.50) and (EC.51), assuming $\theta_{1j} + \mu_{1j}^* > 0$ and $\mu_2 > 0$, we can express the assigned drivers and busy drivers as: $z_{1j}^* = \pi_j^*/(\theta_{1j} + \mu_{1j}^*)$ and $z_{2j}^* = (\mu_{1j}^*/\mu_2)z_{1j}^* = \frac{\mu_{1j}^*}{\mu_2}\frac{\pi_j^*}{\theta_{1j}+\mu_{1j}^*}$. Substituting these expressions for z_{1j}^* and z_{2j}^* into the conservation equation (EC.52) gives $1 = z_0^* + \sum_{j=1}^2 \pi_j^* \left(\frac{1}{\theta_{1j}+\mu_{1j}^*}(1 + \frac{\mu_{1j}^*}{\mu_2})\right)$. Let $k_j = \frac{1+\mu_{1j}^*/\mu_2}{\theta_{1j}+\mu_{1j}^*}$. Then $1 = z_0^* + \sum_{j=1}^2 k_j\pi_j^*$. Using $\pi_j^* = \lambda_j - \theta_{0j}q_j^*$ from (EC.48), and then substituting the expression for q_j^*: $1 = z_0^* + \sum_{j=1}^2 k_j \left(\lambda_j - \theta_{0j}(\mu_{1j}^*/C)^{1/\alpha}(z_0^*)^{-1}\right)$. Multiplying by z_0^* (since $z_0^* > 0$ at this type of equilibrium): $z_0^* = (z_0^*)^2 + \sum_{j=1}^2 k_j\lambda_j z_0^* - \sum_{j=1}^2 k_j\theta_{0j}(\mu_{1j}^*/C)^{1/\alpha}$. Rearranging gives a quadratic equation for z_0^*:

$$(z_0^*)^2 + \left(\sum_{j=1}^2 k_j\lambda_j - 1\right)z_0^* - \sum_{j=1}^2 k_j\theta_{0j}(\mu_{1j}^*/C)^{1/\alpha} = 0. \tag{EC.54}$$

Let $A = 1$, $B = \sum_{j=1}^2 k_j\lambda_j - 1$, and $D = -\sum_{j=1}^2 k_j\theta_{0j}(\mu_{1j}^*/C)^{1/\alpha}$. Assume all system parameters $(\lambda_j, \theta_{0j}, \theta_{1j}, \mu_{1j}, \mu_2, C, \mu_{1j}^*)$ are positive. Then $k_j > 0$. If $\theta_{0j} > 0$ and $\mu_{1j}^* > 0$ for at least one j, then $D < 0$. If $D < 0$, the quadratic equation $A(z_0^*)^2 + Bz_0^* + D = 0$ has exactly one positive real root given by $z_0^* = (-B + \sqrt{B^2 - 4AD})/(2A)$. This provides a unique candidate for z_0^*. (If $D = 0$, which would imply $\theta_{0j}(\mu_{1j}^*/C)^{1/\alpha} = 0$ for all j, then $z_0^*(z_0^* + B) = 0$. A positive solution $z_0^* = -B$ exists if $B < 0$.)

Given this unique positive z_0^*:

- $q_i^* = (\mu_{1i}^*/C)^{1/\alpha}(z_0^*)^{-1}$ is uniquely determined and positive.
- $\pi_i^* = \lambda_i - \theta_{0i}q_i^*$ is uniquely determined. For a physical equilibrium of this type (both thresholds active), we require $\pi_i^* \geq 0$. This imposes conditions on the system parameters, namely $\lambda_i \geq \theta_{0i}q_i^*$.
- $z_{1i}^* = \pi_i^*/(\theta_{1i} + \mu_{1i}^*)$ is uniquely determined and non-negative if $\pi_i^* \geq 0$.
- $z_{2i}^* = (\mu_{1i}^*/\mu_2)z_{1i}^*$ is uniquely determined and non-negative if $z_{1i}^* \geq 0$.

Thus, if system parameters ensure $D < 0$ (or $D = 0$ and $B < 0$ for a positive z_0^*) and the resulting matching rates m_i^* are non-negative, there exists a unique non-negative solution $(q_1^*, q_2^*, z_0^*, z_{11}^*, z_{12}^*, z_{21}^*, z_{22}^*)$ to the equilibrium equations (EC.48)-(EC.53).

2. The Solution is the Unique Equilibrium of the Fluid Model (under active threshold conditions)

The solution derived in Part 1, by its construction, satisfies the conditions for an equilibrium point of the fluid model where all time derivatives are zero and both matching thresholds μ_{1i}^* are active. Proposition EC.2 establishes that the fluid model equations (EC.53)-(EC.56), under the specified threshold matching policy, have a unique solution trajectory given any initial conditions. As the system parameters and initial conditions lead the system to an equilibrium where both matching thresholds are active (i.e., the system operates in Stage 2, as described in the proof of Proposition EC.2, for $t \to \infty$), then this equilibrium point must be the unique point found in Part 1.

3. Convergence to the Equilibrium

To show convergence of the fluid process $(q_i(t), z_0(t), z_{1i}(t), z_{2i}(t))$ to the equilibrium point $X^* = (q_1^*, q_2^*, z_0^*, z_{11}^*, z_{12}^*, z_{21}^*, z_{22}^*)$ found in Part 1, we analyze the dynamics of the system under the assumption

that both matching thresholds are active. This means $\mu_{1i}(t) = Cq_i(t)^\alpha z_0(t)^\alpha = \mu_{1i}^* > 0$ for $i = 1, 2$, so $q_i(t)z_0(t) = \sqrt[\alpha]{\mu_{1i}^*/C} =: K_i$, where K_i is a positive constant for $i = 1, 2$. This implies $q_i'(t)z_0(t) + q_i(t)z_0'(t) = 0$.

The argument for convergence proceeds in two main steps: (a) First, we argue that $z_0(t)$ (and consequently $q_i(t) = K_i/z_0(t)$) converges to its equilibrium value z_0^* (and q_i^*). (b) Second, given the convergence of $z_0(t)$ and $q_i(t)$ to their equilibrium values, we analyze the subsystem for $(z_{11}(t), z_{12}(t), z_{21}(t), z_{22}(t))$. We show that this subsystem, when z_0, q_i are at equilibrium, is governed by a system of linear autonomous ODEs whose Jacobian matrix (denoted J) has all eigenvalues with strictly negative real parts. This implies exponential convergence for these components, which is key to the "exponentially fast" claim for the overall system convergence.

(a) Convergence of $z_0(t)$ and $q_i(t)$: Let $A_{tot}(t) = \sum_{j=1}^2(\lambda_j - \theta_{0j}q_j(t))$ and $B_{state}(t) = \sum_{j=1}^2(\theta_{1j}z_{1j}(t) + \mu_2 z_{2j}(t))$. The total matching rate $\Pi(t) = \pi_1(t) + \pi_2(t)$, as derived in Proposition EC.2 (Stage 2), is a weighted average of $A_{tot}(t)$ and $B_{state}(t)$ when both thresholds are active:

$$\Pi(t)(z_0(t) + \sum_{j=1}^2 q_j(t)) = A_{tot}(t)z_0(t) + B_{state}(t)(\sum_{j=1}^2 q_j(t))$$

This implies $\Pi(t)$ lies between $A_{tot}(t)$ and $B_{state}(t)$. The dynamics for $z_0(t)$ are $z_0'(t) = B_{state}(t) - \Pi(t)$. Let $Q(t) = \sum_{j=1}^2 q_j(t)$. From $q_j'(t)z_0(t) + q_j(t)z_0'(t) = 0$, summing over j gives $Q'(t)z_0(t) + Q(t)z_0'(t) = 0$. Also, $Q'(t) = \sum q_j'(t) = \sum(\lambda_j - \theta_{0j}q_j(t) - \pi_j(t)) = A_{tot}(t) - \Pi(t)$. Thus, $z_0'(t)$ and $Q'(t)$ have opposite signs if $A_{tot}(t) \neq B_{state}(t)$ (assuming $z_0(t) > 0, Q(t) > 0$).

Case i: $A_{tot}(t) - B_{state}(t)$ has a fixed sign for $t \geq t_{start}$ (e.g., $A_{tot}(t) \geq B_{state}(t)$). Then $A_{tot}(t) \geq \Pi(t) \geq B_{state}(t)$. This implies $z_0'(t) = B_{state}(t) - \Pi(t) \leq 0$. Since $z_0(t)$ is bounded below by $z_0^* > 0$ (as per Part 1, assuming a physical equilibrium with $z_0^* > 0$), $z_0(t)$ is non-increasing and converges to some limit $\tilde{z}_0 \geq z_0^*$. Consequently, $q_i(t) = K_i/z_0(t)$ is non-decreasing and converges to $\tilde{q}_i = K_i/\tilde{z}_0$. The state variables $z_{1j}(t), z_{2j}(t)$ are bounded (e.g., between 0 and 1). As $t \to \infty$, if the system converges to an equilibrium, then $z_0'(t) \to 0$, which implies $B_{state}(t) \to \Pi(t)$. Also, $A_{tot}(t) \to \Pi(t)$. This means $A_{tot}(\tilde{q}) = B_{state}(\tilde{z}_1, \tilde{z}_2)$. Given the uniqueness of the overall equilibrium X^* (from Part 1) where all derivatives are zero, it must be that this limit point is indeed X^*, so $\tilde{z}_0 = z_0^*$ and $\tilde{q}_i = q_i^*$. Thus, $z_0(t) \to z_0^*$ and $q_i(t) \to q_i^*$. The case $A_{tot}(t) \leq B_{state}(t)$ is analogous, leading to $z_0(t)$ being non-decreasing and converging to z_0^*.

Case ii: $A_{tot}(t) - B_{state}(t)$ changes sign or reaches 0. Suppose at some time t_1, $f(t_1) = A_{tot}(t_1) - B_{state}(t_1) = 0$. If $X(t_1)$ is the equilibrium point X^*, then $f(t) \equiv 0$ for $t \geq t_1$ as all derivatives are zero. If $X(t_1) \neq X^*$, then $q_j'(t_1) = 0$ and $z_0'(t_1) = 0$ (as shown in the reasoning for Case i when $A_{tot} = B_{state} = \Pi$). However, since $X(t_1) \neq X^*$, at least one of $z_{1j}'(t_1)$ or $z_{2j}'(t_1)$ must be non-zero. Let $Z_{sub}'(t_1)$ be the vector of these derivatives. We analyze the behavior of $f(t)$ in the neighborhood of t_1 by looking at its derivatives. $f'(t) = A_{tot}'(t) - B_{state}'(t)$. At $t = t_1$: $A_{tot}'(t_1) = \sum_{j=1}^2(-\theta_{0j}q_j'(t_1)) = 0$. So, $f'(t_1) = -B_{state}'(t_1) = -\sum_{j=1}^2(\theta_{1j}z_{1j}'(t_1) + \mu_2 z_{2j}'(t_1))$.

If $f'(t_1) \neq 0$: Since $f(t_1) = 0$, $f(t)$ will change sign locally around t_1. Specifically, for $t > t_1$ and t close to t_1, $f(t)$ will have the same sign as $f'(t_1)$. For example, if $f'(t_1) > 0$, then $f(t) > 0$ for $t \in (t_1, t_1 + \delta)$ for some $\delta > 0$. This means $A_{tot}(t) > B_{state}(t)$ in this interval, which is the scenario of Case i. Consequently, $z_0(t)$ becomes monotonic (non-increasing in this example) and converges to z_0^*.

If $f'(t_1) = 0$: We then examine $f''(t_1)$. $A''_{tot}(t_1) = \sum_{j=1}^{2}(-\theta_{0j} q''_j(t_1))$. The condition $q'_j(t) z_0(t) + q_j(t) z'_0(t) = 0$, when differentiated and evaluated at t_1 (where $q'_j(t_1) = 0, z'_0(t_1) = 0$), yields $q''_j(t_1) z_0(t_1) + q_j(t_1) z''_0(t_1) = 0$. Also, $z'_0(t) = B_{state}(t) - \Pi(t)$. When $A_{tot}(t_1) = B_{state}(t_1)$, $\Pi(t_1) = A_{tot}(t_1)$, so $z'_0(t_1) = B_{state}(t_1) - A_{tot}(t_1) = -f(t_1)$. Thus $z''_0(t_1) = -f'(t_1)$. Since we are in the subcase $f'(t_1) = 0$, we have $z''_0(t_1) = 0$. If $z_0(t_1) > 0$, then $q''_j(t_1) z_0(t_1) = 0$ which implies $q''_j(t_1) = 0$. So $A''_{tot}(t_1) = 0$. Therefore, $f''(t_1) = -B''_{state}(t_1) = -\sum_{j=1}^{2}(\theta_{1j} z''_{1j}(t_1) + \mu_2 z''_{2j}(t_1))$. If $f''(t_1) \neq 0$: Since $f(t_1) = 0$ and $f'(t_1) = 0$, $f(t)$ has a local extremum at t_1. For $t \neq t_1$ near t_1, $f(t)$ will have the sign of $f''(t_1)$. For example, if $f''(t_1) > 0$, then $f(t) > 0$ for $t \in (t_1 - \delta, t_1) \cup (t_1, t_1 + \delta)$. This means $A_{tot}(t) \geq B_{state}(t)$ locally, which again leads to the monotonic behavior of $z_0(t)$ as in Case i (allowing for $A_{tot}(t) = B_{state}(t)$ at t_1).

This process can be continued. If $f(t_1) = f'(t_1) = \cdots = f^{(k-1)}(t_1) = 0$ and $f^{(k)}(t_1) \neq 0$ for some $k \geq 1$: Then $f(t) \approx f^{(k)}(t_1) \frac{(t-t_1)^k}{k!}$ for t near t_1. If k is odd, $f(t)$ changes sign at t_1. For $t > t_1$ (locally), $f(t)$ has a fixed sign (that of $f^{(k)}(t_1)$), which leads to Case i behavior. If k is even, $f(t)$ does not change sign at t_1; it has the sign of $f^{(k)}(t_1)$ for $t \neq t_1$ locally, which also leads to Case i behavior (where $A_{tot}(t)$ is either $\geq B_{state}(t)$ or $\leq B_{state}(t)$ locally). In both situations, for $t > t_1$ and t sufficiently close to t_1, $f(t)$ maintains a fixed sign (or is zero only at t_1 if k is even). This ensures $z_0(t)$ is monotonic in $(t_1, t_1 + \delta)$ and thus converges to z_0^*.

The only scenario not covered is if $f^{(k)}(t_1) = 0$ for all $k \geq 1$. This would imply $f(t) \equiv 0$ in a neighborhood of t_1. If $A_{tot}(t) \equiv B_{state}(t)$ in a neighborhood, then $z'_0(t) = -f(t) \equiv 0$, so $z_0(t)$ is constant. This implies $q_j(t) = K_j / z_0(t)$ are also constant. So $q'_j(t) \equiv 0$. Thus $A_{tot}(t)$ is constant, which means $B_{state}(t)$ must also be constant. $B'_{state}(t) = \sum_{j=1}^{2}(\theta_{1j} z'_{1j}(t) + \mu_2 z'_{2j}(t)) \equiv 0$. The dynamics for $Z_{sub}(t) = (z_{11}(t), z_{12}(t), z_{21}(t), z_{22}(t))$ are $Z'_{sub}(t) = J_{sub}(Z_{sub}(t) - Z_{sub,eq})$, where J_{sub} and $Z_{sub,eq}$ are constant because q_j, z_0 are constant. If $Z_{sub}(t_1) \neq Z_{sub,eq}$ (which must be true if $X(t_1) \neq X^*$), then $Z_{sub}(t) - Z_{sub,eq}$ is a non-zero solution to $x' = J_{sub} x$. The condition $B'_{state}(t) \equiv 0$ means $v_B \cdot Z'_{sub}(t) \equiv 0$, where $v_B = (\theta_{11}, \theta_{12}, \mu_2, \mu_2)$ (matching the order of components in Z_{sub}). So $v_B \cdot J_{sub}(Z_{sub}(t) - Z_{sub,eq}) \equiv 0$, which implies $v_B J_{sub}^k (Z_{sub}(t_1) - Z_{sub,eq}) = 0$ for all $k \geq 0$. For a stable and observable system (J_{sub}, v_B), this implies $Z_{sub}(t_1) - Z_{sub,eq} = 0$ (observability means that $\cap_{k=0}^{3} \ker(v_B J_{sub}^k) = \{0\}$). Thus $Z_{sub}(t_1) = Z_{sub,eq}$, meaning $X(t_1)$ satisfies all equilibrium conditions, so $X(t_1) = X^*$. This contradicts the assumption that $X(t_1) \neq X^*$. Therefore, if $X(t_1) \neq X^*$, there must be some $k \geq 1$ such that $f^{(k)}(t_1) \neq 0$. As shown above, this leads to $z_0(t)$ becoming locally monotonic and thus continuing its convergence towards z_0^*.

Combining these arguments, if the system state $X(t)$ reaches a point where $A_{tot}(t) = B_{state}(t)$ but $X(t) \neq X^*$, the system dynamics will necessarily drive $A_{tot}(t) - B_{state}(t)$ to be non-zero (or at least one-sided) in the subsequent interval, re-establishing the conditions for monotonic convergence of $z_0(t)$ (and $q_i(t)$) as

described in Case i. Sustained oscillations of $A_{tot}(t) - B_{state}(t)$ around zero are not possible because X^* is the unique equilibrium.

Based on these arguments, we conclude that $z_0(t) \to z_0^*$ and $q_i(t) \to q_i^*$ as $t \to \infty$.

(b) Exponential Convergence of $z_{1i}(t)$ and $z_{2i}(t)$ given $z_0(t) \to z_0^*, q_i(t) \to q_i^*$: Once $z_0(t)$ and $q_i(t)$ are close to their equilibrium values z_0^* and q_i^*, we analyze the dynamics for the subsystem $Z_{sub}(t) = (z_{11}(t), z_{12}(t), z_{21}(t), z_{22}(t))^T$. The governing equations become approximately:

$$z_{1i}'(t) = \pi_i(z_0^*, q_1^*, q_2^*, Z_{sub}(t)) - (\theta_{1i} + \mu_{1i}^*)z_{1i}(t),$$
$$z_{2i}'(t) = \mu_{1i}^* z_{1i}(t) - \mu_2 z_{2i}(t).$$

Let $s(Z_{sub}) = \sum_{j=1}^{2}(\theta_{1j}z_{1j} + \mu_2 z_{2j})$. The matching rates π_i can be expressed as linear functions of $s(Z_{sub})$ when z_0, q_1, q_2 are fixed at their equilibrium values:

$$\pi_1(Z_{sub}) = \frac{(\lambda_1 - \theta_{01}q_1^*)(z_0^* + q_2^*) + (s(Z_{sub}) - (\lambda_2 - \theta_{02}q_2^*))q_1^*}{z_0^* + q_1^* + q_2^*} =: c_{11} + c_{12}s(Z_{sub}),$$
$$\pi_2(Z_{sub}) = \frac{(\lambda_2 - \theta_{02}q_2^*)(z_0^* + q_1^*) + (s(Z_{sub}) - (\lambda_1 - \theta_{01}q_1^*))q_2^*}{z_0^* + q_1^* + q_2^*} =: c_{21} + c_{22}s(Z_{sub}).$$

The coefficients c_{ij} are constants determined by the equilibrium values z_0^*, q_1^*, q_2^* and system parameters λ_k, θ_{0k}. Specifically, $c_{12} = \frac{q_1^*}{z_0^*+q_1^*+q_2^*}$ and $c_{22} = \frac{q_2^*}{z_0^*+q_1^*+q_2^*}$. Given $q_i^* > 0$ and $z_0^* > 0$, these satisfy $0 < c_{12} < 1$, $0 < c_{22} < 1$. Crucially, $c_{12} + c_{22} = \frac{q_1^*+q_2^*}{z_0^*+q_1^*+q_2^*} < 1$. The system for Z_{sub} is $Z_{sub}' = JZ_{sub} + K$, where $K = [c_{11}, c_{21}, 0, 0]^T$ and J is the 4×4 Jacobian matrix:

$$J = \begin{pmatrix} c_{12}\theta_{11} - \theta_{11} - \mu_{11}^* & c_{12}\theta_{12} & c_{12}\mu_2 & c_{12}\mu_2 \\ c_{22}\theta_{11} & c_{22}\theta_{12} - \theta_{12} - \mu_{12}^* & c_{22}\mu_2 & c_{22}\mu_2 \\ \mu_{11}^* & 0 & -\mu_2 & 0 \\ 0 & \mu_{12}^* & 0 & -\mu_2 \end{pmatrix}.$$

The equilibrium for this subsystem is $Z_{sub}^* = (z_{11}^*, z_{12}^*, z_{21}^*, z_{22}^*)$. We now prove that all eigenvalues of J have strictly negative real parts.

Proof that eigenvalues of J have strictly negative real parts: The matrix J is a Metzler matrix (all off-diagonal elements are non-negative, as $c_{ij} \geq 0$ and all θ, μ parameters are positive). For a Metzler matrix, stability (all eigenvalues λ satisfying $\mathrm{Re}(\lambda) < 0$) is equivalent to the existence of a strictly positive vector $v \gg 0$ (all components $v_k > 0$) such that $Jv \ll 0$ (all components of Jv are strictly negative). Let $v = [v_1, v_2, v_3, v_4]^T$. The components of Jv are:

$$(Jv)_1 = [(c_{12} - 1)\theta_{11} - \mu_{11}^*]v_1 + c_{12}\theta_{12}v_2 + c_{12}\mu_2 v_3 + c_{12}\mu_2 v_4, \tag{EC.55}$$
$$(Jv)_2 = c_{22}\theta_{11}v_1 + [(c_{22} - 1)\theta_{12} - \mu_{12}^*]v_2 + c_{22}\mu_2 v_3 + c_{22}\mu_2 v_4, \tag{EC.56}$$
$$(Jv)_3 = \mu_{11}^* v_1 - \mu_2 v_3, \tag{EC.57}$$
$$(Jv)_4 = \mu_{12}^* v_2 - \mu_2 v_4. \tag{EC.58}$$

We need to find $v_k > 0$ such that $(Jv)_k < 0$ for $k = 1, 2, 3, 4$.

From (EC.57), we require $\mu_{11}^* v_1 - \mu_2 v_3 < 0$. Choose $v_3 = (1+\epsilon)\frac{\mu_{11}^*}{\mu_2} v_1$ for some small $\epsilon > 0$. Then,

$$(Jv)_3 = \mu_{11}^* v_1 - \mu_2(1+\epsilon)\frac{\mu_{11}^*}{\mu_2} v_1 = \mu_{11}^* v_1 - (1+\epsilon)\mu_{11}^* v_1 = -\epsilon\mu_{11}^* v_1.$$

Since $v_1 > 0$ and $\epsilon > 0$, $(Jv)_3 < 0$.

Similarly, from (EC.58), we require $\mu_{12}^* v_2 - \mu_2 v_4 < 0$. Choose $v_4 = (1+\epsilon)\frac{\mu_{12}^*}{\mu_2} v_2$ for the same $\epsilon > 0$. Then,

$$(Jv)_4 = \mu_{12}^* v_2 - \mu_2(1+\epsilon)\frac{\mu_{12}^*}{\mu_2} v_2 = -\epsilon\mu_{12}^* v_2.$$

Since $v_2 > 0$ and $\epsilon > 0$, $(Jv)_4 < 0$.

Substitute these choices for v_3 and v_4 into (EC.55) and (EC.56). For (EC.55):

$$\begin{aligned}(Jv)_1 &= [(c_{12}-1)\theta_{11} - \mu_{11}^*]v_1 + c_{12}\theta_{12}v_2 + c_{12}\mu_2(1+\epsilon)\frac{\mu_{11}^*}{\mu_2}v_1 + c_{12}\mu_2(1+\epsilon)\frac{\mu_{12}^*}{\mu_2}v_2 \\ &= [(c_{12}-1)\theta_{11} - \mu_{11}^* + c_{12}(1+\epsilon)\mu_{11}^*]v_1 + [c_{12}\theta_{12} + c_{12}(1+\epsilon)\mu_{12}^*]v_2.\end{aligned}$$

Let $P_1 = \theta_{11} + \mu_{11}^* > 0$ and $P_2 = \theta_{12} + \mu_{12}^* > 0$. Then $(Jv)_1$ becomes:

$$(Jv)_1 = [(c_{12}-1)P_1 + c_{12}\epsilon\mu_{11}^*]v_1 + [c_{12}P_2 + c_{12}\epsilon\mu_{12}^*]v_2. \tag{EC.59}$$

Similarly, $(Jv)_2$ becomes:

$$(Jv)_2 = [c_{22}P_1 + c_{22}\epsilon\mu_{11}^*]v_1 + [(c_{22}-1)P_2 + c_{22}\epsilon\mu_{12}^*]v_2. \tag{EC.60}$$

We first choose $v_1, v_2 > 0$ such that the main terms (without ϵ) in (EC.59) and (EC.60) are negative:

$$(c_{12}-1)P_1 v_1 + c_{12}P_2 v_2 < 0, \tag{EC.61}$$

$$c_{22}P_1 v_1 + (c_{22}-1)P_2 v_2 < 0. \tag{EC.62}$$

Since $0 < c_{12} < 1$ and $0 < c_{22} < 1$ (as $q_i^*, z_0^* > 0$), we have $c_{12} - 1 < 0$ and $c_{22} - 1 < 0$. From (EC.61), $\frac{v_2}{v_1} < \frac{(1-c_{12})P_1}{c_{12}P_2}$. From (EC.62), $\frac{v_2}{v_1} > \frac{c_{22}P_1}{(1-c_{22})P_2}$. A valid ratio v_2/v_1 exists if $\frac{c_{22}P_1}{(1-c_{22})P_2} < \frac{(1-c_{12})P_1}{c_{12}P_2}$, which simplifies to $\frac{c_{22}}{1-c_{22}} < \frac{1-c_{12}}{c_{12}}$ (since $P_1, P_2 > 0$). This further simplifies to $c_{12}c_{22} < (1-c_{12})(1-c_{22})$, implying $c_{12} + c_{22} < 1$. To show this:

$$c_{12} + c_{22} = \frac{q_1^*}{z_0^* + q_1^* + q_2^*} + \frac{q_2^*}{z_0^* + q_1^* + q_2^*} = \frac{q_1^* + q_2^*}{z_0^* + q_1^* + q_2^*}.$$

Since the equilibrium assumes active matching thresholds $\mu_{1i}^* > 0$, it implies $C(q_i^*)^\alpha (z_0^*)^\alpha = \mu_{1i}^* > 0$. This requires $q_i^* > 0$ and $z_0^* > 0$. Therefore, the denominator $z_0^* + q_1^* + q_2^*$ is strictly greater than the numerator $q_1^* + q_2^*$ (as $z_0^* > 0$). Thus, $\frac{q_1^* + q_2^*}{z_0^* + q_1^* + q_2^*} < 1$. This condition is satisfied.

Thus, we can choose $v_1, v_2 > 0$ such that for some $\delta_1, \delta_2 > 0$:

$$(c_{12} - 1)P_1 v_1 + c_{12} P_2 v_2 = -\delta_1,$$
$$c_{22} P_1 v_1 + (c_{22} - 1)P_2 v_2 = -\delta_2.$$

Then, (EC.59) and (EC.60) become:

$$(Jv)_1 = -\delta_1 + \epsilon[c_{12}\mu_{11}^* v_1 + c_{12}\mu_{12}^* v_2],$$
$$(Jv)_2 = -\delta_2 + \epsilon[c_{22}\mu_{11}^* v_1 + c_{22}\mu_{12}^* v_2].$$

Since $\delta_1, \delta_2 > 0$, and the terms multiplied by ϵ are positive (as $v_1, v_2 > 0$, $c_{ij} \geq 0$, $\mu_{ij}^* > 0$), we can choose a sufficiently small $\epsilon > 0$ such that $(Jv)_1 < 0$ and $(Jv)_2 < 0$. With such $v_1, v_2 > 0$ and $\epsilon > 0$, we have constructed $v \gg 0$ such that $Jv \ll 0$. Therefore, all eigenvalues of J have strictly negative real parts. *End of proof for eigenvalues of J.*

Since all eigenvalues of J have strictly negative real parts, the solution $Z_{sub}(t)$ of the linear system $Z'_{sub} = JZ_{sub} + K$ converges exponentially fast to its unique equilibrium $Z^*_{sub} = -J^{-1}K = (z_{11}^*, z_{12}^*, z_{21}^*, z_{22}^*)$.

Combining parts (a) and (b): $z_0(t)$ and $q_i(t)$ converge to their equilibrium values. As they approach these values, the dynamics for $z_{1i}(t)$ and $z_{2i}(t)$ become those of a stable linear system, ensuring that these components also converge exponentially to their respective equilibrium values. This establishes the exponential convergence of the overall fluid process $(q_i(t), z_0(t), z_{1i}(t), z_{2i}(t))$ to the unique equilibrium point X^*. $\square$